Complex Numbers

06 November 2025

Revision: 4989

Aziz Manva

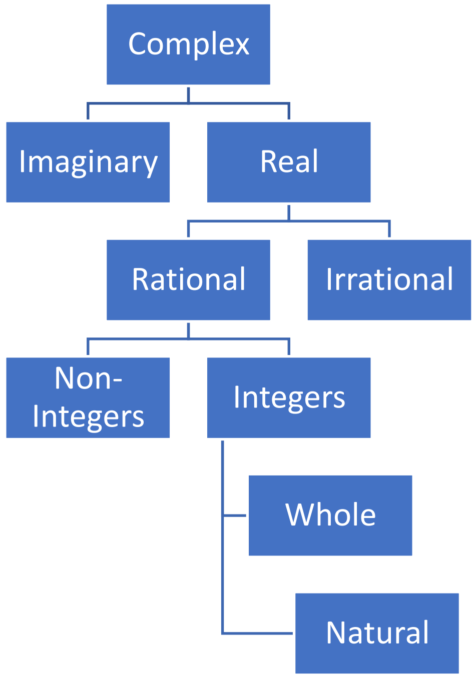
azizmanva@gmail.com

Table of contents

1. Complex Numbers: Algebra

1.1 Imaginary Numbers

A. Development of the Number System



1.1: Natural Numbers
Natural can be defined as the positive integers:

$$\mathbb{N} = \{1,2,3,4 \dots\}$$

You would have learnt about the natural numbers as a child.

1.2: Whole Numbers
Whole numbers can be defined as the nonnegative integers:

$$\mathbb{W} = \{0,1,2,3 \dots\}$$

At some stage, the number zero was introduced giving rise to the set of whole numbers. The concept of zero seems natural but it has not always been around. The Romans, for example, did not use zero.
The set of whole numbers is closed under addition. This means that if we add two whole numbers, we will always get a whole number.
However, the set of whole numbers is not closed under subtraction.
To solve this problem, the negative numbers are introduced.

1.3: Integers

$$\underbrace{\mathbb{Z}}_{Preferred} = \mathbb{I} = \{\dots -3, -2, -1, 0, 1, 2, 3 \dots\}$$

The set of integers is closed under both addition and subtraction. It is closed under multiplication as well (as in fact, are the natural numbers). But the set of integers is not closed under division. To solve this problem, we introduced rational numbers, defined as

$$\pi$$

1.4: Rational Numbers

$$\mathbb{Q} = \left\{x | x = \frac{p}{q}, \qquad p, q \in \mathbb{Z}, \qquad q \neq 0\right\}$$

However, the development of the number system is not yet complete. The square roots of prime numbers are not rational (as can be proved, this was known to the Ancient Greeks). Similarly, numbers like π and e are not rational. To accommodate, them we introduce the set of irrational numbers, and call the collection of rational and irrational numbers as real numbers.

1.5: Irrational Numbers
Numbers which are not rational are irrational

1.6: Real Numbers
The union of rational numbers and irrational numbers form the real numbers.
Real numbers are found on the real number line.

We have now reached the next step in the development of the number system, and will add the imaginary numbers. This will lead to complex numbers.

B. Imaginary Numbers: Background

Imaginary numbers were first encountered in the study of quadratic equations. For a long time, they were thought to not be meaningful. Consider the equation:

$$x^2+1=0 \Rightarrow \underbrace{x^2=-1}_{\text{Subtract 1 from both sides}} \Rightarrow \underbrace{x=\pm\sqrt{-1}}_{\text{Take square roots both sides}} \Rightarrow \underbrace{x=\pm i}_{\text{Let } \sqrt{-1}=i=\text{iota}}$$

We cannot proceed beyond this using the real number system. And hence the subject was ignored by many mathematicians. Finally, they were studied by Cardano in arriving at a general formula for the solution of a quadratic equation.

Even today, there is no solution for the expression

$$\sqrt{-1}$$

Instead, we treat i as a variable, and apply various algebraic properties to it.

C. Complex Numbers: Usefulness

Complex numbers are a subject of study in their own right, useful in

- Finding solutions to equations of polynomials
- In trigonometry
- In engineering (especially electrical engineering)

D. Defining i

$$x^2=1 \Leftrightarrow x=\pm 1$$

However, the corresponding equation

$$x^2=-1 \Leftrightarrow x=\pm\sqrt{-1}$$

Does not have any solution in terms of real numbers, since the square of a positive quantity, as well as the square of a negative quantity, is always positive. The equation cannot be assigned any meaning in terms of real numbers.

(As an example, consider the fraction $\frac{3}{2}$. It does not have a meaning in terms of integers – unless you extend integers in such a way as to include fractions. Similarly, complex numbers do not have meaning, or cannot be explained only by using real numbers.)

1.7: Definition of i

$$i=\sqrt{-1}$$

The i is called *iota*.

The rules for manipulating imaginary numbers depend on algebra. By treating the quantity i like a variable. In certain places, we can eliminate i, letting us assign a meaning in the real number system to certain (not all) expressions.

1.8: Imaginary Number

An imaginary number is a number of the form

$$bi, \qquad i=\sqrt{-1}, \qquad b\in\mathbb{R}$$

1.9: Addition and Subtraction

When adding and subtracting imaginary numbers, you can treat the i like an unknown variable. Just as

$$4\ Apples + 3\ Apples = 7\ Apples$$

Similarly

$$4i + 3i = 7i$$

In particular

$$i + 0 = i + 0i = i$$

Example 1.10: Addition and Subtraction

Simplify:

A. $4i - 3i$
B. $7i - 7i$
C. $22i + 30i$
D. $5i - 9i$
E. $\frac{1}{2}i + \frac{1}{3}i$
F. $2.3i + 0.5i$

$$4i - 3i = i$$
$$7i - 7i = 0$$
$$22i + 30i = 52i$$
$$2.3i + 0.5i = 2.8i$$
$$4i - 3i = i$$
$$7i - 7i = 0i = 0$$

1.11: Multiplication

You can multiply by a real number by multiplying it with the real number coefficient:

$$k \times bi = (kb)i$$

The familiar property that any multiple of zero is zero is preserved in the imaginary numbers:

$$0 \times i = 0$$

Example 1.12: Multiplication

Simplify:

A. $2 \times 3i$
B. $3 \times \frac{1}{3}i$
C. $\frac{1}{2} \times \frac{3}{5}i$
D. $\frac{2}{3} \times \left(-\frac{1}{4}\right)i$
E. $0 \times 4i$
F. $3i \times 0$

$$2 \times 3i = (2 \times 3)i = 6i$$
$$3 \times \frac{1}{3}i = i$$
$$\frac{1}{2} \times \frac{3}{5}i = \frac{3}{10}i = 0.3i$$

$$\frac{5i}{7 + i}$$

1.13: Powers of iota

The squaring operation and the square root operation cancel as can be seen below:

$$x = \sqrt{7} \Rightarrow x^2 = 7, \qquad x = \sqrt{y} \Rightarrow x^2 = y$$

We use this property to find powers of i.

$$i^2 = -1$$
$$i^3 = -i$$
$$i^4 = 1$$

And as already defined:

$$i = \sqrt{-1}$$

Square of *i*

$$i = \sqrt{-1}$$

Square both sides of the above to find that $i = -1$. This is not surprising at all, since this precisely the equation that we started with:

$$i^2 = \sqrt{-1} \times \sqrt{-1} = \left(\sqrt{-1}\right)^2 = -1$$

This is similar to the expression below where the square and the square root "cancel":

$$\left(\sqrt{4}\right)^2 = 4$$

Cube of *i*

Before we find the cube, recall that we can break exponents using the properties of exponents:

$$2^3 = 8 = 2 \times 4 \Rightarrow 2^3 = 2 \times 2^2$$

Simplify i^3 by expanding:

$$i^3 = \underbrace{i \cdot i^2}_{\textit{Break up } i^3} = \underbrace{i \cdot -1}_{\textit{Substitute } i^2 = -1} = -i$$

Fourth Power of *i*

Simplify i^4 by expanding:

$$i^4 = \underbrace{i^2 \cdot i^2}_{\textit{Break up } i^4} = \underbrace{(-1)(-1)}_{\textit{Substitute } i^2 = -1} = 1$$

Example 1.14

Find the value of each expression below

Square of *i*

A. $(-i)^2$
B. $(3i)^2$
C. $\left(-\frac{2}{3}i\right)^2$

Cube of *i*

D. $(3i)^3$
E. $(-2i)^3$
F. $\left(-\frac{3}{4}i\right)^3$

Fourth Power of *i*

G. $(2i)^4$
H. $(-3i)^4$
I. $\left(\frac{2}{5}i\right)^4$

Expressions

J. $a + a^2 + a^3 + a^4, a = -2i$

K. $2b - b^2 + 3b^3 - b^4, b = \frac{1}{2}i$

Square of i

$$(-i)^2 = (-i)(-i) = (-1)(-1)(i)(i) = 1 \times i^2 = -1$$
$$(3i)^2 = 9i^2 = 9(-1) = -9$$
$$\left(-\frac{2}{3}i\right)^2 = \left(-\frac{2}{3}\right)^2 i^2 = \frac{4}{9}(-1) = -\frac{4}{9}$$

Cube of i

$$(3i)^3 = (3^3)(i^3) = (27)(-i) = -27i$$
$$(-2i)^3 = (-2)^3(i^3) = (-8)(-i) = 8i$$
$$\left(-\frac{3}{4}i\right)^3 = \left(-\frac{3}{4}\right)^3 (i^3) = -\frac{27}{64}(-i) = \frac{27}{64}i$$

Fourth Power of i

$$(2i)^4 = (2^4)(i^4) = (16)(1) = 16$$
$$(-3i)^4 = (-3)^4(i^4) = (81)(1) = 81$$

Expressions

$$a = -2i$$
$$a^2 = (-2i)^2 = (-2)^2(i^2) = (4)(-1) = -4$$
$$a^3 = (-2i)^3 = (-2)^3(i^3) = (-8)(-i) = 8i$$
$$a^4 = (-2i)^4 = (-2)^4(i^4) = (16)(1) = 16$$

$$(-2i) + (-2i)^2 + (-2i)^3 + (-2i)^4$$
$$= -2i - 4 + 8i + 16 = 12 + 6i$$

$$2b = 2\left(\frac{1}{2}i\right) = i$$
$$-b^2 = -\left(\frac{1}{2}i\right)^2 = -\left(\frac{1}{4}(-1)\right) = \frac{1}{4}$$
$$3b^3 = 3\left(\frac{1}{2}i\right)^3 = 3\left(\frac{1}{8}\right)(-i) = -\frac{3}{8}i$$
$$-b^4 = -\left(\frac{1}{2}i\right)^4 = -\left(\frac{1}{16}\right)(1) = -\frac{1}{16}$$

Substitute the above values in $2b - b^2 + 3b^3 - b^4$ to get:

$$i + \frac{1}{4} - \frac{3}{8}i - \frac{1}{16} = \frac{5}{8}i + \frac{3}{16}$$

E. Reducing Powers of iota

1.15: Reducing powers of iota

Any power of iota does not change upon replacing its exponent by the remainder when the power is divided by 4.

$$\boldsymbol{i^p = i^r}, \qquad \textbf{where } \boldsymbol{r} \textbf{ is the remainder on dividing } \boldsymbol{p} \textbf{ by 4}$$

Higher powers of i can be simplified by rewriting the power as a multiple of 4, and some remainder.

$$i^p = i^{4q} \times i^r = (i^4)^q \times i^r = 1^q \times i^r = 1 \times i^r = i^r$$

Example 1.16

Simplify:

A. i^{97}
B. i^{54}
C. i^{87}
D. i^{2023}
E. $-10i^{30}+8i^{57}+i^{88}+9i^{34}$

97 has remainder 1 when divided by 4. Use this to rewrite:

$$i^{97}=\underbrace{i^{96}\times i}_{97=24\times 4+1}=(i^4)^{24}\times i=1^{24}\times i=1\times i=i$$

$$i^{54}=(i^4)^{13}\times i^2=1^{13}\times(-1)=-1$$
$$i^{87}=(i^4)^{21}\times i^3=-i$$

2023 has remainder 3 when divided by 4. Use this to rewrite:

$$i^{2023}=i^{2020}\times i^3=(i^4)^{505}\times i^3=(1)^{505}\times i^3=1\times i^3=i^3=-i$$

$$Shortcut: i^{2023}=i^3=-i$$

$$-10i^{30}+8i^{57}+i^{88}+10i^{34}$$
$$=-10i^2+8i^1+i^0+9i^2$$
$$=8i+1-i^2$$
$$=8i+1-(-1)$$
$$=8i+2$$

Example 1.17

Simplify each element of the sets below:

A. $i+i^2$
B. $i+i^2+i^3$
C. $i+i^2+i^3+i^4$

$$i+i^2=i+(-1)=i-1$$

$$i+i^2+i^3=i-1-i=-1$$
$$i+i^2+i^3+i^4=i-1-i+1=0$$

Example 1.18

$$\frac{i^2+i^4+i^6+\cdots+i^{100}}{i^2\times i^4\times i^6\times\ldots\times i^{100}}$$

Rewrite the numerator:

$$i^2+(i^2)^2+(i^2)^3+(i^2)^4+\cdots+(i^2)^{49}+(i^2)^{50}$$

Substitute $i^2=-1$:

$$(-1)+(-1)^2+(-1)^3+(-1)^4+\cdots+(-1)^{49}+(-1)^{50}$$

Even powers of -1 are -1, and odd powers of -1 are 1:

$$\underbrace{-1+1}_{Add\ to\ zero}+\underbrace{(-1+1)}_{Add\ to\ zero}+\cdots+\underbrace{(-1+1)}_{Add\ to\ zero}$$

Two consecutive terms of the above expression add to zero. We have 50 pairs. So, the entire expression is zero.

Hence, we get:

$$\frac{i^2 + i^4 + i^6 + \cdots + i^{100}}{i^2 \times i^4 \times i^6 \times \ldots \times i^{100}} = \frac{0}{i^2 \times i^4 \times i^6 \times \ldots \times i^{100}} = 0$$

We did not need the denominator at all. However, we can still simplify it:

$$i^2 \times (i^2)^2 \times (i^2)^3 \times \ldots \times (i^2)^{50}$$

Substitute $i^2 = -1$:

$$(-1) \times (-1)^2 \times (-1)^3 \times \ldots \times (-1)^{50} = (-1)^{1+2+3+\cdots+50} = (-1)^{\frac{50\times 51}{2}} = (-1)^{25\times 51} = -1$$

1.19: Sum of four consecutive powers of i is zero

The sum of any four consecutive, integral powers of $\boldsymbol{i}$ is zero

$$\boldsymbol{i^x + i^{x+1} + i^{x+2} + i^{x+3} = 0, \qquad x \in \mathbb{Z}}$$

$$\underbrace{i^x(1 + i + i^2 + i^3\ )}_{Factoring\ out\ i^x} = \underbrace{i^x(1 + i - 1 - i\ )}_{Substituting\ values} = i^x(0) = 0$$

Since x can be negative, this above identity is true for negative values of x as well.

Example 1.20

Show that $i^{-666} + i^{-667} + i^{-668} + i^{-669} = 0$

Take the lowest power of iota common, and factor the LHS:

$$i^{-669}(i^3 + i^2 + i^1 + 1) = i^{-669}(0) = 0$$

F. Sequences with iota

Example 1.21

If n is a multiple of 4, the sum $s = 1 + 2i + 3i^2 + \cdots + (n+1)i^n$, where $i = \sqrt{-1}$ equals **(AHSME 1964/34)**

Since i has a cyclicity of 4, consider groups of 4 terms:

$$x + (x+1)i + (x+2)i^2 + (x+3)i^3$$
$$= x + (xi + i) + (-x - 2) + (-xi - 3i)$$
$$= -2 - 2i$$

If $n = 0$, then the sum

$$= S_1 = 1$$

If $n = 4$, then the sum to 5 five terms:

$$= S_4 + T_5 = -2 - 2i + 5i^4 = -2 - 2i + 5 = 3 - 2i$$

If $n = 8$, then the number of terms is 9. Sum to 9 five terms:

$$= S_8 + T_9 = 2(-2 - 2i) + 9i^8 = -4 - 4i + 9 = 5 - 4i$$

If $n = 12$,

$$= S_{12} + T_{13} = 7 - 6i$$

The real part and imaginary part each form a sequence:

$$3,5,7,9, \ldots \Rightarrow \frac{n}{2} + 1$$

$$-2i, -4i, -6i \Rightarrow -\frac{n}{2}i$$

$$\underbrace{\left(\frac{n}{2}+1\right)}_{\textit{Real Part}} + \underbrace{\left(-\frac{n}{2}i\right)}_{\textit{Imaginary Part}}$$

Example 1.22

If $i^2 = -1$, then the sum

$$\cos 45° + i\cos 135° + \cdots + i^n \cos(45+90n)^{\circ} + \cdots + i^{40}\cos 3645°$$

can be written in the form $\frac{\sqrt{2}}{2}x$. Find x. (AHSME 1977/16)

Powers of i cycle every 4 terms.
The cos terms also cycle every

$$= \frac{360}{90} = 4$$

$$t_1 = t_5 = t_9 = t_{4n+1}$$
$$t_2 = t_6 = t_{10} = t_{4n+2}$$
$$t_3 = t_7 = t_{11} = t_{4n+3}$$
$$t_4 = t_8 = t_{12} = t_{4n}$$

$$t_1 + t_3 = \cos 45° + i^2 \cos 225° = \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$$
$$t_2 + t_4 = i\cos 135° + i^3 \cos 315° = i\left(-\frac{1}{\sqrt{2}}\right) - i\left(\frac{1}{\sqrt{2}}\right) = -i\sqrt{2}$$

Note that each group of four terms has the same total.

$$t_1 + t_2 + t_3 + t_4 = t_5 + t_6 + t_7 + t_8 = t_{4n+1} + t_{4n+2} + t_{4n+3} + t_{4n}$$

The first 40 terms will have a total of:

$$S_{40} = 10\,[t_1 + t_2 + t_3 + t_4] = 10\left[\sqrt{2} - i\sqrt{2}\right] = 10\sqrt{2} - 10i\sqrt{2}$$
$$t_{41} = t_1 = \cos 45° = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Final answer:

$$S_{41} = S_{40} + t_{41} = 10\sqrt{2} - 10i\sqrt{2} + \frac{\sqrt{2}}{2} = \frac{21\sqrt{2}}{2} - 10i\sqrt{2} = \frac{\sqrt{2}}{2}(21 - 20i)$$

Example 1.23

Determine the cyclicity of

$$\cos 45° + i\cos 105° + \cdots + i^n \cos(45+60n)^{\circ} + \cdots +$$

(AHSME 1977/16, Adapted)

Powers of i cycle every 4 terms.
The cos terms cycle every

$$\frac{360}{60} = 6\ \textit{terms}$$

We want a complete cycle of i. And we also want a complete cycle of cos. Hence, the number of terms that we can group together will be

$$LCM(4,6) = 12$$

G. Domain

Example 1.24

For how many real numbers x is $\sqrt{-(x+1)^2}$ a real number? **(AHSME 1976/2)**

For the expression to be a real number, the quantity inside the square root must be nonnegative. That is:

$$-(x+1)^2 \geq 0$$

Multiply both sides by -1, and reverse the sign of the inequalities:

$$(x+1)^2 \leq 0$$

Since the LHS is a perfect square, it cannot be less than zero. The inequality converts into an equality:

$$(x+1)^2 = 0$$

Take the square root both sides:

$$x + 1 = 0$$
$$x = -1$$
$$One\ Solution$$
$$One\ Real\ Number$$

1.2 Complex Numbers

A. Complex Numbers: Algebraic Definition

1.25: Set of Complex Numbers

The set of complex numbers is denoted $\mathbb{C}$.

1.26: Complex Number Definition

A complex number is a number which has a real part, and an imaginary part, and can be written in the form

$$z \in \mathbb{C}$$
$$z = a + bi, \qquad a, b \in \mathbb{R}, \qquad i = \sqrt{-1}$$

Notes:

- a and b can be zero
- The variable z is often used to represent complex numbers.

Example 1.27

Mark all correct options

Is the number -1:

A. A real number
B. An imaginary number
C. A Complex Number

$$-1\ is\ Real$$

$$-1 \text{ is not imaginary}$$
$$-1 = -1 + 0i \text{ is complex}$$

$$\text{Options } A \text{ and } C$$

1.28: Functions

- Functions are a rule that assign an output for every valid input.
- The set of valid inputs is called the domain.
- The set of valid outputs is called the range.
- Functions can have complex numbers for their domain and range.

Example 1.29

Determine the domain and range of the constant function
$$f(z) = z$$

$f: \mathbb{C} \to \mathbb{C}$ is a function with
$$Domain = Range = All\ Complex\ Numbers$$

1.30: Standard Form

The standard form of a complex number is:
$$z = a + bi$$

1.31: Real and Imaginary Part

In a complex number $a + bi$,
$$Real\ part = Re(z) = Re(a + bi) = a$$
$$Imaginary\ part = Im(z) = Im(a + bi) = b$$

$$Re(z) \text{ and } Im(z) \text{ are functions}$$

Example 1.32

Identify the domain and range of the functions $Re(z)$ and $Im(z)$

Domain is the set of acceptable input values. Hence,
$$Domain\ of\ Re(z) = Domain\ of\ Im(Z) = \mathbb{C}$$
$$Range\ of\ Re(z) = Range\ of\ Im(Z) = \mathbb{R}$$

$Re(z)$ is a function $f: \mathbb{C} \to \mathbb{R}$
$Im(z)$ is a function $f: \mathbb{C} \to \mathbb{R}$

Example 1.33

For each complex number below, identify the real part and the imaginary part of each:

A. $3 + 4i$
B. $5 + 7i$
C. $\frac{2}{5} + \frac{3}{4}i$

$$3 + 4i \Rightarrow Re(3 + 4i) = 3, \qquad Im(3 + 4i) = 4$$
$$5 + 7i \Rightarrow Re(5 + 7i) = 5, \qquad Im(5 + 7i) = 7$$

$$\frac{2}{5}+\frac{3}{4}i \Rightarrow Re\left(\frac{2}{5}+\frac{3}{4}i\right)=\frac{2}{5}, \qquad Im\left(\frac{2}{5}+\frac{3}{4}i\right)=\frac{3}{4}$$

Example 1.34

For each complex number below, identify the real part and the imaginary part of each:

A. $9i$
B. 7

$$9i \Rightarrow Re(9i)=0, \qquad Im(9i)=9i$$
$$7 \Rightarrow Re(7)=7, \qquad Im(\quad)=9i$$

Example 1.35

True or False

A. If $z=2+7i$, then $Re(z)=2$, and $Im(z)=7i$

$$False. Im(z)=7, not\ 7i$$

B. Complex Numbers: Geometric Definition

Complex numbers can be represented on a complex plane.

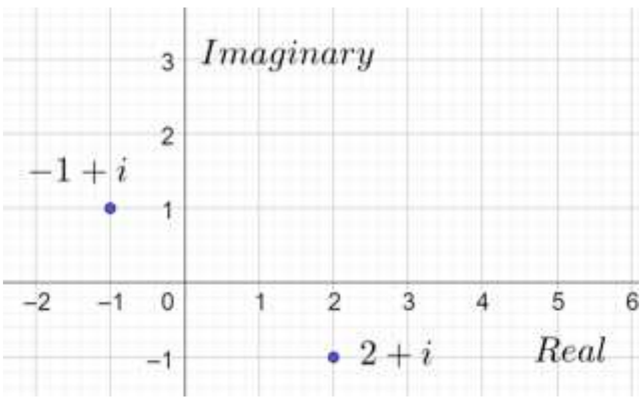
1.36: Complex Plane

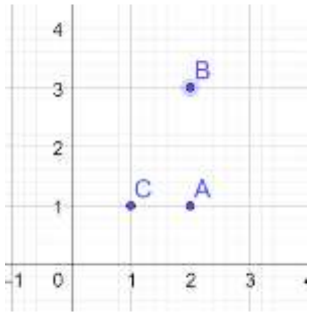
- A coordinate plane with a on the x-axis, and b on the y-axis is a complex plane.
- A number in the plane with coordinates (a,b) is the complex number $a+bi$
- Every complex point on the complex plane is uniquely associated with a complex number, and vice versa.
- The x-axis measures the real part of the complex number, and the y-axis measures the imaginary part of the complex number.

Example 1.37

Plot the following numbers in the complex plane:

A. $z_1=2-i$
B. $z_2=-1+i$





Example 1.38

Identify the values of the following points plotted in the complex plane.

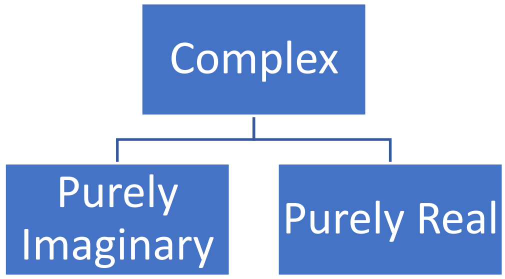
$$A = 2 + i$$
$$B = 2 + 3i$$
$$C = 1 + i$$

C. Purely Real and Purely Imaginary Numbers

1.39: Classifying Complex Numbers

Numbers for which the:

- real part is zero are called purely imaginary.
- imaginary part is zero are called purely real.



A purely real number must be of the form

$$z = a + 0i$$

A purely imaginary number must be of the form

$$z = 0 + bi$$

Example 1.40

Identify which numbers are purely real, purely imaginary, or neither.

A. $3i$
B. 4
C. $2 + 3i$

$$3i \Rightarrow Purely\ Imaginary$$
$$4 \Rightarrow Purely\ Real$$
$$2 + 3i \Rightarrow Neither$$

Example 1.41

Given the complex numbers $\{3 + 4i, 2i, 7\}$:

A. Identify the real part and the imaginary part of each
B. State numbers, if any, which are purely real, and purely imaginary

$$Part\ A:\ \underbrace{3}_{\substack{Real\\Part}} + \underbrace{4}_{\substack{Imaginary\\Part}} i, \quad \underbrace{0}_{\substack{Real\\Part}} + \underbrace{2}_{\substack{Imaginary\\Part}} i, \quad \underbrace{7}_{\substack{Real\\Part}} + \underbrace{0}_{\substack{Imaginary\\Part}} i$$

$$Part\ B: Purely\ Real: 7, \qquad Purely\ Complex\ Numbers: 2i$$

Example 1.42

The number zero is:

A. Purely real
B. Purely imaginary
C. Both of the above

D. None of the above

Option C

1.43: Purely Real and Purely Imaginary Numbers

A number lies on the real axis in the complex plane if and only if it is purely real.
A number lies on the imaginary axis in the complex plane if and only if it is purely imaginary.

Example 1.44

A. A number z_1 lies on both the imaginary and the real axis in the complex plane. What is its value?
B. A number z_2 lies on the real axis three units away from the origin. What are the possible values of the number?
C. A number z_3 lies on the imaginary axis two units away from the origin. What are the possible values of the number?

$$z_1 = 0 + 0i$$
$$|z_2| = 3 \Rightarrow z_2 = \pm 3$$
$$|z_3| = 2 \Rightarrow z_2 = \pm 2i$$

MCMC 1.45

Mark All Correct Options

A number lies in the complex plane on neither the real axis nor the imaginary axis. Then:

A. It is purely real.
B. It is not purely real.
C. It is purely imaginary
D. It is not purely imaginary

$$Options\ B, D$$

Example 1.46

A number on the real axis in the complex plane must have a:

A. non-zero Real part
B. non-zero Imaginary part
C. Both of the above
D. None of the above

Consider the number $0 + 0i$, which lies on the real axis. It has a zero real part and a zero imaginary part. Hence:

$$Option\ D\ is\ correct$$

D. Equality of Numbers

1.47: Equality of Complex Numbers: Algebraic

Given two complex numbers $z_1 = a + bi$ and $z_2 = c + di$, $z_1 = z_2$ if and only if

$$a = c$$
$$b = d$$

Two complex numbers are equal *if and only if*:

- Their real parts are equal
- Their imaginary parts are also equal

Example 1.48

Find the values of a and b for each part below:

A. $a + bi = 4 + 7i$
B. $a + 4i = 3 + bi$
C. $\frac{2}{9} + bi = a + \frac{2}{3}i$
D. $0.3 + bi = a + 1.1i$

Since the numbers are equal, their real and imaginary parts must be equal

$$Part\ A: a = 4, b = 7$$
$$Part\ B: a = 3, b = 4$$

1.49: Equality of Complex Numbers: Geometric

Two complex numbers are equal if they represent the same point in the complex plane.

Example 1.50

When is $a + bi = b + ai$

Equate the real parts

$$a = b$$

Equate the imaginary parts

$$b = a$$

E. Well-Ordering Property

1.51: Well Ordering Property

The well ordering property holds for real numbers, and it lets us compare them.
Given two real numbers a and b, we can say that precisely one of the following three relations holds:

$$a < b$$
$$a = b$$
$$a > b$$

There is no fourth possibility.

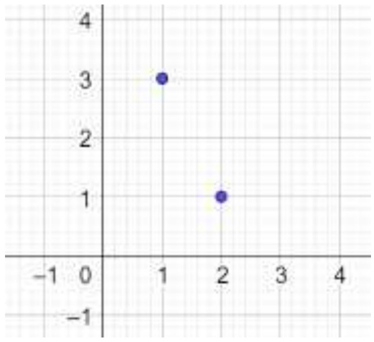
Example 1.52

Compare:

A. $4\ and\ 5$
B. $6\ and\ \frac{12}{2}$
C. $8\ and\ -2$

$$4 < 5$$
$$6 = \frac{12}{2}$$
$$8 > -2$$

1.53: Complex Numbers
The complex numbers are not well-ordered.
Complex numbers cannot be compared to find the greater or the smaller, only to check for equality.



In acquiring the complexity and the sophistication to handle square roots of negative numbers, there is a loss as well. Two complex numbers cannot be compared to find the greater or the smaller.

Example 1.54
Out of $1 + 2i$ and $3 - i$, which one is greater?

$$\textit{Cannot be determined}$$

1.3 Addition, Subtraction and Conjugation

A. Addition and Subtraction: Algebraic Definition

1.55: Addition and Subtraction
To add two complex numbers, we add the real and imaginary parts separately
$$(a + bi) + (c + di) = (a + c) + (b + d)i$$
$$(a + bi) - (c + di) = (a - c) + (b - d)i$$

Example 1.56
Let $x = 3 + 4i$ and $y = 7 + 5i$. Find:
A. $x + y$
B. $x - y$

$$x + y = (3 + 4i) + (7 + 5i) = (3 + 7) + (4i + 5i) = 10 + 9i$$
$$x - y = (3 + 4i) - (7 + 5i) = (3 - 7) + (4i - 5i) = -4 - i$$

Example 1.57
Let $x = \frac{2}{3} + 4i$ and $y = \frac{1}{4} - 0.01i$. Find:
A. $x + y$
B. $x - y$

$$x + y = \frac{2}{3} + 4i + \left(\frac{1}{4} - 0.01i\right) = \frac{2}{3} + \frac{1}{4} + 4i - 0.01i = \frac{8}{12} + \frac{3}{12} - 3.99i = \frac{11}{12} - 3.99i$$
$$x - y = \frac{2}{3} + 4i - \left(\frac{1}{4} - 0.01i\right) = \frac{2}{3} - \frac{1}{4} + 4i + 0.01i = \frac{8}{12} - \frac{3}{12} + 4.01i = \frac{5}{12} + 4.01i$$

Example 1.58
Let $x = \sqrt[3]{16} + \sqrt{8}i$ and $y = \sqrt[3]{128} - 0.01i$. Find:
A. $x + y$
B. $x - y$

Example 1.59: Linear Equations

A. When a number is added to $2+7i$, the result is $1+4i$. Find the number.
B. When a number is added to $4+3i$, the result is $2+9i$. Find the number.
C. When a number is subtracted from $3+5i$, the result is $-2+7i$. Find the number.
D. When a number is subtracted from $-2+12i$, the result is $5+4i$. Find the number.

Part A

Let the complex number be z. Then, by the condition given in the question:

$$z+2+7i=1+4i \Rightarrow z=-1-3i$$
$$z+4+3i=2+9i \Rightarrow z=-2+6i$$
$$3+5i-z=-2+7i \Rightarrow 5-2i=z$$
$$-2+12i-z=5+4i \Rightarrow -7+8i=z$$

B. Conjugate

1.60: Conjugate

The conjugate of a complex number $a+bi$ is the number $a-bi$. That is

$$z=a+bi \Leftrightarrow Conjugate\ of\ z=\bar{z}=a-bi$$

- You would have seen conjugates in the context of surds.
- The conjugate of $2+\sqrt{3}$ is $2-\sqrt{3}$
- Conjugates are used in rationalization of the denominator.
- Since $i=\sqrt{-1}$ is also a radical, one of the uses of the conjugate is help simplify the denominator.

Example 1.61

Find the conjugate of the following numbers:

A. $2+3i$
B. $-3-2i$
C. $-\frac{2}{3}+\sqrt{4}i$

$$2-3i$$
$$-3+2i$$
$$-\frac{2}{3}-2i$$

Example 1.62

A complex number is equal to its conjugate. What kind of number is it?

$$z=\bar{z}$$
$$a+bi=a-bi$$
$$bi=-bi$$
$$b=-b$$
$$b=0$$

Hence, the number is purely real.

1.63: Self-Inverse Property of the Conjugate

The conjugate of the conjugate of a complex number is equal to the original complex number.

$$\overline{(\bar{z})}=z$$

If $z = a + bi$ then:

$$\overline{(\bar{z})} = \overline{(a - b\imath)} = a + bi$$

Example 1.64

Show that $f(z) = \bar{z}, z \in \mathbb{C}$ is a self-inverse function.

$$f(z) = \bar{z}$$
$$f(\bar{z}) = \overline{(\bar{z})} = z$$

Since applying the function twice results in the same original number, it is a self-inverse function.

1.65: Conjugation is commutative with addition

The conjugate of a sum is the sum of the conjugates.

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$LHS = \overline{z_1 + z_2}$$

Let $z_1 = a + bi,\ z_2 = c + di$ giving us:

$$= \overline{(a + b\imath) + (c + d\imath)} = \overline{(a + c) + (b + d)\imath}$$

Find the conjugate:

$$= (a + c) - (b + d)i = a - bi + c - di = \bar{z}_1 + \bar{z}_2 = RHS$$

Example 1.66

$$\overline{(2 + 3\imath) + (3 - \imath)} = \overline{(2 + 3\imath)} + \overline{(3 - \imath)}$$

$$\overline{(2 + 3\imath) + (3 - \imath)} = \overline{5 + 2\imath} = 5 - 2i$$
$$\overline{(2 + 3\imath) + (3 - \imath)} = \overline{(2 + 3\imath)} + \overline{(3 - \imath)} = 2 - 3i + 3 + i = 5 - 2i$$

1.67: Conjugation is commutative with subtraction

The conjugate of a difference is the difference of the conjugates.

$$\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$$

The proof of this is very similar to the proof on commutativity of addition

C. Midpoint in the Complex Plane

1.68: Midpoint of a Line Segment

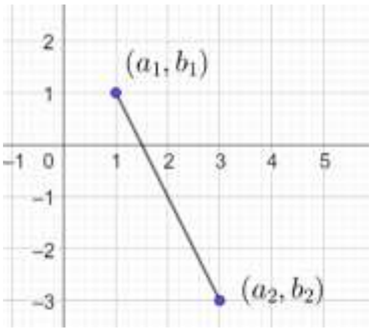
In coordinate geometry, the midpoint of a line segment with points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by the average of the x and the y coordinates respectively:

$$Midpoint_{AB} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

1.69: Midpoint of a Line Segment

The midpoint of the line segment given by points in the complex plane z_1 and z_2 is given by:

$$\frac{z_1 + z_2}{2}$$



Let $z_1 = a + bi, z_2 = c + d$

$$\frac{z_1 + z_2}{2} = \frac{a + bi + c + di}{2} = \underbrace{\frac{a + c}{2}}_{\text{Real Part}} + \underbrace{\frac{(b + d)}{2}}_{\text{Imaginary Part}} i$$

Example 1.70

Romeo and Juliet are on the complex plane, where Romeo resides at $2 - 3i$, and Juliet stays at $-4 + 2i$. If they decide to meet at the midpoint of where they stay, find the distance of that point from the origin.

Let the midpoint be:

$$m = \frac{z_1 + z_2}{2} = \frac{(2 - 3i) + (-4 + 2i)}{2} = \frac{-2 - i}{2} = -\frac{2}{2} - \frac{i}{2} = -1 - 0.5i$$

$$|m| = |-1 - 0.5i| = \sqrt{(-1)^2 + (-0.5)^2} = \sqrt{1 + 0.25} = \sqrt{1.25}$$

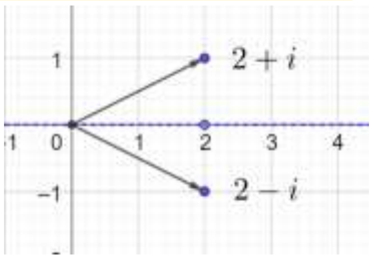
1.71: Conjugate (Reflection across the x axis)

The conjugate of a complex number z is the reflecting of z across the real axis.

The conjugate of a complex number $a + bi$ is the number $a - bi$.

Example 1.72

Plot the number $2 + i$ and its conjugate on the complex plane by reflecting it across the x-axis.



1.73: Purely Real Numbers

A purely real number is equal to its conjugate.

$$z \in \mathbb{R} \Leftrightarrow z = \bar{z}$$

Graphical Method

A purely real number lies on the real axis.
The conjugate of a number is its reflection across the real axis.

If a number lies on the real axis, then its reflection across the real axis is the number itself.

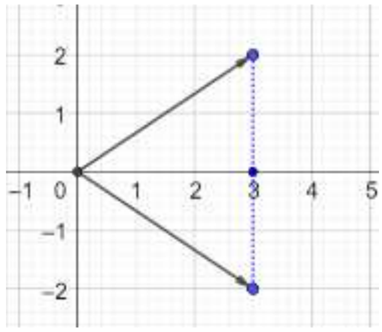
Algebraic Method

$$a + 0i = a - 0i$$

1.74: Real Part of z

The real part of z is the midpoint of a complex number and its conjugate on the complex plane.

$$\frac{z+\bar{z}}{2} = Re(z)$$



$$\frac{z+\bar{z}}{2} = \frac{(a+bi)+(a-bi)}{2} = \frac{2a}{2} = a$$

- The sum of a complex number and its conjugate is a number with imaginary part zero. That is, it is purely imaginary.
- The average of a complex number with its conjugate is the real part of the complex number.

Example 1.75

Mark the Correct Option

The real part of the sum of a number and its conjugate is zero. Then, the number is:

A. Purely Real
B. Purely Imaginary
C. Zero
D. Neither purely real nor purely imaginary

$$2 \times Re(z) = z + \bar{z} = 0$$

Think in terms of the coordinate plane (not the complex plane). The real part of a complex number corresponds to the real axis, which corresponds to the x −axis.

In terms of coordinate geometry:

$$x = 0$$

The set of points that satisfy $x = 0$ in the coordinate plane is the $y - axis$.

The set of points in the complex plane corresponding to the y-axis is the imaginary axis.

If the number lies on the imaginary axis, the number is purely imaginary. Hence:

$$Option\ B\ is\ correct$$

1.76: Reflection across the y axis

Reflecting across the y axis is equivalent to converting the complex number $a+bi$ into the complex number $-a+bi$

1.77: Difference of a Number and its Conjugate

$$\frac{z-\bar{z}}{2} = Im(z)i$$

$$\frac{z-\bar{z}}{2} = \frac{(a+bi)-(a-bi)}{2} = \frac{2bi}{2} = bi$$

- The difference of a complex number and its conjugate is a number with real part zero. That is, it is purely imaginary.

Example 1.78

Mark the Correct Option

The imaginary part of the difference of a number and its conjugate is zero. Then, the number is:

A. Purely Real
B. Purely Imaginary
C. Zero
D. Neither purely real nor purely imaginary

$$2bi = z - \bar{z} = 0$$

Think in terms of the coordinate plane (not the complex plane). The real part of a complex number corresponds to the real axis, which corresponds to the x −axis.
In terms of coordinate geometry:

$$y = 0$$

The set of points that satisfy y= 0 in the coordinate plane is the $x - axis$.
The set of points in the complex plane corresponding to the x-axis is the real axis.

If the number lies on the real axis, the number is purely real. Hence:

$$Option\ A\ is\ correct$$

1.4 Multiplication and Division

A. Multiplication

1.79: Binomial Multiplication

To multiply two complex numbers, we treat like them two binomials, and multiply out, using the distributive property:

$$(a + bi)(c + di) = \underbrace{ac + adi + bci + bdi^2}_{\textbf{Expand like a binomial}} = ac - bd + (ad + dc)i$$

Don't memorize this formula. Carry out the binomial multiplication every time.

Example 1.80

Multiply:

A. $(3 + 4i)(2 + 5i)$
B. $(3 + 4i)(2 - 5i)$
C. $(3 - 4i)(2 + 5i)$
D. $(3 - 4i)(2 - 5i)$

$$(3 + 4i)(2 + 5i) = 6 + 15i + 8i - \underbrace{20}_{20i^2=-20} = -14 + 23i$$
$$(3 + 4i)(2 - 5i) = 6 - 15i + 8i + 20 = 26 - 7i$$
$$(3 - 4i)(2 + 5i) = 6 + 15i - 8i + 20 = 26 + 7i$$
$$(3 - 4i)(2 - 5i) = 6 - 15i - 8i - 20 = -14 - 23i$$

Example 1.81

$$[x - (1 + i)][x - (1 - i)]$$

$$[x - 1 - i][x - 1 + i]$$

$$\Big(\underbrace{x-1}_{a}-\underbrace{i}_{b}\Big)\Big(\underbrace{x-1}_{a}+\underbrace{i}_{b}\Big)$$

Use the property:

$$(a+b)(a-b)=a^2-b^2=(x-1)^2-(i^2)=x^2-2x+1+1=x^2-2x+2$$

Example 1.82

A. When a number is divided by $2+6i$, the result is $4-2i$. Find the number.
B. When a number is divided by $3+7i$, the result is $1-3i$. Find the number.

$$\frac{z}{2+6i}=4-2i\Rightarrow z=(4-2i)(2+6i)=8+24i-4i+12=20+20i$$
$$\frac{z}{3+7i}=1-3i\Rightarrow z=(1-3i)(3+7i)=3-21i-9i+21=24-30i$$

Example 1.83

Multiply:

A. $(x+2i)(x-2i)$
B. $(x+3i)(x-3i)$

$$(x+2i)(x-2i)=x^2-(4i^2)=x^2+4$$
$$(x+3i)(x-3i)=x^2+9$$

1.84: Squaring

The square of a complex number $z=a+bi$ is

$$(a+bi)^2=\underbrace{a^2+2abi+(bi)^2}_{Using\ (x+y)^2=x^2+2xy=y^2}=a^2-b^2+2abi$$

Again, don't memorize. Use the formula for square of a sum every time.

Example 1.85

Expand

A. $(6+2i)^2$
B. $(6-2i)^2$

$$(6+2i)^2=36+24i+4i^2=32+24i$$
$$(6-2i)^2=36-24i+4i^2=32-24i$$

Example 1.86

Solve:

A. $(a+bi)^2=1,\ a,b\in\mathbb{R}$
B. $(a+bi)^2=1,\ a,b\in\mathbb{C}$

Part A

Expand the LHS:

$$\underbrace{a^2-b^2}_{Real\ Part}+\underbrace{2ab}_{Imaginary\ Part}i=1+0i$$

Equate the imaginary parts:

$$2abi=0$$

Divide by $2i$ both sides:

$$ab=0$$

Use the zero-product property:

$$a=0\ OR\ b=0$$

Equate the real parts:

$$a^2 - b^2 = 1$$

If $b = 0$:

$$a^2 = 1 \Rightarrow a = \pm 1$$

If $a = 0$:

$$-b^2 = 1 \Rightarrow b^2 = -1 \Rightarrow b = \pm i$$

$$b \in \mathbb{R} \Rightarrow Reject \pm i$$

Hence, this case does not have valid solutions.

$$a + bi \in \{-1 + 0i, 1 + 0i\}$$

Alternate Method

Take the square root of both sides:

$$a + bi = \pm 1$$

$$b = 0$$

$$a = \pm 1$$

Part B

The solution proceeds as before, but we do not need to reject the imaginary values for b since it is a complex number. Hence, the final answer is:

$$a + bi \in \{-i, i, -1, 1\}$$

Example 1.87

Let n denote the number of solutions of the equation $z^2 + 3\bar{z} = 0$, where z is a complex number. Then, the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ (JEE Main, 22 July 2021, Shift-II)

Substitute $z = x + yi \Rightarrow \bar{z} = x - yi$ in the given equation:

$$(x + iy)^2 + 3(x - iy) = 0$$

Expand:

$$x^2 - y^2 + 2xyi + 3x - 3yi = 0$$

Rearrange:

$$(x^2 - y^2 + 3x) + i(2xy - 3y) = 0$$

Equate the imaginary parts:

$$y(2x - 3) = 0 \Rightarrow y = 0 \text{ or } x = \frac{3}{2}$$

Equate the real parts:

$$x^2 - y^2 + 3x = 0$$

$$Case\ I: y = 0 \Rightarrow x^2 + 3x = 0 \Rightarrow x \in \{-3, 0\} \Rightarrow (x, y) \in \{(0,0)(-3,0)\} \Rightarrow Two\ Solutions$$

$$Case\ II: x = \frac{3}{2} \Rightarrow \frac{9}{4} - y^2 + \frac{9}{2} = 0 \Rightarrow y^2 = \frac{27}{4} \Rightarrow y = \pm\frac{3\sqrt{3}}{2} \Rightarrow (x, y) \in \left\{\left(\frac{3}{2}, \pm\frac{3\sqrt{3}}{2}\right)\right\} \Rightarrow Two\ Solutions$$

There are 4 solutions.

$$n = 4 \Rightarrow \sum_{k=0}^{\infty} \frac{1}{n^k} = 1 + \frac{1}{4} + \frac{1}{4^2} + \cdots$$

Use the formula for the sum of a geometric series. Substitute $a = 1, r = \frac{1}{4}$:

$$S = \frac{a}{1 - r} = \frac{1}{1 - \frac{1}{4}} = \frac{3}{4}$$

1.88: Product of two conjugates

The product of two conjugates is the sum of the squares of their real and imaginary parts.

$$(a + bi)(a - bi) = a^2 + b^2$$

$$(a + bi)(a - bi)$$

This is a difference of squares. Use the formula $(x + y)(x - y) = x^2 - y^2$:

$$= a^2 - b^2 i^2 = a^2 + b^2$$

Example 1.89

If a complex number z is such that $(7+i)(z+\bar{z})-(4+i)(z-\bar{z})+116i=0$, then $z\cdot\bar{z}=$ **(AP EAPCET, 21 April 2019, Shift-II)**

Let,

$$z=a+bi\Rightarrow\bar{z}=a-bi\Rightarrow\ z+\bar{z}=2a, z-\bar{z}=2bi$$

Substitute the above in the LHS of the given equation:

$$(7+i)(2a)-(4+i)(2bi)+116i=0$$

Expand:

$$14a+2ia-(8bi-2b)+116i=0$$

Rearrange:

$$(14a+2b)+i(2a-8b+116)=0+0i$$

Recall that two complex numbers are equal if and only if their real and their imaginary parts are equal.
Equate the real parts:

$$14a+2b=0\Rightarrow b=-7a$$

Equate the imaginary parts:

$$2a-8b+116\ =0$$

Substitute $b=-7a$:

$$2a-8(-7a)+116=0$$
$$58a=-116$$
$$a=-2$$
$$b=-7a=14$$
$$z\cdot\bar{z}=a^2+b^2=(-2)^2+(14)^2=4+196=200$$

1.90: Square of a Conjugate

For a complex number $z=a+bi$, the square of its conjugate $\bar{z}=a-bi$ is:

$$(a-bi)^2=\underbrace{a^2-2abi+(bi)^2}_{Using\ (x+y)^2=x^2+2xy=y^2}=a^2-b^2-2abi$$

Again, don't memorize. Use the formula for square of a sum every time.

Example 1.91

Let $z=a+bi$, where a and b are integers and $i=\sqrt{-1}$. The area of the rectangle whose vertices are the roots of the equation $\bar{z}z^3+z(\bar{z})^3=350$ is: **(AP EAPCET 18 Sep 2020, Shift-II, Adapted)**

Factor $z\bar{z}$ on the LHS:

$$z\bar{z}[z^2+(\bar{z})^2]=350$$

Substitute $z=a+bi$:

$$\underbrace{(a^2+b^2)}_{z\bar{z}}\left[\underbrace{a^2-b^2+2abi}_{z^2}+\underbrace{(a^2-b^2-2abi)}_{(\bar{z})^2}\right]=350$$

Simplify, and factor on the RHS (since $a,b\in\mathbb{Z}$):

$$(a^2+b^2)(a^2-b^2)=175$$

Case I

$$\underbrace{a^2+b^2=35}_{Equation\ I},\underbrace{a^2-b^2=5}_{Equation\ II}\Rightarrow 2a^2=40\Rightarrow a^2=20\Rightarrow a=\sqrt{20}=2\sqrt{5}\notin\mathbb{Z}\Rightarrow Not\ Valid$$

Case II

$$\underbrace{a^2+b^2=25}_{\textit{Equation I}}, \qquad \underbrace{a^2-b^2=7}_{\textit{Equation II}}$$

Add Equations I and II:

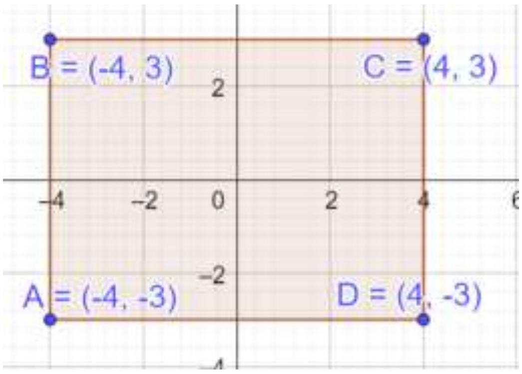
$$2a^2=32\Rightarrow a^2=16\Rightarrow a=\pm4$$
$$b^2=16-7=9\Rightarrow b=\pm3$$

The coordinates of the rectangle are:

$$\{(-4,-3)(-4,3)(4,-3)(4,3)\}$$

The area of the rectangle is:

$$[4-(-4)][3-(-3)]=(8)(6)=48$$



1.92: Conjugation is commutative with multiplication

The conjugate of a product is the product of the conjugates.

$$\overline{z_1z_2}=\overline{z_1}\,\overline{z_2}$$

Let $z_1=a+bi,\ z_2=c+di$ giving us:

$$LHS=\overline{z_1z_2}=\overline{(a+bi)(c+di)}=\overline{ac-bd+(bc+da)i}=ac-bd-(bc+da)i$$
$$RHS=\overline{z_1}\,\overline{z_2}=\left(\overline{a+bi}\right)\left(\overline{c+di}\right)=(a-bi)(c-di)=ac-bd-(bc+da)i=LHS$$

B. Division

1.93: Division by i

Example 1.94

Simplify

A. $\frac{4}{3i}$
B. $\frac{7}{5i}$
C. $3\div\frac{1}{2}i$
D. $4\div\frac{2}{5}i$

Part A

Eliminate the imaginary number in the denominator by multiplying by i:

$$\frac{4}{3i}\times\frac{i}{i}=\frac{4i}{3i^2}=\frac{4i}{-3}=-\frac{4i}{3}$$

Part B

$$\frac{7}{5i}\times\frac{i}{i}=-\frac{7i}{5}$$

Part C

$$3\div\frac{1}{2}i=3\times\frac{2}{i}=\frac{6}{i}\times\frac{i}{i}=\frac{6i}{i^2}=-6i$$

Part D

$$4\times\frac{5}{2i}=\frac{10}{i}\times\frac{i}{i}=-10i$$

Example 1.95: Counting

If $S=i^n+i^{-n}$, where $i=\sqrt{-1}$ and n is an integer, then the total number of possible distinct values for S is: (AHSME 1957/42)

$$S = i^n + i^{-n} = i^n + \left(\frac{1}{i}\right)^n = i^n + (-i)^n$$

Because the powers of i have a cyclicity of 4, we only need to check the first four values.

$$S_1 = i^1 + (-i)^1 = i - y = 0$$
$$S_2 = i^2 + (-i)^2 = -1 - 1 = -2$$
$$S_3 = i^3 + (-i)^3 = -i + i = 0$$
$$S_3 = i^4 + (-i)^4 = 1 + 1 = 2$$

$$S \in \{-2,0,2\} \Rightarrow 3\ Values$$

1.96: Reciprocal of a complex number

To find the reciprocal of the complex number $a + bi$, we multiply $\frac{1}{a+bi}$ with the conjugate of the denominator, which is $a - bi$.

Example 1.97

A. $\frac{1}{2+3i}$
B. $\frac{1}{3+2i}$

$$\frac{1}{2+3i} \times \frac{2-3i}{2-3i} = \frac{2-3i}{4+9} = \frac{2-3i}{13} = \frac{2}{13} - \frac{3}{13}i$$

1.98: Division by a complex number

To determine $a + bi$ divided by $c + di$, we rewrite the division as a fraction, and multiply numerator and denominator by the *conjugate of the denominator*:

$$\frac{a+bi}{c+di} \times \frac{c-di}{c-di} = \frac{(\quad)(\quad)}{c^2+d^2}$$

Example 1.99

A. $\frac{1+2i}{2-5i}$
B. Find $Re\left(\frac{2+3i}{4+5i}\right)$ and $Im\left(\frac{2+3i}{4+5i}\right)$
C. If $a + bi = \frac{i}{1-i}$, then $(a, b) =$ **(AP EAPCET 17 Sep 2020, Shift-I)**

Part A

$$\frac{1+2i}{2-5i} \times \frac{2+5i}{2+5i} = \frac{2+5i+4i-10}{4+25} = \frac{-8+9i}{29} = -\frac{8}{29} + \frac{9}{29}i$$

Part B

Convert to standard form by carrying out the division.
Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{2+3i}{4+5i} \times \frac{4-5i}{4-5i} = \frac{8-10i+12i-15i^2}{16+25} = \frac{23+2i}{41} = \underbrace{\frac{23}{41}}_{\substack{Real\\Part}} + \underbrace{\frac{2}{41}}_{\substack{Imaginary\\Part}}i$$

Part C

$$\frac{i}{1-i}\times\frac{1+i}{1+i}=\frac{1+i}{1+1}=\frac{1+i}{2}=\frac{1}{2}+\frac{1}{2}i\Rightarrow(a,b)=\left(\frac{1}{2},\frac{1}{2}\right)$$

Example 1.100

A. When a number is multiplied by $2+6i$, the result is $4-2i$. Find the number.
B. When a number is multiplied by $3+7i$, the result is $1-3i$. Find the number.

$$z(2+6i)=4-2i\Rightarrow z(1+3i)=2-i$$

$$z=\frac{4-2i}{2+6i}\times\frac{2-6i}{2-6i}=\frac{8-24i-4i-12}{4+36}=\frac{-4-28i}{40}=\frac{-1-7i}{10}$$

Example 1.101

A. Find the conjugate of $\left(\frac{5i}{7+i}\right)$ **(AP EAPCET, 22 Sep. 2020, Shift-II)**
B. The conjugate of a complex number is $\frac{1}{1-i}$. Then, the complex number is **(AIEEE 2008)**

Part A
Multiply the numerator and denominator by the complex conjugate of the denominator:

$$z=\left(\frac{5i}{7+i}\right)\times\left(\frac{7-i}{7-i}\right)=\left(\frac{5i(7-i)}{7^2+1^2}\right)=\left(\frac{5(7i-i^2)}{50}\right)=\frac{1+7i}{10}=\frac{1}{10}+\frac{7}{10}i$$

$$Conjugate\ of\, z=\bar{z}=\frac{1}{10}-\frac{7}{10}i$$

Part B

$$\bar{z}=\frac{1}{1-i}\times\frac{1+i}{1+i}=\frac{1+i}{1+1}=\frac{1}{2}+\frac{i}{2}\Rightarrow z=Conjugate\ of\ \bar{z}=\frac{1}{2}-\frac{i}{2}$$

Example 1.102

Solve for $z, z\in\mathbb{C}$

$$2-3iz+3i=3+5i+2z$$

$$2+3i-3-5i=2z+3iz$$
$$-1-2i=z(2+3i)$$
$$z=\frac{-1-2i}{2+3i}$$

1.103: Dividing zero

- If 0 is divided by 0, it is not defined
- If 0 is divided by a non-zero complex number, it is zero.

Example 1.104

If $f(x)=\frac{x^4+x^2}{x+1}$, then $f(i)$, where $i=\sqrt{-1}$ is equal to: **(AHSME 1970/5)**

$$f(i)=\frac{i^4+i^2}{i+1}=\frac{1-1}{i+1}=\frac{0}{i+1}=0$$

1.105: Conjugation is commutative with division

The conjugate of a quotient is the quotient of the conjugates.

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$$

Let

$$\frac{z_1}{z_2} = z_3 \Rightarrow z_1 = z_2 z_3$$

Take the conjugate on both sides:

$$\overline{z_1} = \overline{z_2 z_3}$$

Use the property $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$:

$$\overline{z_1} = \overline{z_2}\,\overline{z_3}$$

Divide both sides by $\overline{z_2}$:

$$\frac{\overline{z_1}}{\overline{z_2}} = \overline{z_3}$$

But note that $\frac{z_1}{z_2} = z_3 \Rightarrow \overline{\left(\frac{z_1}{z_2}\right)} = \overline{z_3}$:

$$\frac{\overline{z_1}}{\overline{z_2}} = \overline{\left(\frac{z_1}{z_2}\right)}$$

Which was what we wanted to prove.

Example 1.106

A. If z is a complex number such that $\frac{1-z}{1+z}$ is a real number, then determine $Im(z)$.
B. Is the converse of the statement from Part A true?

Part A

The question tells us that:

$$\frac{1-z}{1+z} \in \mathbb{R}$$

Use the property that a real number is equal to its conjugate.

$$\frac{1-z}{1+z} = \overline{\left(\frac{1-z}{1+z}\right)}$$

Use the conjugate division property $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$:

$$\frac{1-z}{1+z} = \frac{\overline{1-z}}{\overline{1+z}}$$

Use the subtraction and addition property of conjugates:

$$\frac{1-z}{1+z} = \frac{1-\bar{z}}{1+\bar{z}}$$

Cross multiply:

$$(1-z)(1+\bar{z}) = (1-\bar{z})(1+z)$$

Expand:

$$1+\bar{z}-z-z\bar{z} = 1-\bar{z}+z-z\bar{z}$$

Simplify:

$$2\bar{z} = 2z$$
$$z = \bar{z}$$
$$z \in \mathbb{R}$$
$$Im(z) = 0$$

Part B

$$Im(z) = 0 \Rightarrow z \in \mathbb{R} \Rightarrow \frac{1-z}{1+z} \in \mathbb{R}$$

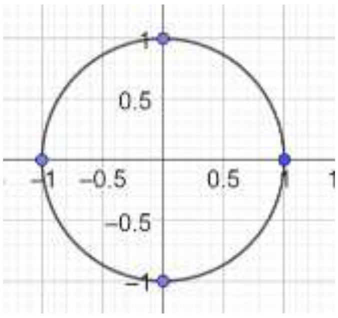
C. Rotation

1.107: Multiplication as Rotation

- Multiplication by i is equivalent to rotation by $90°$ about the origin.
- By convention, positive angles are in the counter-clockwise direction.

Example 1.108

Given that $f(z) = iz$, determine $f(1), f\big(f(1)\big), f^3(1), f^4(1)$ where $f^n(z)$ indicates composition n times, not an exponent.



$$f(1) = i$$
$$f(f(1)) = f(i) = i^2 = -1$$
$$f^3(1) = f(-1) = -i$$
$$f^4(1) = f(-i) = 1$$

Example 1.109

A. The complex number $z_1 = 2 + 3i$ is rotated 270° clockwise about the origin to give point $z_2 = a + bi$. Find $a + b$.
B. The point z_2 is obtained by rotating the number $z_1 = 3 + 5i$ about $1 + i$ in the complex plane by 90°. Find $z_2\bar{z}_2$.
C. The point P in the real plane has coordinates $(x, y) = (-2,4)$. Find the coordinates of P' obtained by rotating P 180° anti-clockwise about the origin.

Part A

Rotating 270° clockwise is the same as rotating 90° counter-clockwise:

$$z_2 = iz_1 = i(2 + 3i) = 2i + 3i^2 = 2i - 3 \Rightarrow a + b = 2 - 3 = -1$$

Part B

We know how to rotate about the origin, not with respect to a general number.
Hence, use a change of coordinates. Introduce an origin at $1 + i$. Then, in the new coordinate system:

$$z_{1-New} = 3 + 5i - (1 + i) = 2 + 4i$$

Rotate the new number 90° by multiplying it by i:

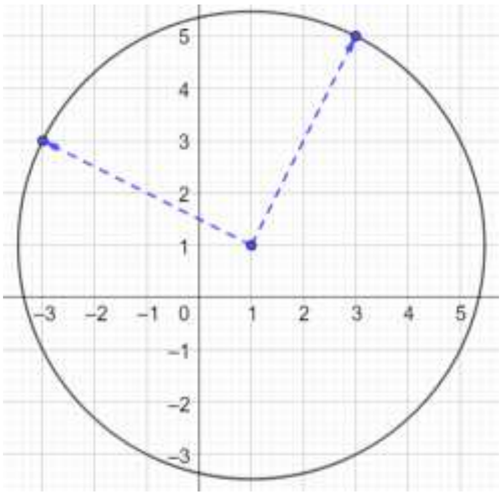
$$z_{2-New} = i(2 + 4i) = 2i - 4 = -4 + 2i$$

Switch back to the original coordinate system:

$$z_2 = -4 + 2i + (1 + i) = -3 + 3i$$

And we need to find:

$$z_2\bar{z}_2 = (a + bi)(a - bi) = a^2 + b^2 = (-3)^2 + 3^2 = 9 + 9 = 18$$



Part C

Convert $(x, y) = (-2,4)$ to the corresponding point on the complex plane

$$z = -2 + 4i$$

Rotating by 180° is equivalent to rotating by 90° twice:

$$z' = (i^2)(-2 + 4i) = (-1)(-2 + 4i) = 2 - 4i$$

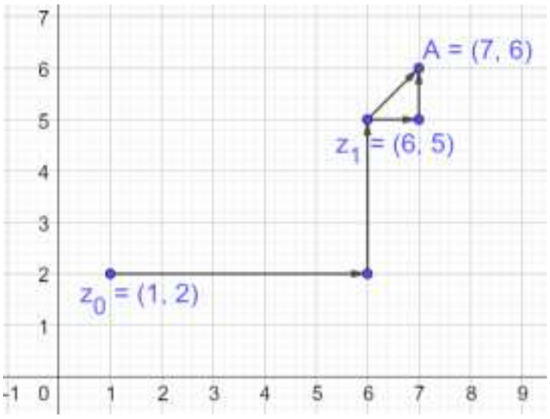
Convert back to the real plane:

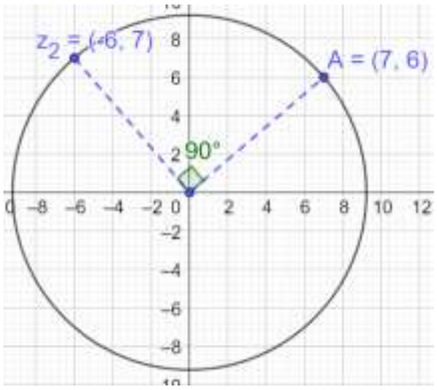
$$P' = (2, -4)$$

Example 1.110

A particle P starts from the point $z_0 = 1 + 2i$ where $i = \sqrt{-1}$. It moves horizontally away from the origin by 5 units, and then vertically away from the origin by 3 units to reach a point z_1. From z_1, the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{\imath} + \hat{\jmath}$ and then it moves through an angle $\frac{\pi}{2}$ in the anticlockwise direction on a circle with center at the origin, to reach point z_2. The point z_2 is given by: (JEE Adv. 2008)

Since the particle moves horizontally *away*, and vertically *away*, it must travel right and up. (If it travels left or down, it will move closer to the origin before it moves away).





Move five units to the right and 3 units up from $z_0 = 1 + 2i$ to reach:

$$z_1 = 6 + 5i$$

We get an isosceles right-angled triangle with side length 1, giving us:

$$A = 7 + 6i$$

Rotate A $90°$ by multiplying it by i:

$$z_2 = -6 + 7i$$

1.111: Division

- Division by i is equivalent to rotation by $-90°$.
- By convention, negative angles are in the clockwise direction.

1.5 Equations

A. Trivial Quadratic Equations

1.112: Trivial Quadratics

Quadratics of the form $x^2 - c = 0$ are trivial quadratics since the middle term is missing and they can solved directly:

$$x^2 - c = 0 \Rightarrow x^2 = c \Rightarrow x = \pm\sqrt{c}$$

Example 1.113: Trivial Quadratics

A. $x^2 - 9 = 0$
B. $x^2 + 9 = 0$
C. $x^2 + 8 = 0$
D. $x^2 + 72 = 0$

$$x^2 = 9 \Rightarrow x = \pm 3$$
$$x^2 + 9 = 0 \Rightarrow (x + 3i)(x - 3i) = 0 \Rightarrow x = \pm 3i$$
$$x^2 = -8 \Rightarrow x = \pm\sqrt{-8} = \pm\sqrt{8}i = \pm 2\sqrt{2}i$$
$$x^2 = -72 \Rightarrow x = \pm\sqrt{-72} = \pm\sqrt{72}i = \pm 6\sqrt{2}i$$

Example 1.114

The real factors of x^2+4 are:

A. $(x^2+2)(x^2+2)$
B. $(x^2+2)(x^2-2)$
C. $x^2(x^2+4)$
D. $(x^2-2x+2)(x^2+2x+2)$
E. Non-existent **(AHSME 1950/15)**

$$x^2=-4 \Rightarrow x=\pm\sqrt{-4}=\pm\sqrt{4}i=\pm 2i$$
$$Option\ E$$

1.115: Factoring Trivial Quadratics

$$x^2+c=0, c>0$$
$$\left(x+\sqrt{c}i\right)\left(x-\sqrt{c}i\right)=0$$

$$x^2+c=0$$
$$x^2-(-c)=0$$

This is in the form a^2-b^2 with:

$$\left(x+\sqrt{c}i\right)\left(x-\sqrt{c}i\right)=0$$

Example 1.116

Solve the following equations by factoring method only:

A. $x^2+9=0$
B. $x^2+4=0$

Part A

$$x^2-9=0 \Rightarrow (x+3)(x-3)=0 \Rightarrow x=\pm 3$$

Part B

$$x^2-(-4)=0$$

This is in the form a^2-b^2 with:

$$a^2=x^2 \Rightarrow a=x$$
$$b^2=-4 \Rightarrow b=\pm\sqrt{-4}=\pm\sqrt{4}i=\pm 2i$$

Factorize $x^2-(-4)$ using $a^2-b^2=(a+b)(a-b)$:

$$\underbrace{x^2}_{a^2}-\underbrace{(-4)}_{b^2}=\left(\underbrace{x}_{a}+\underbrace{2i}_{b}\right)\left(\underbrace{x}_{a}-\underbrace{2i}_{b}\right)$$

$$(x+2i)(x-2i)=0$$
$$x=-2i\ OR\ x=2i$$
$$x\in\{\pm 2i\}$$

B. Non-Trivial Quadratic Equations

1.117: Solving by Completing the Square

Example 1.118

Solve by completing the square

A. $x^2+2x+6=0$

$$x^2+2x+6=0$$

$$x^2 + 2x + 1 + 5 = 0$$
$$(x+1)^2 + 5 = 0$$
$$(x+1)^2 = -5$$
$$x + 1 = \pm\sqrt{5}i$$
$$x = -1 \pm 5i$$

1.119: Factoring

Example 1.120

Solve $x^2 + x + 1 = 0$ by factoring method only.

Hint: Complete the square. Introduce i^2 to create a negative sign, and factor using a difference of squares. Apply the zero-product property, as usual.

Complete the square:
$$x^2 + x + \frac{1}{4} + \frac{3}{4} = 0$$
Factor:
$$\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = 0$$
Rewrite:
$$\left(x + \frac{1}{2}\right)^2 - \left(-\frac{3}{4}\right) = 0$$
Substitute $i^2 = -1$
$$\left(x + \frac{1}{2}\right)^2 - \left(\frac{3}{4}i^2\right) = 0$$

Factor the expression as a difference of squares $a^2 - b^2 = (a+b)(a-b)$:
$$\left(x + \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\left(x + \frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 0$$
Use the zero-product property to find the roots:
$$x = -\frac{1}{2} - \frac{\sqrt{3}}{2}i \ OR\ x = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$
The final answer is:
$$x \in \left\{= -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i\right\}$$
These are the primitive cube roots of unity.

1.121: Solving using the Quadratic Formula

The quadratic formula works for complex coefficients. For a quadratic equation $ax^2 + bx + c$ with complex coefficients $a, b, c \in \mathbb{C}$ and $a \neq 0$:
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The proof is the same for complex coefficients as it is for real coefficients.
Divide throughout by a to make the leading coefficient one:
$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$
Add and subtract the square of half the second term in preparation to complete the square:
$$\left[x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2\right] - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$$
Rewrite the terms inside the square brackets as a perfect square.
$$\left[x + \frac{b}{2a}\right]^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$
Isolate the perfect square term on the LHS:
$$\left[x + \frac{b}{2a}\right]^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$
Add the two fractions in the RHS by taking the LCM, and then take the square root on both sides:

$$x+\frac{b}{2a}=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$$

Simplify the denominator in the RHS, and isolate x on the LHS to arrive at the standard quadratic formula.

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Example 1.122

Find the roots of:

A. $x^2+x+1=0$
B. $ix^2-3x-2i=0$ (AP EAPCET, 23 Sep. 2020, Shift-I)

Part A

Substitute $a=1, b=1, c=1$ in the quadratic formula:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-1\pm\sqrt{1-4}}{2}=\frac{-1\pm\sqrt{-3}}{2}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$$

Part B

Substitute $a=i, b=-3, c=-2i$ in the quadratic formula:

$$x=\frac{3\pm\sqrt{(-3)^2-(4)(i)(-2i)}}{2i}=\frac{3\pm\sqrt{9-8}}{2i}=\frac{3\pm 1}{2i}=\frac{(3\pm 1)i}{-2}=-\frac{4i}{2}\ or-\frac{2i}{2}=-2i\ or-i$$

1.123: Nature of Roots

The quantity b^2-4ac in the quadratic formula is called the discriminant.

The discriminant is used to determine the nature of roots of the quadratic:

- If the discriminant is positive, the quadratic has two real roots.
  - ✓ The graph of the quadratic will intersect the x axis in two distinct places.
- If the discriminant is zero, the quadratic has a single repeated, real, real root.
  - ✓ The graph of the quadratic will touch, but not cross the x axis.
- If the discriminant is negative, the quadratic has roots which are complex conjugates of each other.
  - ✓ The graph of the quadratic will not touch or cross the x axis.

C. Complex Roots

1.124: Roots occur in pairs

If a quadratic equation has:

- real coefficients, then its complex roots occur in pairs. The pairs are complex conjugates of each other.
- has complex coefficients, then it is not necessary that its complex roots occur in pairs.

Example 1.125

A. A quadratic equation ax^2+bx+c where $a,b,c\in\mathbb{R}$ has a root $2+3i$. Find the other root, if possible.
B. A quadratic equation ax^2+bx+c where $a,b,c\in\mathbb{C}$ has a root $5-7i$. Find the other root, if possible.

Part A

$$2-3i$$

Part B

We can't say anything because the coefficients are complex.

Example 1.126

Mark all correct options

A quadratic equation with real coefficients has a complex root with non-zero imaginary part. Then, which of the following may be true:

A. The quadratic has two distinct roots
B. The quadratic has a repeated root
C. The roots of the quadratic are complex conjugates of each other.
D. The real part of the root(s) is non-zero.
E. The real part of the root(s) is zero.

$$Options\ A, C, D, E$$

1.127: Relation between Roots (Vieta's Formulas)

For a quadratic equation, $ax^2 + bx + c = 0, a \neq 0$, the roots α and β, satisfy the relations:

$$\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$$

Notes:

- The Note on Quadratics covers the relation between roots for a quadratic in detail.

Example 1.128

It is given that one root of $2x^2 + rx + s = 0$ with r and s real numbers is $3 + 2i\left(i = \sqrt{-1}\right)$. The value of s is: **(AHSME 1962/21)**

Since the coefficients are real, and one root is complex, the roots are complex conjugates of each other. The second root of the quadratic

$$= Complex\ conjugate\ of\ given\ root = 3 - 2i$$

Using Vieta's Formulas, the product of the roots is:

$$(3 + 2i)(3 - 2i) = \frac{s}{2}$$
$$9 + 4 = \frac{s}{2}$$
$$s = 26$$

Example 1.129

If $2 + 4i$ is one of the roots of $x^2 + bx + c = 0$ with $b, c \in \mathbb{R}$ then $(b, c) =$ **(AP EAPCET, 21 Sep. 2020, Shift-I)**

Since the roots are complex conjugates of each other, the other root

$$= 2 - 4i$$

Use Vieta's Formulas, for the sum and product of roots is:

$$Sum = (2 + 4i) + (2 - 4i) = \frac{-b}{1} \Rightarrow b = -4$$
$$Product = (2 + 4i)(2 - 4i) = c \Rightarrow c = 20$$
$$(b, c) = (-4, 20)$$

Example 1.130

Which one of the following is not true for the equation $ix^2 - x + 2i = 0$, where $i = \sqrt{-1}$.

A. The sum of the roots is 2
B. The discriminant is 9
C. The roots are imaginary
D. The roots can be found using the quadratic formula
E. The roots can be found by factoring, using imaginary numbers **(AHSME 1959/27)**

Note: Apart from finding the correct option justify each of the other options as true.

Multiply the given equation by i:

$$-x^2 - ix - 2 = 0$$
$$x^2 + ix + 2 = 0$$

Option A

Apply Vieta's Formulas:

$$Sum\ of\ Roots = -i \Rightarrow Option\ A\ Correct$$

Option B

$$b^2 - 4ac = i^2 - (4)(1)(2) = -1 - 8 = -9$$
$$b^2 - 4ac = (-1)^2 - (4)(i)(2i) = 1 + 8 = 9$$

Option C

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} =$$

Option E

$$x^2 + ix + 2 = 0$$
$$Product = 2 = (2i)(-i), Sum = i$$
$$(x - 2i)(x + i) = 0$$

Example 1.131

Mark the correct option

The equation $x + \sqrt{x - 2} = 4$ has:

A. 2 real roots
B. 1 real and 1 imaginary root
C. 2 imaginary roots
D. No roots
E. 1 real root **(AHSME 1950/24)**

Rearrange to isolate the radical:

$$\sqrt{x - 2} = 4 - x$$

Square both sides:

$$x - 2 = 16 - 8x + x^2$$
$$x^2 - 9x + 18 = 0$$
$$x \in \{3,6\}$$

Since we squared the equation, we have check for "extraneous" roots:

$$3\ works$$
$$6\ does\ not\ work$$
$$Option\ E$$

D. Cubic Equations

1.132: Roots occur in pairs

If a polynomial equation has:

- real coefficients, then its complex roots occur in pairs. The pairs are complex conjugates of each other.
- has complex coefficients, then it is not necessary that its complex roots occur in pairs.

1.133: Relation between Roots (Vieta's Formulas)

For a cubic equation, $ax^3 + bx^2 + cx + d = 0$, the roots α, β, γ, satisfy the relations:

$$Sum\ of\ Roots = \alpha + \beta + \gamma = -\frac{b}{a}$$

$$Sum\ of Product\ of\ Roots\ (2\ at\ a\ time): \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$Product\ of\ Roots = \alpha\beta\gamma = -\frac{d}{a}$$

Example 1.134

If $(2 + i)$ is a root of the equation $x^3 - 5x^2 + 9x - 5 = 0$ then the other roots are **(AP EAPCET, 21 Sep. 2020, Shift-I)**

Since complex roots of a polynomial with real coefficients occur in pairs, the second root will be the complex conjugate of the given root:

$$2nd\ Root = 2 - i$$

Let the third root be α.

Use Vieta's Formulas, for the product of roots is

$$(2 + i)(2 - i)\alpha = (2^2 + 1^2)\alpha = 5\alpha = 5 \Rightarrow \alpha = 1$$

The other roots are:

$$\{2 - i, 1\}$$

Example 1.135

If $a \pm bi (b \neq 0)$ are imaginary roots of the equation $x^3 + qx + r = 0$, where a, b, q and r are real numbers, then q in terms of a and b is: **(AHSME 1972/22)**

Since the coefficient of the x^2 term is zero, the sum of the roots must be zero:

$$\alpha + \beta + \gamma = 0$$
$$(a + bi) + (a - bi) + \gamma = 0$$
$$\gamma = -2a$$

The product of the roots taken two at a time must be:

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{q}{1}$$
$$[(a + bi)(a - bi) + (a + bi)(-2a) + (a - bi)(-2a)] = q$$
$$a^2 + b^2 + (a + bi + a - bi)(-2a) = q$$
$$a^2 + b^2 + (2a)(-2a) = q$$
$$b^2 - 3a^2 = q$$

E. Forming Equations

1.136: Forming Equations

Example 1.137

Find the complex polynomial with least degree with roots:

A. $\pm 5i$
B. $-2, 0, -1 \pm i$
C. $-1+3i, -1+\sqrt{2}$
D. $-\frac{3}{2}, \sqrt{2}$

Part A

$$x = 5i \Rightarrow \underbrace{x - 5i = 0}_{Equation\ I}$$
$$x = -5i \Rightarrow \underbrace{x + 5i = 0}_{Equation\ II}$$

Multiply Equation I and Equation II:

$$(x-5i)(x+5i) = 0$$
$$x^2 + 25 = 0$$

Part B

$$x = -1 + i \Rightarrow x + 1 - i = 0$$
$$x = -1 - i \Rightarrow x + 1 + i = 0$$
$$(x+2)(x)(x+1-i)(x+1+i) = 0$$
$$(x^2+2x)[(x+1)^2+1] = 0$$
$$(x^2+2x)[x^2+2x+1+1] = 0$$
$$(x^2+2x)[x^2+2x+2] = 0$$

Part C

$$(x+1+3i)(x+1-3i)\left(x+1-\sqrt{2}\right)\left(x+1+\sqrt{2}\right) = 0$$
$$[(x+1)^2-(3i)^2]\left[(x+1)^2-\left(\sqrt{2}\right)^2\right] = 0$$
$$[x^2+2x+10][x^2+2x-1] = 0$$

Part D

$$x - \frac{3}{2} = 0 \Rightarrow x = \frac{3}{2} \Rightarrow 2x = 3 \Rightarrow 2x - 3 = 0$$
$$(2x-3)\left(x-\sqrt{2}\right)\left(x+\sqrt{2}\right) = 0$$

F. Quartic Equations

Example 1.138

Factor $x^4 + 1$ over the real numbers.

Hint: Complete the square and use a difference of squares

Note that the middle term is missing, and we add and subtract it to get a perfect square:

$$x^4 + 2x^2 + 1 - 2x^2$$

Factor:

$$= (x^2+1)^2 - 2x^2$$

Factor using a difference of squares:

$$= \left(x^2 + \sqrt{2}x + 1\right)\left(x^2 - \sqrt{2}x + 1\right)$$

Example 1.139

If $2i$ is a root of $z^4 + z^3 + 2z^2 + 4z - 8$, then what are the other roots? (AP EAPCET, 22 Sep 2020, Shift-I, Adapted)

Since roots occur in pairs for equation with real coefficients, $-2i$ is also a root. Combining the two roots gives us:

$$(z+2i)(z-2i)=0$$
$$z^2+4=0$$

Hence, z^2+4 is a factor of $z^4+z^3+2z^2+4z-8$. By polynomial long division:

$$(z^2+4)(z^2+z-2)=0$$
$$(z^2+4)(z+2)(z-1)=0$$

Hence, the other roots are

$$z \in \{-2,+1,-2i\}$$

$$\begin{array}{r|l}
 & z^2+z-2 \\ \hline
z^2+0z+4 & z^4+z^3+2z^2+4z-8 \\
 & -z^4-0-4z^2 \\ \hline
 & +z^3-2z^2+4z \\
 & -z^3-0-4z \\ \hline
 & -2z^2+0-8 \\
 & +2z^2-0+8 \\ \hline
 & 0
\end{array}$$

G. Seventh Degree Equations

Example 1.140

Let $\alpha_1, \alpha_2, \dots, \alpha_7$ be the roots of $x^7+3x^5-13x^3-15x=0$, and $|\alpha_1| \ge |\alpha_2| \ge \dots \ge |\alpha_7|$. Then $\alpha_1\alpha_2-\alpha_3\alpha_4+\alpha_5\alpha_6$ is equal to: (JEE Main, Jan 29, 2023-II)

Factor out x on the RHS:

$$x(x^6+3x^4-13x^2-15)=0$$
$$x=0\ OR\ x^6+3x^4-13x^2-15=0$$

$$|z| \ge 0 \Rightarrow |\alpha_7|=0 \Rightarrow \alpha_7=0$$

Hence, we need consider only the second equation. Use a change of variable. Let $t=x^2$:

$$t^3+3t^2-13t-15=0 \Rightarrow (t-3)(t+1)(t+5)=0$$

From the cubic, we get:

$$t=x^2 \in \{3,-1,-5\} \Rightarrow x \in \{\pm\sqrt{3}, \pm i, \pm\sqrt{5}i\}$$

$$|\alpha_1|=|\alpha_2|=\sqrt{5}$$
$$|\alpha_3|=|\alpha_4|=\sqrt{3}$$
$$|\alpha_5|=|\alpha_6|=1$$

$$\alpha_1\alpha_2-\alpha_3\alpha_4+\alpha_5\alpha_6=(\sqrt{5}i)(-\sqrt{5}i)-(\sqrt{3})(-\sqrt{3})+(i)(-i)=5-(-3)-(-1)=5+3+1=9$$

H. Sum of Squares

1.141: Revision: Sum of Squares

If the sum of squares of real numbers a and b is zero, then the numbers are themselves zero:

$$a^2+b^2=0 \Leftrightarrow a=b=0$$

For a real number, a, we must have

$$a^2 \ge 0, \qquad b^2 \ge 0$$

1.142: Sum of Squares

If the sum of squares of complex numbers z and w is zero, then we cannot conclude the numbers are themselves zero:

$$z^2+w^2=0 \nRightarrow z=w=0$$

$$z^2+w^2=0$$

The sum of squares property that holds for real numbers does not hold for complex numbers.

Example 1.143

Give an example of complex numbers z and w that satisfy

$$z^2 + w^2 = 0, \qquad z \neq 0, w \neq 0$$

$$-1 + 1 = 0$$
$$z^2 = i^2 = -1 \Rightarrow z = i$$
$$w = 1$$

I. Resources

Further Practice 1.144

Alcumus > Algebra > Complex Number Arithmetic covers topics done so far:

- Imaginary Numbers: Powers, Series, Cyclicity
- Complex Numbers:
  - ✓ Arithmetic: Addition, Subtraction, Multiplication, Division
  - ✓ Squaring
  - ✓ Linear Equations in Complex Numbers

1.6 Exponentiation

A. Positive Integer Powers

1.145: Positive Integer Powers

$$z^n = \underbrace{z \cdot z \cdot \ldots \cdot z}_{n\ times}$$

Example 1.146

If $i^2 = -1$, then $(1+i)^{20} - (1-i)^{20}$ equals **(AHSME 1974/17)**

Rather than exponentiating 20 times, note that squaring it makes the expressions much simpler:

$$(1+i)^2 = 2i$$
$$(1-i)^2 = -2i$$

$$(1+i)^{20} - (1-i)^{20} = (2i)^{10} - (-2i)^{10} = (2i)^{10} - (2i)^{10} = 0$$

Example 1.147

Suppose two real numbers a and b are both unity. Let z_1 be a complex number equal to $a + bi$. Let z_2 be the conjugate of z_1. Let z_3 be the quotient of z_1 and z_2. Let z_4 be the n^{th} power of z_3, where n is a natural number. Find the smallest possible value of n such that z_4 is 1.

$$z_3 = \frac{z_1}{z_2} = \frac{a+bi}{a-bi} = \frac{1+i}{1-i} = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1-1+2i}{1+1} = \frac{2i}{2} = i$$

We now need to find:

$$z_4 = z_3^n = i^n = 1 \Rightarrow Smallest\ n = 4$$

Example 1.148: Counting

For how many natural numbers 'n' such that $1 \le n \le 2021$ is $\left(\frac{1+i}{1-i}\right)^n = 1$?**(AP EAPCET, 21 Sep. 2020, Shift-II)**

Multiply the numerator and denominator by the complex conjugate of the denominator:

$$\frac{1+i}{1-i} \times \frac{1-i}{1-i} = \frac{1+i^2+2i}{1+1} = \frac{2i}{2} = i$$

$$\left(\frac{1+i}{1-i}\right)^n = i^n = 1$$

If n is an integral multiple of 4, then $i^n = 1$. The possible values of n are:

$$\{4,8,\dots,2020\} = \{4 \times 1, 4 \times 2, \dots, 4 \times 505\} \Rightarrow 505\ Terms \Rightarrow 505\ numbers$$

Example 1.149: Number Theory

If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{i-1}\right)^{\frac{n}{3}} = 1, (m, n \in N)$, then the greatest common divisor of the least values of m and n is **(JEE Main 3 Sep. 2020, Shift-I)**

$$\frac{1+i}{1-i} = i, \qquad \frac{1+i}{i-1} = -\frac{1+i}{1-i} = -i$$

Substitute the above in given equality:

$$i^{\frac{m}{2}} = (-i)^{\frac{n}{3}} = 1$$

Note that $\frac{m}{2}$ must be a multiple of 4:

$$\frac{m}{2} = 4k \Rightarrow m = 8k$$

Note that $\frac{n}{3}$ must be a multiple of 4:

$$\frac{n}{3} = 4k \Rightarrow n = 12k$$

Smallest possible value of k is 1.

$$GCD\{m,n\} = GCD\{8,12\} = 4$$

B. Square Roots

1.150: Square roots

Example 1.151

Find the square root of $15 + 8i$ using the method of undetermined coefficients.

We want to find a complex number z such that:

$$z = a + bi = \sqrt{15 + 8i}$$

Square both sides:

$$\underbrace{a^2 - b^2}_{Real} + \underbrace{2ab}_{Complex}\ i = \underbrace{15}_{Real} + \underbrace{8}_{Complex}\ i$$

Use the method of undetermined coefficients. Equating the real and imaginary parts, we get:

$$a^2 - b^2 = 15$$

$$2ab = 8 \Rightarrow ab = 4 = 1 \times 4 = 2 \times 2$$

To solve the system of equations above, try:

$$a = 4, b = 1 \Rightarrow a^2 - b^2 = 16 - 1 = 15 \Rightarrow Works$$

$$\sqrt{15 + 8i} = a + bi = 4 + i$$

Example 1.152

Find the square root of i using the method of undetermined coefficients.

Let

$$\sqrt{i} = a + bi \Rightarrow i = a^2 - b^2 + 2abi$$

Equate the real parts:

$$a^2 - b^2 = 0 \Rightarrow a^2 = b^2 \Rightarrow a = \pm b$$

Equate the imaginary parts:

$$\underbrace{2ab}_{+ve} = \underbrace{1}_{+ve}$$

Since the RHS is positive, both a and b must be positive, or both must be negative. Hence, $a = b$, which we substitute:

$$2(b)b = 1 \Rightarrow b = \pm\sqrt{\frac{1}{2}} = \pm\frac{1}{\sqrt{2}}$$

$$\sqrt{i} \in \left\{-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i, \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right\}$$

Example 1.153

Find the square roots of 1 in the complex number system using the method of undetermined coefficients.

$$\sqrt{1} = a + bi$$

Square both sides:

$$1 = a^2 - b^2 + 2abi$$

Equate the imaginary parts:

$$2ab = 0 \Rightarrow a = 0\ OR\ b = 0$$

Equate the real parts:

$$a^2 - b^2 = 1$$

$$Case\ I: a = 0 \Rightarrow -b^2 = 1 \Rightarrow b = \sqrt{-1} \Rightarrow Not\ Valid$$

$$Case\ II: b = 0 \Rightarrow a^2 = 1 \Rightarrow a = \pm 1$$

$$a + bi = \pm 1 + 0i$$

C. Cube Roots

Example 1.154

Find the cube roots of 1 by cubing $\sqrt[3]{1} = a + bi$, and then using the method of undetermined coefficients.[1]

Cube $\sqrt[3]{1} = a + bi$:

$$1 = a^3 - 3ab^2 + (3a^2b - b^3)i$$

Equate the imaginary parts:

$$b(3a^2 - b^2) = 0$$

$$b = 0\ OR\ 3a^2 - b^2 = 0 \Rightarrow a = \pm\frac{b}{\sqrt{3}}$$

Equate the real parts:

$$a^3 - 3ab^2 = 1$$

Case I: $b = 0$

$$a^3 = 1 \Rightarrow a = 1$$

Case II: $a = \frac{b}{\sqrt{3}}$

$$\left(\frac{b}{\sqrt{3}}\right)^3 - 3\left(\frac{b}{\sqrt{3}}\right)b^2 = 1$$

$$\frac{b^3}{3\sqrt{3}} - \frac{3b^3}{\sqrt{3}} = 1$$

$$\frac{b^3 - 9b^3}{3\sqrt{3}} = 1$$

[1] This is not the most efficient method of finding the cube roots. We will see a faster method later on.

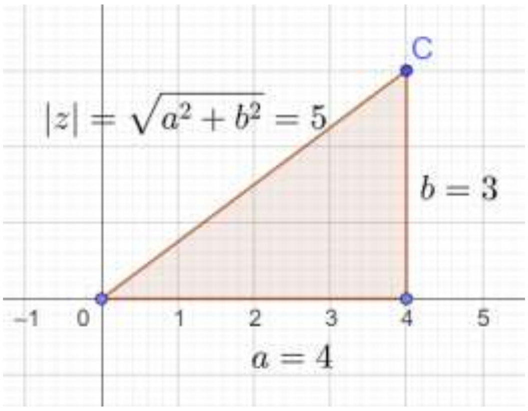
$$-8b^3 = 3\sqrt{3}$$
$$b^3 = -\frac{3\sqrt{3}}{8}$$
Take the cube root of both sides:
$$b = -\frac{\sqrt{3}}{2}$$

$$3\sqrt{3} = \sqrt{27}$$
Take the cube root:
$$\sqrt[3]{\sqrt{27}} = \left(27^{\frac{1}{2}}\right)^{\frac{1}{3}} = (3^3)^{\frac{1}{6}} = 3^{\frac{1}{2}} = \sqrt{3}$$

2. Complex Numbers: Geometry

2.1 Distance (Modulus)

A. Distance from Origin



2.1: Modulus/Absolute Value of a complex number

The modulus or absolute value of a number is the distance of that number in the complex plane from the origin, given by:

$$|z| = |a + bi| = \sqrt{a^2 + b^2}$$

- The distance concept of absolute value can be related to known concepts in a number of ways.
- The modulus of a complex number is always a nonnegative real number.
- The modulus of a complex number is also called its magnitude.

Example 2.2: Basics

Find the magnitude of the following complex numbers:

A. $3 + 4i$
B. $5 + 12i$
C. $8 - 15i$
D. $\sqrt{3} + \sqrt{4}i$
E. $\frac{1+i}{1-i}$

Part A

$$|3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Part B

Recognize that (5,12,13) is a Pythagorean Triplet and hence:

$$|5 + 12i| = 13$$

Part C

Recognize that (8,15,17) is a Pythagorean Triplet and hence:

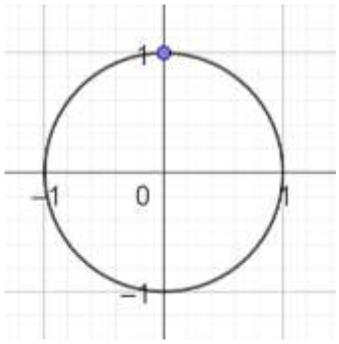
$$|8 - 15i| = 17$$

Example 2.3

Find the magnitude of the following complex numbers:

$$z = \frac{1+i}{1-i}$$

$$z = \frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{1 + 2i - 1}{1 - i^2} = \frac{2i}{2} = i$$



Algebraic Method
We can do this algebraically:

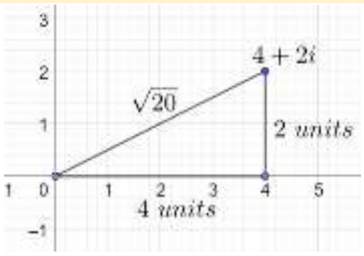
$$i = 0 + i = \sqrt{0^2 + 1^2} = \sqrt{1} = 1$$

Geometric Method
We can also do this on the complex plane. The number i lies on the unit circle, and hence has distance 1 from the origin.

$$|i| = 1$$

Example 2.4

A. Find the distance from the origin of the complex number $z_1 = 4 + 2i$
B. Find the absolute value of the complex number $z_2 = 3 + 4i$
C. Find $|z_1| + |z_2|$
D. Find $|z_1| - |z_2|$



Part A

$$|z_1| = \sqrt{4^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$

Part B

$$|z_2| = |3 + 4i| = \sqrt{3^2 + 4^2} = 5$$

Part C

$$|z_1| + |z_2| = 2\sqrt{5} + 5$$

Part D

$$|z_1| - |z_2| = 2\sqrt{5} - 5$$

Concept 2.5: Distance on the Real Plane

Show how the absolute value of a complex number compares with the distance formula for real numbers.

Distance on the real plane between two points (x_1, y_1) and (x_2, y_2) is measured using the distance formula, given by:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

In this context, we require distance from the origin $(0,0)$, and hence the distance is:

$$\sqrt{(x_2)^2 + (y_2)^2}$$

In our case $a = x, b = y$, we get:

$$\sqrt{(x_2)^2 + (y_2)^2} = \sqrt{a^2 + b^2}$$

Concept 2.6: Generalization of Distance on the Real Number Line

Show how the absolute value of a complex number generalizes the absolute value of a real number.

The absolute value of a real number gives the distance of the number from zero. Correspondingly, the absolute value of a complex number gives the distance of the number from the origin.
Hence, the absolute value of a complex number is a generalization of the absolute value of a real number.

We can see this by using the formula for absolute value of a complex number on a purely real number:

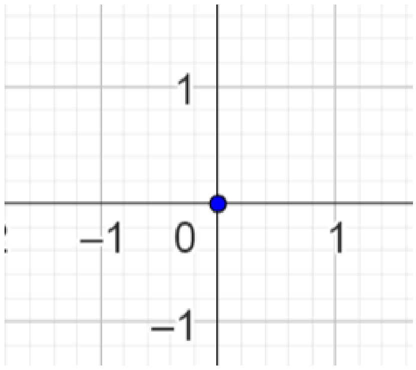
Substitute $b = 0$ in $|z| = |a + bi| = \sqrt{a^2 + b^2}$

$$|a + 0i| = \sqrt{a^2 + 0^2} = \sqrt{a^2} = |a|$$

2.7: Magnitude of zero

For a complex number z:

$$|z| = 0 \Leftrightarrow z = 0 = 0 + 0i$$



Graphical Method

$|z| = 0$ means that the distance of z from the origin is zero.
Which means that z is the origin itself.
Which means that

$$z = 0 + 0i = 0$$

Algebraic Method

$$|z| = 0$$
$$a^2 + b^2 = 0$$
$$a = b = 0$$

B. Distance between two points

2.8: Distance in the Complex Plane

The distance between two complex numbers z_2 and z_1 is given by:

$$|z_2 - z_1|$$

The distance between two real numbers r_1 and r_2 on the real number line is given by:

$$|r_2 - r_1|$$

The same formula holds for complex numbers. The distance between two complex numbers z_1 and z_2 on the complex plane number line is given by

$$|z_2 - z_1|$$

Example 2.9

A cat on the complex plane at $z_1 = 4 + 2i$ is trying to catch a rat at $z_2 = 3 + 4i$. Find the minimum distance that it must travel if the rat is frozen in terror, and remains stationary.

Distance on the complex plane

$$= |z_2 - z_1| = |(3 + 4i) - (4 + 2i)| = |-1 + 2i| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

Example 2.10

Is $|z_2| - |z_1| = |z_2 - z_1|$ in general? If yes, prove it. If no, find a counterexample.

$$z_2 = 4, z_1 = -3$$
$$LHS = |z_2| - |z_1| = 4 - 3 = 1$$
$$RHS = |4 - (-3)| = |7| = 7$$

2.11: Distance is non-negative

$$|z| \geq 0$$

Example 2.12

If z is a complex number, find the solution set to:

A. $|z| = -3$
B. $|z| < -2$

Since distance is never negative, the modulus of a complex number is negative, and hence each of the parts has:

$$No\ Solutions$$

$$z \in \phi$$

C. Properties of Modulus

2.13: Distribution over Multiplication

The modulus operation is commutative (or distributive) over multiplication.

$$|z_1 z_2| = |z_1| \cdot |z_2|$$

Let $z_1 = a + bi, z_2 = c + di$. Then:

$$LHS = |z_1 z_2| = |(a+bi)(c+di)| = |ac - bd + i(cb + ad)|$$

Use the definition of modulus:

$$= \sqrt{(ac-bd)^2 + (cb+ad)^2}$$

Expand:

$$= \sqrt{a^2c^2 - 2abcd + b^2d^2 + c^2b^2 + 2abcd + a^2d^2}$$

Simplify:

$$= \sqrt{a^2c^2 + b^2d^2 + c^2b^2 + a^2d^2}$$

Factor:

$$= \sqrt{(a^2+b^2)(c^2+d^2)}$$

Separate into two radicals:

$$= \sqrt{a^2+b^2} \cdot \sqrt{c^2+d^2} = |a+bi| \cdot |c+di| = |z_1| \cdot |z_2| = RHS$$

Example 2.14

What is the modulus of the complex number $(1+2i)(-2+i)$ (AP EAPCET 17 Sep. 2020, Shift-II)

Apply the mod:

$$|(1+2i)(-2+i)|$$

Use the distributive property over multiplication:

$$= |1+2i||-2+i|$$

Calculate the mod of each term separately:

$$= \sqrt{1^2+2^2}\sqrt{2^2+1^2} = \sqrt{5}\sqrt{5} = 5$$

Example 2.15

Consider the complex number $z = \frac{1-3i}{2+i}$. Let A be the conjugate of the modulus of the reciprocal of z, and B be the reciprocal of the conjugate of the modulus of z. Find $A - B$.

2.16: Distribution over Division

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$$

Example 2.17

If $\left|\frac{z-25}{z-1}\right| = 5$, then $|z| =$ (AP EAPCET, 22 Sep. 2020, Shift-II)

Let $z = x + iy$:

$$\left|\frac{x+iy-25}{x+iy-1}\right| = 5$$

Use the property $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$

$$\frac{|x+iy-25|}{|x+iy-1|}=5$$

Use the definition of modulus:

$$\frac{\sqrt{(x-25)^2+y^2}}{\sqrt{(x-1)^2+y^2}}=5$$

Square both sides of the above and cross-multiply to eliminate fractions:

$$(x-25)^2+y^2=25[(x-1)^2+y^2]$$

Expand:

$$x^2-50x+625\ +\ y^2=25x^2-50x+25+25y^2$$
$$24x^2+24y^2=600$$

Divide both sides by 24:

$$x^2+y^2=\frac{600}{24}=25$$

Take square roots both sides:

$$|z|=\sqrt{x^2+y^2}=\sqrt{25}=5$$

Example 2.18

If $x+iy=\frac{(3+2i)(4-7i)(12+13i)}{(13-12i)(2-3i)(11+3i)}$ then $x^2+y^2=$ **(AP EAPCET 17 Sep. 2020, Shift-II)**

Taking the mod on the LHS gives us:

$$LHS=|x+iy|=\sqrt{x^2+y^2}$$

Taking the mod on the RHS gives us:

$$RHS=\left|\frac{(3+2i)(4-7i)(12+13i)}{(13-12i)(2-3i)(11+3i)}\right|$$

Distribute the mod over division:

$$=\frac{|(3+2i)(4-7i)(12+13i)|}{|(13-12i)(2-3i)(11+3i)|}$$

Distribute the mod over multiplication:

$$=\frac{|3+2i||4-7i||12+13i|}{|13-12i||2-3i||11+3i|}$$

Apply the definition of mod:

$$=\frac{\sqrt{3^2+2^2}\sqrt{4^2+7^2}\sqrt{12^2+13^2}}{\sqrt{13^2+12^2}\sqrt{2^2+3^2}\sqrt{11^2+3^2}}$$

Cancel:

$$=\frac{\sqrt{4^2+7^2}}{\sqrt{11^2+3^2}}=\frac{\sqrt{16+49}}{\sqrt{121+9}}=\frac{\sqrt{65}}{\sqrt{130}}=\sqrt{\frac{65}{130}}=\frac{1}{\sqrt{2}}$$

Finally equate LHS and RHS:

$$\sqrt{x^2+y^2}=\frac{1}{\sqrt{2}}\Rightarrow x^2+y^2=\frac{1}{2}$$

2.19: Distribution over Exponentiation

$$|z^n|=|z|^n$$

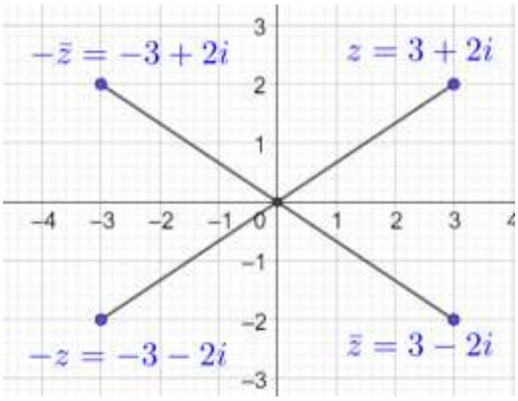
The multiplication property for modulus can be extended to exponentiation.

D. Conjugate Properties

2.20: Mod of a number and its conjugate

The distance from the origin of a number is the same as the distance of its conjugate.

$$|z|=|-z|=|\bar{z}|=|-\bar{z}|$$



The adjoining diagram explains this geometrically:

- $-z$ is the reflection of z across the origin, and it will have same distance from the origin as z.
- $\bar{z}$ is the reflection of z across the x-axis, and it will have the same distance from the origin as z.
- $-\bar{z}$ is the reflection of z across the y-axis, and it will have the same distance from the origin as z.

2.21: Product of a number and its conjugate is square of the modulus

$$z\bar{z}=|z|^2$$

Algebraic Proof

$$z\bar{z}=(a+bi)(a-bi)=a^2-b^2i^2=a-(-b^2)=a^2+b^2=|z|^2$$

Geometric Interpretation

You can interpret this geometrically as well.

Example 2.22

Given that z is a non-zero complex number such that $\bar{z}=iz^2$ then $z^3=$ is (AP EAPCET, 21 April 2019, Shift-I, Adapted)

Take mod on both sides of the given equation:

$$|\bar{z}|=|iz^2|$$

Use the distributive property of modulus:

$$|\bar{z}|=|i||z^2|$$

Substitute $|i|=1$

$$|\bar{z}|=|z|^2$$

Substitute $|\bar{z}|=|z|$:

$$|z|=|z|^2\Rightarrow|z|\in\{0,1\}$$

But since z in non-zero, we reject $z=0$:

$$|z| = 1$$

Multiply both sides of $\bar{z} = iz^2$ with z:

$$z\bar{z} = iz^3 \Rightarrow |z|^2 = iz^3 \Rightarrow \frac{1}{i} = z^3 \Rightarrow z^3 = -i$$

Example 2.23

A. Can you solve the above question by using $z = a + bi$.
B. Should you solve using this method?

$$a - bi = i(a + bi)^2$$
$$a - bi = i(a^2 + 2abi - b^2)$$
$$a - bi = a^2 i - 2ab - b^2 i$$
$$0 = a^2 i - b^2 i + bi - 2ab - a$$
$$-2ab - a = 0$$

Equate the real parts:

$$2ab + a = 0$$
$$a(2b + 1) = 0$$
$$a = 0\ OR\ b = -\frac{1}{2}$$

Equate the imaginary parts

$$a^2 - b^2 + b = 0$$
$$a = 0 \Rightarrow -b^2 + b = 0 \Rightarrow b^2 = b \Rightarrow b \in \{0,1\}$$
$$z = a + bi = 0 + i$$
$$z^3 = -i$$

$$a^2 - \frac{1}{4} - \frac{1}{2} = 0$$
$$a^2 - \frac{3}{4} = 0$$
$$a^2 = \frac{3}{4}$$
$$a = \pm\frac{\sqrt{3}}{2}$$

$$z = \left\{\frac{\sqrt{3}}{2} - \frac{1}{2}i, -\frac{\sqrt{3}}{2} - \frac{1}{2}i\right\} \Rightarrow z^3 = -i$$

2.24: Conjugate Property

If a complex number z lies on the unit circle, its conjugate $\bar{z}$ is equal to its reciprocal $\frac{1}{z}$. That is:

$$|z| = 1 \Rightarrow \bar{z} = \frac{1}{z}$$

$$|z| = 1$$

Square both sides:

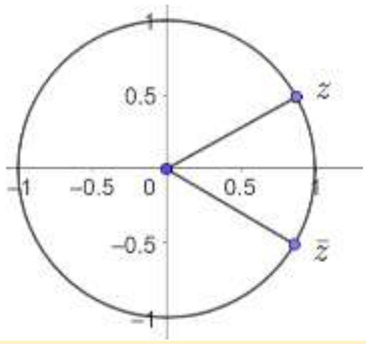
$$|z|^2 = 1$$

Substitute $z\bar{z} = |z|^2$:

$$z\bar{z} = 1$$

Solve for $\bar{z}$:

$$\bar{z} = \frac{1}{z}$$



Example 2.25

If z_1, z_2 and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, then $|z_1 + z_2 + z_3|$ is (JEE Advanced 2000S)

From the property $|z| = 1 \Rightarrow \bar{z} = \frac{1}{z}$, we must have:

$$\bar{z_1} = \frac{1}{z_1}, \bar{z_2} = \frac{1}{z_2}, \bar{z_3} = \frac{1}{z_3}$$

Make the substitution to get:

$$1 = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = |\bar{z_1} + \bar{z_2} + \bar{z_3}|$$

Use the distributive property of conjugation over addition:

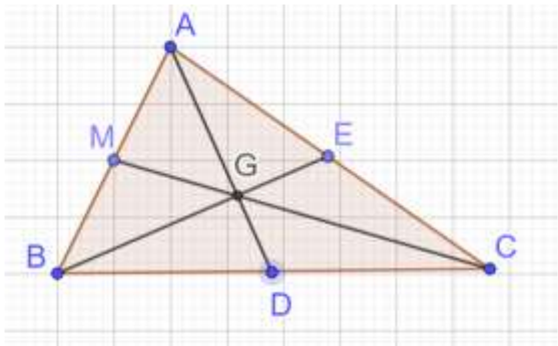
$$1 = |\overline{z_1 + z_2 + z_3}|$$

Use the property that $|z| = |\bar{z}|$

$$1 = |z_1 + z_2 + z_3|$$

Hence, the quantity to be found is also 1.

E. Geometry



2.26: Centroid of a Triangle (Real Plane)

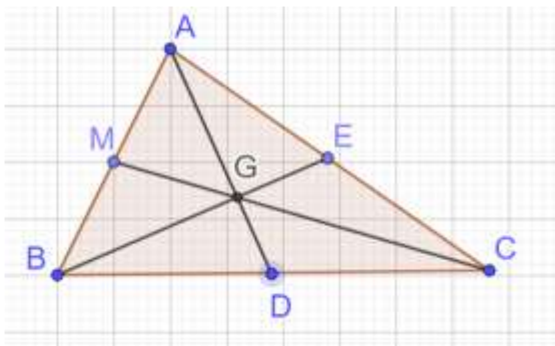
The three medians of a triangle are concurrent at its centroid. The centroid of a triangle with vertices $(x_1, y_1)(x_2, y_2)(x_3, y_3)$ in the real plane is:

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

2.27: Centroid of a Triangle (Complex Plane)

The centroid of a triangle with vertices (z_1, z_2, z_3) in the complex plane is:

$$\frac{z_1 + z_2 + z_3}{3}$$



2.28: Centroid of a Triangle

The centroid of a triangle divides its medians in the ratio 2: 1, with the longer side towards the vertex and the shorter side towards the base.

$$AG = 2 \cdot GD$$
$$BG = 2 \cdot GE$$

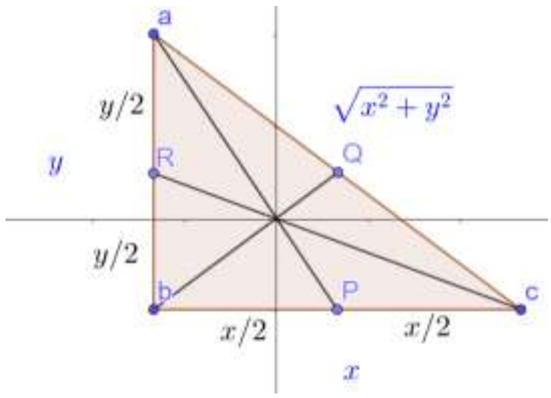
$$CG = 2 \cdot GM$$

Example 2.29

Complex numbers a, b, and c are zeros of a polynomial $P(z) = z^3 + qz + r$, and $|a|^2 + |b|^2 + |c|^2 = 250$. The points corresponding to a, b, and c in the complex plane are the vertices of a right triangle with hypotenuse h. Find h^2. **(AIME-I 2012/14)**

Apply Vieta's Formulas to the polynomial $P(z) = z^3 + 0z^2 + qz + r$.

$$Sum\ of\ Roots = a + b + c = -\frac{0}{1} = 0$$



The centroid of the triangle is:

$$\frac{a+b+c}{3} = \frac{0}{3} = 0 \Rightarrow Centroid\ is\ the\ origin$$

Draw the triangle, and without loss of generality, let ac be the hypotenuse.

In right ΔabP, by Pythagoras Theorem, the hypotenuse

$$= Length\ of\ median\ from\ a = \sqrt{\left(\frac{x}{2}\right)^2 + y^2} = \sqrt{\frac{x^2+4y^2}{4}}$$

Since the centroid divides the medians in the ratio 2: 1, the magnitude of a is $\frac{2}{3}$ the length of the median:

$$|a| = \frac{2}{3}\sqrt{\frac{x^2+4y^2}{4}} \Rightarrow |a|^2 = \frac{4}{9}\left(\frac{x^2+4y^2}{4}\right) = \frac{x^2+4y^2}{9}$$

Similarly:

$$|c| = \frac{2}{3}\sqrt{x^2 + \left(\frac{y}{2}\right)^2} = \frac{2}{3}\sqrt{\frac{4x^2+y^2}{4}} \Rightarrow |c|^2 = \frac{4}{9}\left(\frac{4x^2+y^2}{4}\right) = \frac{4x^2+y^2}{9}$$

The median from b is half the length of the hypotenuse:

$$|b| = \frac{2}{3} \cdot \frac{\sqrt{x^2+y^2}}{2} \Rightarrow |b|^2 = \frac{4}{9} \cdot \frac{x^2+y^2}{4} = \frac{x^2+y^2}{9}$$

Substitute

$$|a|^2 + |b|^2 + |c|^2 = 250$$
$$\underbrace{\frac{x^2+4y^2}{9}}_{|a|^2} + \underbrace{\frac{x^2+y^2}{9}}_{|b|^2} + \underbrace{\frac{4x^2+y^2}{9}}_{|c|^2} = 250$$
$$\frac{6x^2+6y^2}{9} = 250$$
$$\frac{2}{3}(x^2+y^2) = 250$$
$$x^2 + y^2 = 375$$
$$h^2 = 375$$

F. Vectors

Challenge 2.30

Complex numbers a, b, and c are zeros of a polynomial $P(z) = z^3 + qz + r$, and $|a|^2 + |b|^2 + |c|^2 = 250$. The points corresponding to a, b, and c in the complex plane are the vertices of a right triangle with hypotenuse h. Find h^2. **(AIME-I 2012/14)**

Convert to Vector Notation

Convert everything to the real number plane, and apply vector concepts. Call the points in the real plane corresponding to the complex plane points a, b and c, and call them a, b and c as well.

Assume, without loss of generality, that the right triangle is right-angled at b. Then:

$$h^2 = |\vec{\boldsymbol{a}} - \vec{\boldsymbol{c}}|^2 = (\vec{\boldsymbol{a}} - \vec{\boldsymbol{c}}) \cdot (\vec{\boldsymbol{a}} - \vec{\boldsymbol{c}}) = |\vec{\boldsymbol{a}}|^2 - 2\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}} + |\vec{\boldsymbol{c}}|^2$$

$|a|^2 + |b|^2 + |c|^2 = 250$ when written in vector notation is:

$$\underbrace{|\vec{\boldsymbol{a}}|^2 + \left|\vec{\boldsymbol{b}}\right|^2 + |\vec{\boldsymbol{c}}|^2 = 250}_{\textit{Equation I}}$$

Right-Angle Condition

Since the triangle is right-angled, the legs are perpendicular, which means the dot product of the legs is zero:

$$\left(\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}\right) \cdot \left(\vec{\boldsymbol{c}} - \vec{\boldsymbol{b}}\right) = 0$$

$$\underbrace{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}} - \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} - \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} + \left|\vec{\boldsymbol{b}}\right|^2 = 0}_{\textit{Equation II}}$$

Polynomial Condition

Apply Vieta's Formulas to the polynomial $P(z) = z^3 + qz + r$.

$$\textit{Sum of Roots} = a + b + c = 0$$

Write the above in vector form:

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = \vec{\boldsymbol{0}}$$

Since we have information on the magnitudes, take the dot product of the above with itself:

$$\left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}\right) \cdot \left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}\right) = 0$$

Expand:

$$|\vec{\boldsymbol{a}}|^2 + \left|\vec{\boldsymbol{b}}\right|^2 + |\vec{\boldsymbol{c}}|^2 + 2\left(\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} + \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} + \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}}\right) = \vec{\boldsymbol{0}}$$

Substitute Equation I in the above:

$$250 + 2\left(\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} + \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} + \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}}\right) = 0$$

$$\underbrace{\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} + \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} + \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}} = -125}_{\textit{Equation III}}$$

Combining Conditions

We want to find $|\vec{\boldsymbol{a}}|^2 - 2\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}} + |\vec{\boldsymbol{c}}|^2$. Adding equations II and III gives us the middle term from what we want:

$$2\vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}} + \left|\vec{\boldsymbol{b}}\right|^2 = -125$$

$$\underbrace{-2\vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}} - \left|\vec{\boldsymbol{b}}\right|^2 = 125}_{\textit{Equation IV}}$$

Add Equations I and IV:

$$h^2 = |\vec{\boldsymbol{a}}|^2 - 2\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}} + |\vec{\boldsymbol{c}}|^2 = 375$$

2.2 Vector Operations and Triangle Inequality

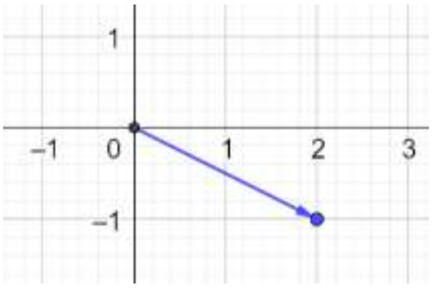
A. Complex Numbers as Vectors

2.31: Complex Numbers as Vectors

A complex number $z = a + bi$ can be represented as a vector (a, b) in the complex plane.

Example 2.32

Represent $2 - i$ as a vector in the complex plane.



2.33: Addition: Algebraic

Two complex numbers (treated as vectors) can be added by adding their individual components:

$$(a, b) + (c, d) = (a + c, b + d)$$

We already know, from algebra, that

$$(a + bi) + (c + di) = a + c + (b + d)i$$

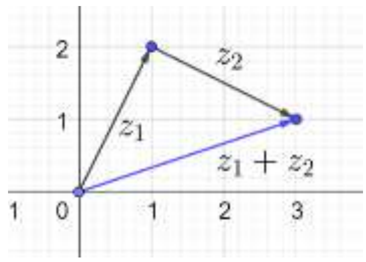
We did examples of algebraic addition in the previous section.

2.34: Addition: Graphical Addition

Two complex numbers (treated as vectors) can be added by representing them as vectors on the complex plane, and carrying out the addition graphically, in the same way that you add vectors on the real plane.

Example 2.35

Represent the numbers $z_1 = 1 + 2i, z_2 = 2 - i$ on the complex plane as vectors, and then add them graphically. Confirm your answer by adding them algebraically.



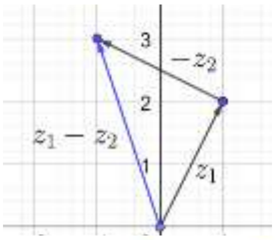
$$z_1 + z_2 = (1 + 2i) + (2 - i) = 3 + i$$

2.36: Subtraction

Two complex numbers (treated as vectors) can be added by representing them as vectors on the complex plane, and adding the inverse of the vector to be subtracted, in the same way that you add vectors on the real plane.

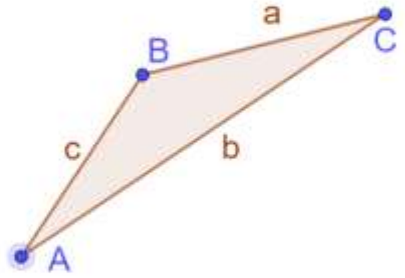
Example 2.37

Represent the numbers $z_1 = 1 + 2i, z_2 = 2 - i$ on the complex plane as vectors, and then find $z_1 - z_2$ graphically. Confirm your answer by adding them algebraically.



$$z_2 = 2 - i \Rightarrow -z_2 = -2 + i$$
$$z_1 - z_2 = (1 + 2i) - (2 - i) = -1 + 3i$$

B. Geometric Interpretation



2.38: Geometry: Triangle Inequality: Sum of Two Sides

In a triangle, any side is less than the sum of the other two sides. In a triangle with sides a, b and c:

$$a \le b + c$$
$$b \le a + c$$
$$c \le a + b$$

- This is a property from geometry, and does not require complex numbers.
- The triangle inequality is a very important property.
- It has some non-obvious applications.

Example 2.39

A. When is the equality achieved in the triangle inequality?
B. What geometric property do we recognize (which will we useful when we consider vectors)?

Part A
Equality is achieved when you have a degenerate triangle.

Part B
In a degenerate triangle, the lines are parallel.
The vertices of the triangle are collinear.

2.40: Triangle Inequality: Difference of Two Sides

In a triangle, the difference of any two sides is less than the third side. In a triangle with sides a, b and c:

$$a - b \le c$$
$$b - a \le c$$
$$a - c \le b$$
$$c - a \le b$$
$$b - c \le a$$
$$c - b \le a$$

$$a \le b + c$$
$$a - c \le b$$

Example 2.41

Use the triangle inequality from geometry to show that if the sides of a triangle are real numbers a,b and c then:

$$|b - c| \le |a| \le |b + c|$$

From the triangle inequality for the sum of two sides:

$$a \le b + c$$

Since $b > 0, c > 0 \Rightarrow b + c = |b + c|$

$$\underbrace{a \le |b + c|}_{\textit{Inequality I}}$$

From the triangle inequality for the difference of two sides, we know that:

$$b - c \le a, \qquad c - b \le a$$

And $|b - c| = \pm(b - c) = b - c \; OR \; c - b$ we can combine the two inequalities above to:

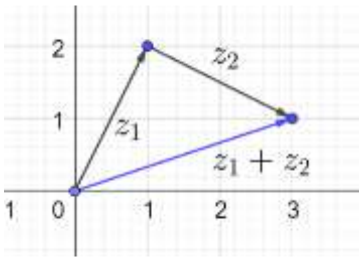
$$\underbrace{|b - c| \le a}_{\textit{Inequality II}}$$

Combine inequalities I and II:

$$|b - c| \le a \le |b + c|$$

Also, $|a| = a$ since $a > 0$:

$$|b - c| \le |a| \le |b + c|$$



2.42: Triangle Inequality: Complex Plane

Given complex numbers z and w, they lie on the complex plane, and satisfy the triangle inequality for complex numbers:

$$\big||z| - |w|\big| \le |z + w| \le |z| + |w|$$

The modulus of the sum of two complex number is

- less than or equal to the sum of the moduli of the individual complex numbers
- greater than or equal to the absolute value of the difference of the moduli of the individual complex numbers.

Example 2.43

$$\big||z| - |w|\big| \le |z + w| \le |z| + |w|$$

In the triangle inequality above, when is the equality achieved?

Trivial Case: Zero Numbers

In the trivial case where $z = w = 0$, we have equality holding throughout:

$$\big||z| - |w|\big| = |0 - 0| = |0| = 0$$
$$|z + w| = |0 + 0| = 0$$
$$|z| + |w| = |0| + |0| = 0 + 0 = 0$$

Purely Real Numbers

If z and w are purely real non-zero numbers:

Case I: $z > 0, w > 0$

In terms of vectors, both numbers are going rightward.

$$|z + w| = |z| + |w|$$

Case II: $z < 0, w < 0$

In terms of vectors, both numbers are going leftward.

$$|z + w| = |z| + |w|$$

The common aspect about Case I and Case II is that the direction of both numbers is the same.

Case III: z and w have opposite sign:

$$|z+w| = \big||z|-|w|\big|$$

2.44: Triangle Inequality: Equality Case

$|z+w| = |z|+|w|$ holds if the direction of z in the complex plane is the same as the direction of w in the complex plane.

Example 2.45

The complex numbers z_1 and z_2 satisfy $|z_1|+|z_2| = |z_1+z_2| = 100$. If $z_1 = 3+5i$ find z_2.

Consider the first and third parts of the equality:

$$|z_1|+|z_2| = 100$$

Solve for $|z_2|$:

$$|z_2| = 100-|z_1| = 100-\sqrt{3^2+5^2} = 100-\sqrt{34}$$

If $|z_1+z_2| = |z_1|+|z_2|$ then z_1 and z_2 must be in the same direction, and have the same $x:y$ ratio.

$$z_1 = 3+5i \Rightarrow x:y = 3:5$$

Divide $100-\sqrt{34}$ in the ratio $3:5$:

$$\frac{3}{8}\left(100-\sqrt{34}\right)+\frac{5}{8}\left(100-\sqrt{34}\right)i$$

2.46: Triangle Inequality: Equality Case

$|z+w| = \big||z|-|w|\big|$ holds if the direction of z in the complex plane is diametrically opposite to the direction of w in the complex plane.

Example 2.47

The complex numbers z_1 and z_2 satisfy $|z_1|-|z_2| = |z_1+z_2| = \pi$. If $z_1 = e+e^2 i$ find z_2.

Consider the first and third parts of the equality:

$$|z_1|-|z_2| = \pi$$

Solve for $|z_2|$:

$$|z_2| = |z_1|-\pi = \sqrt{e^2+e^4}-\pi = e\sqrt{1+e^2}-\pi$$

If $|z_1+z_2| = |z_1|+|z_2|$ then z_1 and z_2 must be in the opposite directions, and have the opposite $x:y$ ratio.

$$z_1 = e+e^2 i \Rightarrow x:y = e:e^2 = 1:e$$

Divide $e\sqrt{1+e^2}-\pi$ in the ratio $1:e$, and take the negative since we want the direction to be opposite.

$$-\left[\frac{1}{1+e}\left(e\sqrt{1+e^2}-\pi\right)+\frac{e}{1+e}\left(e\sqrt{1+e^2}-\pi\right)i\right]$$

2.48: Triangle Inequality: Equality Case

$\big||z|-|w|\big| = |z+w| = |z|+|w|$ holds when

$$z = w = 0$$

Example 2.49

C. Algebraic Proof

To prove this algebraically, we need to build up some properties first.

2.50: Modulus of Real and Imaginary Parts

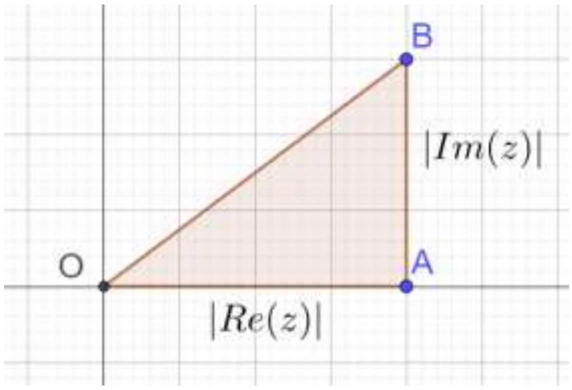
The modulus of the real and imaginary parts of a complex number is less than or equal to the modulus of the number itself. That is:

$$|Re(z)| \le |z|$$
$$|Im(z)| \le |z|$$

Geometric Proof

Consider a complex number on the complex plane

$$|Re(z)| = OA$$
$$|Im(z)| = AB$$
$$|z| = OB$$



And we can see from the diagram that:

$$OA < OB \Rightarrow |Re(z)| \le |z|$$
$$AB < OB \Rightarrow |Im(z)| \le |z|$$

Algebraic Proof

Let $z = a + bi$

$$|z|^2 = |a + bi|^2 = a^2 + b^2$$
$$|Re(z)|^2 = |Re(a + bi)|^2 = |a|^2 = a^2$$
$$|Im(z)|^2 = |Im(a + bi)|^2 = |b|^2 = b^2$$

$$a^2 \le a^2 + b^2$$
$$|Re(z)|^2 \le |z|^2$$
$$|Re(z)| \le |z|$$

Example 2.51

Determine the range of $Re(z)$ if $|Re(z)| \le 3$

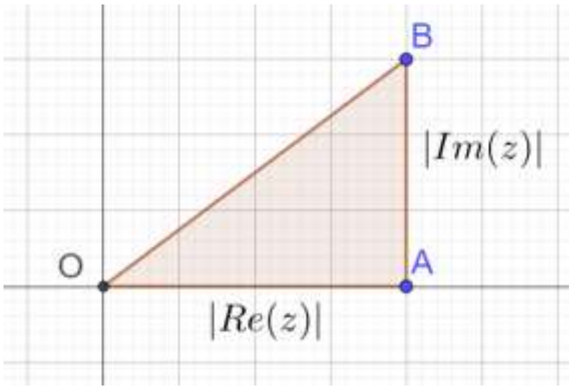
Use the property that $|x| \le a \Rightarrow -a \le x \le a$:

$$-3 \le Re(z) \le 3$$

2.52: Modulus of Real and Imaginary Parts-II

$$|Re(z)|^2 + |Im(z)|^2 = |z|^2$$

$$|z|^2 = |a+bi|^2 = a^2 + b^2$$
$$|Re(z)|^2 = |Re(a+bi)|^2 = |a|^2 = a^2$$
$$|Im(z)|^2 = |Im(a+bi)|^2 = |b|^2 = b^2$$



2.53: Square of the sum of two moduli

$$\left(|z|+|w|\right)^2 = |z|^2 + 2|z||w| + |w|^2$$
$$\left(|z|-|w|\right)^2 = |z|^2 - 2|z||w| + |w|^2$$

Use the property $(a+b)^2 = a^2 + 2ab + b^2$:

$$\left(|z|+|w|\right)^2 = \left(|z|+|w|\right)\left(|z|+|w|\right) = |z|^2 + 2|z||w| + |w|^2$$

2.54: Binomial Expansion of a Modulus[2]

$$|z+w|^2 = |z|^2 + 2Re|z\bar{w}| + |w|^2$$

$$LHS = |z+w|^2$$

Use the property $z\bar{z} = |z|^2$

$$= (z+w)\overline{(z+w)}$$

Use the distributive property of conjugation over addition:

$$= (z+w)(\bar{z}+\bar{w})$$

Multiply:

$$= \underbrace{z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w}}_{\textit{Expression I}}$$

Note that:

$$Re(z\bar{w}) = Re[(a+bi)(c-di)] = Re[ac - \mathbf{adi} + \mathbf{bci} + bd] = ac + bd$$

Consider the two middle terms. Let $z = a+bi, w = c+di$:

$$z\bar{w} + w\bar{z} = (a+bi)(c-di) + (a-bi)(c+di)$$
$$= ac - \mathbf{adi} + \mathbf{bci} + bd + (ac + \mathbf{adi} - \mathbf{bci} + bd)$$
$$= 2(ac+bd)$$
$$= 2Re(z\bar{w})$$

Substitute $z\bar{z} = |z|^2, z\bar{w} + w\bar{z} = 2Re(z\bar{w}), w\bar{w} = |w|^2$ in Expression I:

$$= |z|^2 + 2Re|z\bar{w}| + |w|^2 = RHS$$

2.55: Triangle Inequality

[2] You can think of this property as the complex version of $(a+b)^2 = a^2 + 2ab + b^2$

Given complex numbers z and w, they lie on the complex plane, and satisfy the triangle inequality for complex numbers:

$$\left||z|-|w|\right| \le |z+w| \le |z|+|w|$$

Begin with the expression:

$$|z|^2+2Re|z\overline{w}|+|w|^2$$

Since $-|z\overline{w}| \le Re|z\overline{w}| \le |z\overline{w}|$:

$$|z|^2-2|z\overline{w}|+|w|^2 \le |z|^2+2Re|z\overline{w}|+|w|^2 \le |z|^2+2|z\overline{w}|+|w|^2$$

Note that:

$$\left(\left||z|+|w|\right|\right)^2 = |z|^2+2|z||w|+|w|^2$$
$$\left(\left||z|-|w|\right|\right)^2 = |z|^2-2|z||w|+|w|^2$$
$$|z+w|^2 = |z|^2+2Re|z\overline{w}|+|w|^2$$

Make the substitutions to get:

$$\left(\left||z|-|w|\right|\right)^2 \le |z+w|^2 \le \left(\left||z|+|w|\right|\right)^2$$

Take square roots throughout:

$$\left||z|-|w|\right| \le |z+w| \le |z|+|w|$$

Example 2.56

If a, b are the least and the greatest values respectively of $|z_1+z_2|$, where $z_1 = 12+5i$ and $|z_2| = 9$, then $a^2+b^2 =$ (AP EAPCET, 21 April. 2019, Shift-II)

$$|z_1| = \sqrt{12^2+5^2} = \sqrt{144+25} = \sqrt{169} = 13$$
$$|z_2| = 9$$

$$Greatest\ Value\ of\ |z_1+z_2| = 13+9 = 22 = a$$
$$Least\ Value\ of\ |z_1+z_2| =$$

:east value of $|z_1+z_2| = 13-9 = 4 = b$

Hence, $a^2+b^2 = (22)^2+(4)^2 = 484+16 = 500$

$\therefore a^2+b^2 = 500$

D. Range of Modulus

Example 2.57

If z is a complex number such that $\left|z-\frac{6}{z}\right| = 5$, then the maximum value of $|z|$ is (AP EAPCET, 22 April. 2019, Shift-II)

$$-5 \le |z| - \frac{6}{|z|} \le 5$$

We break the compound inequality above into two separate inequalities

Inequality I: $\boldsymbol{-5 \le |z| - \frac{6}{|z|}}$

$$-5|z| \le |z|^2 - 6$$

Collate all terms on one side:

$$|z|^2 + 5|z| - |z| \ - 6 \ge 0$$

Factor:

$$(|z| - 1)(|z| + 6) \ge 0$$

This is

Since modulus cannot be negative, we ignore the second value and consider:

$$\therefore |z| \ge 1$$

Inequality I: $\boldsymbol{|z| - \frac{6}{|z|} \le 5}$

$$|z|^2 - 6 \le 5|z|$$
$$|z|^2 - 5|z| - 6 \le 0$$
$$(|z| - 6)(|z| + 1) \le 0$$
$$|z| \le 6$$
$$1 \le |z| \le 6$$

So, maximum value of (z) = 6

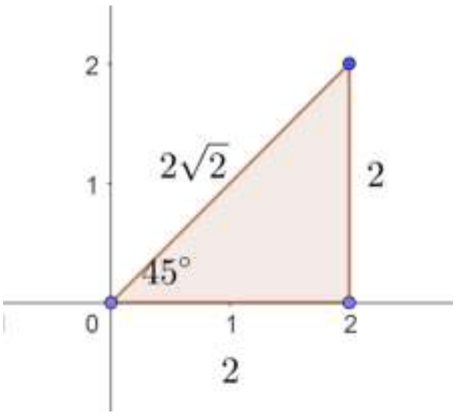
2.3 Polar Form

A. Polar Form

2.58: Rectangular Form

A complex number in rectangular form is of the form:

$$z = a + bi, a \in \mathbb{R}, b \in \mathbb{R}, i = \sqrt{-1}$$



We have seen complex numbers in only the rectangular form so far. However, complex numbers can be written in more than one form. We cover polar form in this section, and exponential form in the next section.

2.59: Polar Form of a complex number

A number z in the complex plane can be uniquely identified as the ordered pair:

$$(r, \theta)$$

A point in the Cartesian (real) plane can be identified using rectangular coordinates (x, y). It can also be identified using polar coordinates (r, θ). Similarly, points in the complex plane can also be written in polar form.

2.60: Magnitude

For a number in polar form, r is the modulus of the complex number. It is also called the magnitude.

$$r = |z|$$

r represents the distance of z from the origin.

2.61: Argument

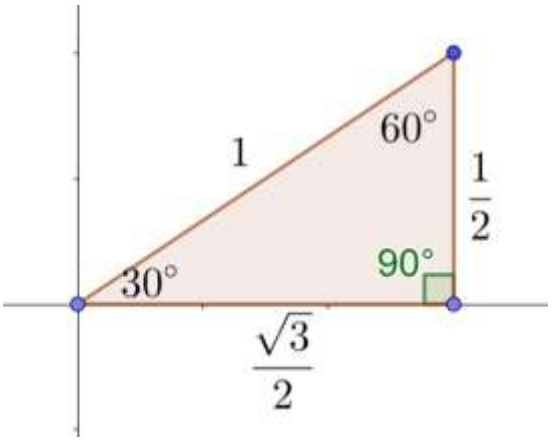
For a number in polar form, θ is the argument of the complex number.

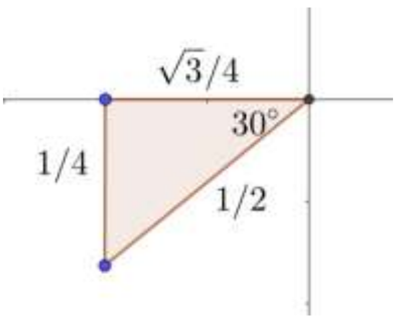
$$\theta = \arg(z)$$

Example 2.62

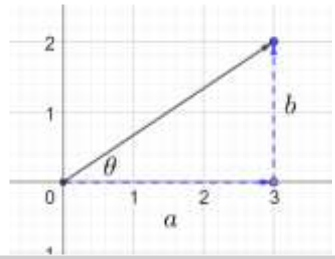
Plot the following points given in polar form (r, θ) as vectors on the complex plane. Then write them in rectangular form.

A. $(1,30°)$
B. $(\sqrt{2}, 60°)$
C. $\left(\frac{1}{2}, 210°\right)$
D. (π, π)
E. $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$





2.63: Converting from Rectangular Form to Polar



$$r = |z| = \sqrt{a^2 + b^2}$$

For a complex number in QI or QIV:

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

For a complex number in QII or QIII:

$$\theta = 180° + \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

The range of $\tan^{-1}\left(\frac{b}{a}\right)$ is

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \Rightarrow QI\ and\ QIV$$

If a complex number lies in QII or QIII, then $\tan^{-1}\left(\frac{b}{a}\right)$ will not give the correct argument. Instead, we must use $\tan(180 + \theta) = \tan\theta$.

Hence, for a complex number that lies in QII or QIII:

$$\theta = 180° + \tan^{-1}\left(\frac{b}{a}\right)$$

Example 2.64

Convert to polar form

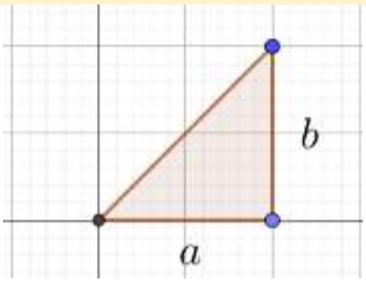
$$z = (-1, -1)$$

$$r = \sqrt{a^2 + b^2} = \sqrt{1+1} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = 45° = \frac{\pi}{4}$$

Example 2.65

Write $z = 1 + \sqrt{3}i$ in polar form.



Substitute $a = 1, b = \sqrt{3}$:

$$r = \sqrt{a^2 + b^2} = \sqrt{1+3} = \sqrt{4} = 2$$

The sides of the triangle are

$$2:1:\sqrt{3} \Rightarrow 30 - 60 - 90\ triangle$$

Note that $\theta = 60°$

$$z = 1 + \sqrt{3}i = \left(2, \frac{\pi}{3}\right)$$

Example 2.66

Write the complex number $\frac{2+3i}{3+2i}$ in the form (r, θ). (ISC 2007, Adapted)

Convert the complex number to the form $a+bi$:

$$\frac{2+3i}{3+2i}\times\frac{3-2i}{3-2i}=\frac{6-4i+9i+6}{13}=\frac{12}{13}+\frac{5}{13}i$$

Substitute $a=\frac{12}{13}, b=\frac{5}{13}$ in $r=\sqrt{a^2+b^2}$ to find the modulus:

$$\sqrt{\left(\frac{12}{13}\right)^2+\left(\frac{5}{13}\right)^2}=\sqrt{\frac{144+25}{169}}=\sqrt{\frac{169}{169}}=\sqrt{1}=1$$

Substitute $a=\frac{12}{13}, b=\frac{5}{13}$ in $\theta=\tan^{-1}\left(\frac{b}{a}\right)$ to find the argument:

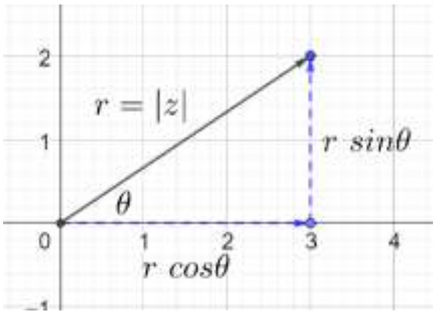
$$\theta=\tan^{-1}\left(\frac{\frac{5}{13}}{\frac{12}{13}}\right)=\tan^{-1}\left(\frac{5}{12}\right)$$

Using the modulus and the argument, $\frac{2+3i}{3+2i}$ can be written in polar form as:

$$r\angle\theta=1,\angle\tan^{-1}\left(\frac{5}{12}\right)$$

2.67: Polar Form

$$z=\underbrace{a+bi}_{\text{Rectangular Form}}=\underbrace{r(\cos\theta+i\sin\theta)}_{\text{Polar Form}}$$



$$\cos\theta=\frac{a}{r}\Rightarrow x=r\cos\theta$$
$$\sin\theta=\frac{b}{r}\Rightarrow y=r\sin\theta$$

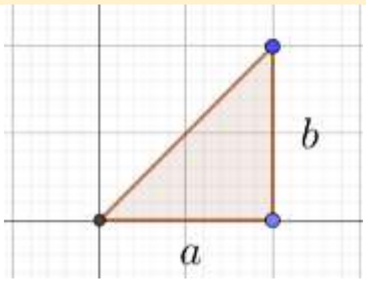
Substitute $a=r\cos\theta, b=r\sin\theta$ in:

$$a+bi=r\cos\theta+ir\sin\theta$$

Factor out r:

$$=r(\cos\theta+i\sin\theta)$$

Example 2.68
Write $z=3+3i$ in trigonometric polar form.



Note that we have a $45-45-90$ triangle.

$$r=3\times\sqrt{2}=3\sqrt{2}$$

Note that $\theta=45°$:

$$z=3+3i=\left(3\sqrt{2},\frac{\pi}{4}\right)=3\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$$

Example 2.69
Write $2\ cis\ 60°$ in rectangular form.

$$2\ cis\ 60°=2\ (\cos 60°+i\sin 60°)$$

$$= 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$
$$= 1 + \sqrt{3}i$$

2.70: cis shortform

$$z = \underbrace{a + bi}_{\substack{\text{Rectangular} \\ \text{Form}}} = \underbrace{r(\text{cis}\,\theta)}_{\text{Polar Form}}$$

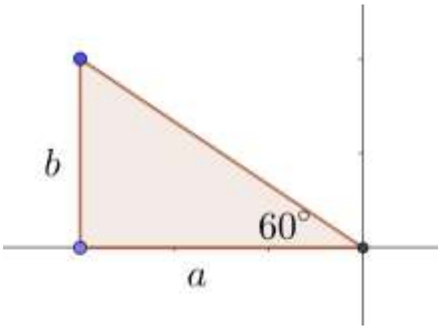
Instead of writing $\cos\theta + i\sin\theta$ every time, we substitute $\text{cis}\,\theta = \cos\theta + i\sin\theta$:

$$a + bi = r(\cos\theta + i\sin\theta) = r\,\text{cis}\,\theta$$

Example 2.71

Write $z = -3 + 3\sqrt{3}i$ in trigonometric polar form.

$$r = \sqrt{(-3)^2 + \left(3\sqrt{3}\right)^2} = \sqrt{9 + 27} = \sqrt{36} = 6$$



The triangle is $30 - 60 - 90$.

$$\theta = 180 - 60 = 120° = \frac{2\pi}{3}$$
$$z = -3 + 3\sqrt{3}i = \left(6, \frac{2\pi}{3}\right) = 6\,cis\,\frac{2\pi}{3}$$

Example 2.72

Calculate

$$cis\,(n\pi), n \in \mathbb{Z}$$

$$cis\,\pi = \cos\pi + i\sin\pi = -1 + i(0) = -1$$
$$cis\,3\pi = cis\,\pi = -1$$
$$cis\,5\pi = cis\,\pi = -1$$

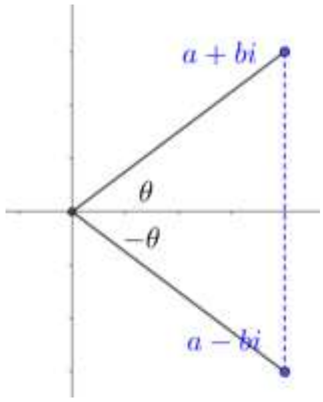
$$n\text{ is odd}: cis\,n\pi = cis\,\pi = -1$$

$$cis\,0 = \cos 0 + i\sin 0 = 1 + i(0) = 1$$
$$cis\,2\pi = cis\,0 = 1$$
$$cis\,4\pi = cis\,0 = 1$$

$$n\text{ is even}: cis\,n\pi = cis\,0 = 1$$

2.73: Conjugate in Polar Form

$$\textit{Polar Form}: z = r \text{ cis } \theta \Leftrightarrow \bar{z} = r \text{ cis } (-\theta)$$
$$\textit{Rectangular Form}: z = a + bi \Leftrightarrow \bar{z} = a - bi$$



The conjugate of z is obtained by reflecting it across the x −axis. This negates the angle θ.

Example 2.74

Find the conjugate of the following numbers in trigonometric polar form. Write the angle in positive radians.

A. $z = 3(cis\ 30°)$
B. $z = p(cis\ 115°)$

$$\bar{z} = 3[cis(-30°)] = 3\left[cis\left(\frac{11\pi}{6}\right)\right]$$
$$\bar{z} = p[cis(-115°)] = p\left[cis\left(-\frac{23\pi}{36}\right)\right] = p\left[cis\left(2\pi - \frac{23\pi}{36}\right)\right] = p\left[cis\left(\frac{49\pi}{36}\right)\right]$$

B. Multiplication in Polar Form

$cis\ \theta$ has some very powerful properties.

2.75: Multiplication property of cis θ

The trigonometric form of a complex number lets you convert multiplication into addition.

$$\text{cis}(\alpha)\ \text{cis}(\beta) = \text{cis}\ (\alpha + \beta)$$

$$LHS = cis(\alpha)cis(\beta)$$

Using the definition of cis:

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)$$

Multiply out:

$$\cos\alpha\cos\beta - \sin\alpha\sin\beta + i(\sin\alpha\cos\beta + \cos\alpha\sin\beta)$$

Use trig identities:

$$= \cos(\alpha + \beta) + i\sin(\alpha + \beta)$$

Note that this is exactly the definition of:

$$cis(\alpha + \beta) = RHS$$

Hence:

$$LHS = RHS$$

2.76: Multiplication in Polar Form

To multiply two complex numbers in polar form, multiply the magnitudes and add the angles

$$(r_1 \text{ cis } \alpha)(r_2 \text{ cis } \beta) = r_1 r_2 \text{ cis}(\alpha + \beta)$$

$$(r_1 \text{ cis } \alpha)(r_2 \text{ cis } \beta) = (r_1 r_2)\text{cis}(\alpha)\ \text{cis}(\beta) = r_1 r_2 \text{ cis}(\alpha + \beta)$$

Example 2.77

Find $z_1 z_2$ given that $z_1 = 3 \text{ cis}\left(\frac{\pi}{4}\right), z_2 = 2 \text{ cis}\left(-\frac{\pi}{6}\right)$

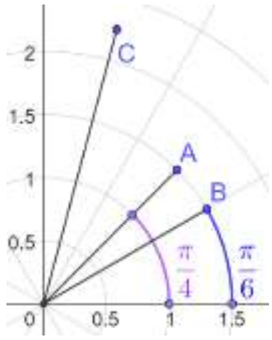
$$z_1z_2=\left[3\text{ cis}\left(\frac{\pi}{4}\right)\right]\left[2\text{ cis}\left(-\frac{\pi}{6}\right)\right]=6\text{ cis}\left(\frac{3\pi}{12}-\frac{2\pi}{12}\right)=6\text{ cis}\left(\frac{\pi}{12}\right)$$

Example 2.78

z_1 and z_2 are complex numbers that both satisfy $|z|=1.5$. z_1 makes an angle of $\frac{\pi}{4}$ with the positive direction of the x −axis, while z_2 makes an angle of $\frac{\pi}{6}$ with the positive direction of the axis. Determine z_1z_2.

$$A=z_1$$
$$B=z_2$$

$$z_1z_2=\left[1.5\text{ cis}\left(\frac{\pi}{4}\right)\right]\left[1.5\text{ cis}\left(\frac{\pi}{6}\right)\right]=2.25\text{ cis}\left(\frac{3\pi}{12}+\frac{2\pi}{12}\right)=2.25\text{ cis}\left(\frac{5\pi}{12}\right)$$



Example 2.79

Convert from rectangular to polar form and then multiply

2.80: Multiplication of more than two numbers

To multiply n numbers in trigonometry form, multiply all the magnitudes, and add all the angles.

$$(r_1\text{ cis }\theta_1)(r_2\text{ cis }\theta_2)\dots(r_n\text{ cis }\theta_n)=r_1r_2\dots r_n\big(\text{cis }(\theta_1+\theta_2+\cdots\theta_n)\big)$$

Method I: Simplification

Begin with the LHS:

$$(r_1\text{ cis }\theta_1)(r_2\text{ cis }\theta_2)(r_3\text{ cis }\theta_3)\dots(r_n\text{ cis }\theta_n)$$

Carry out the first multiplication:

$$[r_1r_2\text{ cis }(\theta_1+\theta_2)](r_3\text{ cis }\theta_3)\dots(r_n\text{ cis }\theta_n)$$

Carry out the second multiplication:

$$[r_1r_2r_3\text{ cis }(\theta_1+\theta_2+\theta_3)]\dots(r_n\text{ cis }\theta_n)$$

And so on, until we get:

$$r_1r_2\dots r_n\big(\text{cis }(\theta_1+\theta_2+\cdots\theta_n)\big)$$

Method II: Induction

Base Case

Note that when $n=2$, we have proved it above that:

$$(r_1\text{ cis }\theta_1)(r_2\text{ cis }\theta_2)=r_1r_2\big(\text{cis }(\theta_1+\theta_2)\big)$$

Inductive Case

Assume it is for some integer $n\geq 2$. Then, we prove that it is true for $n+1$
That is, we must prove that:

$$(r_1\text{ cis }\theta_1)(r_2\text{ cis }\theta_2)\dots(r_n\text{ cis }\theta_n)(r_{n+1}\text{ cis }\theta_{n+1})=r_1r_2\dots r_nr_{n+1}\big(\text{cis }(\theta_1+\theta_2+\cdots+\theta_n+\theta_{n+1})\big)$$

$$LHS=(r_1\text{ cis }\theta_1)(r_2\text{ cis }\theta_2)\dots(r_n\text{ cis }\theta_n)(r_{n+1}\text{ cis }\theta_{n+1})$$

Which using the assumption gives us:

$$r_1 r_2 \ldots r_n\big(\text{cis}\,(\theta_1 + \theta_2 + \cdots \theta_n)\big)(r_{n+1}\,\text{cis}\,\theta_{n+1})$$
$$= r_1 r_2 \ldots r_n r_{n+1}\big(\text{cis}\,(\theta_1 + \theta_2 + \cdots + \theta_n + \theta_{n+1})\big) = RHS$$

Challenge 2.81

$$\prod_{x=1}^{n} z_x$$

Expand and substitute $z_x = x\ cis\ (x\pi)$:

$$\prod_{x=1}^{n} z_x = z_1 \cdot z_2 \cdot z_3 \cdot \ldots \cdot z_n = 1\ cis\ (\pi) \cdot 2\ cis\ (2\pi) \cdot \ldots \cdot n\ cis\ (n\pi)$$

Carry out the multiplication:

$$= (1 \times 2 \times \ldots \times n)\ cis\ \big((1 + 2 + \cdots + n)\pi\big)$$

Substitute $(1 \times 2 \times \ldots \times n) = n!, 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$:

$$= (n!)\ cis\ \big((1 + 2 + \cdots + n)\pi\big)$$

Check the first few values of $1 + 2 + 3 + \cdots + n$:

$$n = 1: 1$$
$$n = 2: 1 + 2 = 3$$
$$n = 3: 1 + 2 + 3 = 6$$
$$n = 4: 1 + 2 + 3 + 4 = 10$$
$$n = 5: 1 + 2 + 3 + 4 + 5 = 15$$
$$n = 6: 1 + 2 + 3 + 4 + 5 + 6 = 21$$

The pattern is:

$$\underbrace{O + O}_{\substack{Two\ odd \\ Numbers}} + \underbrace{E + E}_{\substack{Two\ Even \\ Numbers}} + \underbrace{O + O}_{\substack{Two\ Odd \\ Numbers}} + \cdots$$

$$m\ is\ odd: cis\ (m\pi) = cis\ \pi = -1$$
$$m\ is\ even: cis\ (m\pi) = cis\ 0 = 1$$

Challenge 2.82

$$\sum_{n=1}^{n=2023}\left(\prod_{x=1}^{n} z_x\right)$$

Given that $z_x = x\ cis\ (x\pi)$, write the expression above in the form:

$$1!\,y_1 + 2!\,y_2 + 3!\,y_3 + \cdots 2023!\,y_{2023}$$

And calculate:

$$\left(\sum_{a=1}^{a=2023} y_a\right)\left(\prod_{a=1}^{2023} y_a\right)$$

$$\sum_{n=1}^{n=2023}\left(\prod_{x=1}^{n} z_x\right)=\prod_{x=1}^{1} z_x+\prod_{x=1}^{2} z_x+\cdots+\prod_{x=1}^{2023} z_x$$

$$=-1!-2!+3!+4!-5!-6!+\cdots+2023!$$

Consider $1+2+\cdots+n$

$$n=1 \text{ is first odd number} \Rightarrow Sum\ is\ odd$$
$$n=2 \text{ is first even number} \Rightarrow Sum\ is\ odd$$

In general:

$$n\ is\ a^{th}\ odd\ number\ where\ a\ is\ odd \Rightarrow Sum\ is\ odd$$
$$n\ is\ a^{th}\ odd\ number\ where\ a\ is\ even \Rightarrow Sum\ is\ even$$
$$n\ is\ a^{th}\ even\ number\ where\ a\ is\ odd \Rightarrow Sum\ is\ odd$$
$$n\ is\ a^{th}\ even\ number\ where\ a\ is\ even \Rightarrow Sum\ is\ even$$

Determine 2023 is which odd number:

$$2023=2(1012)-1=2n-1, n=1012$$

$$=-1!-2!+3!+4!-5!-6!+\cdots+2023!$$

$$\sum_{a=1}^{a=2023} y_a=y_1+y_2+\cdots+y_{2023}$$
$$y_1+y_2+y_3+y_4=(-1)+(-1)+(+1)+(+1)=0$$
$$y_5+y_6+y_7+y_8=(-1)+(-1)+(+1)+(+1)=0$$

$$y_1+y_2+\cdots+y_{2020}=0$$
$$y_{2021}+y_{2022}+y_{2023}=-1-1+1=-1$$

$$\prod_{a=1}^{2023} y_a=(y_1)(y_2)\ldots(y_{2023})$$

$$(y_1)(y_2)(y_3)(y_4)=(-1)(-1)(+1)(+1)=1$$
$$(y_5)(y_6)(y_7)(y_8)=(-1)(-1)(+1)(+1)=1$$

$$(y_{2021})(y_{2022})(y_{2023})=(-1)(-1)(1)=1$$

$$\left(\sum_{a=1}^{a=2023} y_a\right)\left(\prod_{a=1}^{2023} y_a\right)=(-1)(1)=-1$$

2.83: Multiplication as Rotation and Scaling

Let z be a complex number such that

$$z=r\operatorname{cis}\theta$$

Multiplying a complex number by z is equivalent to:

- Scaling the magnitude by a factor of r.
- Rotating by an angle θ in the counter-clockwise direction

Let w and z be two complex numbers. Then:

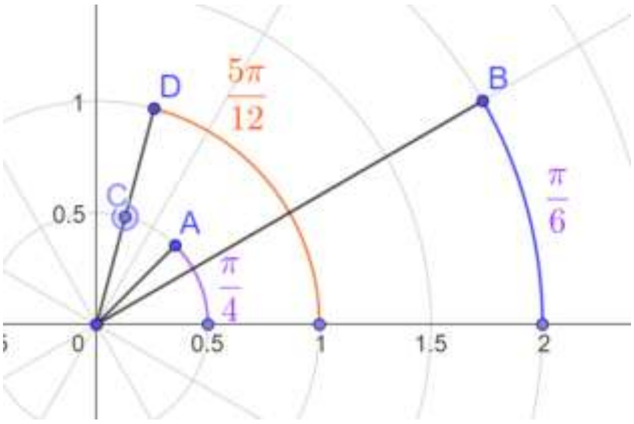
$$wz = (r_1 \text{ cis } \alpha)(r_2 \text{ cis } \beta) = r_1 r_2 \text{ cis}(\alpha + \beta)$$

$$Arg(w) = \alpha$$
$$Arg(z) = \beta$$
$$\arg(wz) = \alpha + \beta$$

Example 2.84

A. Multiplication from rotation and scaling
B. Identify rotation and scaling factor

$$A = 0.5\, cis\left(\frac{\pi}{4}\right)$$
$$B = 2\, cis\left(\frac{\pi}{6}\right)$$



$$AB = \left[0.5\, cis\left(\frac{\pi}{4}\right)\right]\left[2\, cis\left(\frac{\pi}{6}\right)\right]$$
$$= 0.5\ \underbrace{cis\left(\frac{\pi}{4}+\frac{\pi}{6}\right)}_{\textit{Argument}}[2]$$

$$= \underbrace{(0.5 \times 2)}_{\textit{Magnitude}}\ \underbrace{cis\left(\frac{\pi}{4}+\frac{\pi}{6}\right)}_{\textit{Argument}}$$

$$= \underbrace{1}_{\textit{Magnitude}}\ \underbrace{cis\left(\frac{5\pi}{12}\right)}_{\textit{Argument}}$$

The point A is rotated $\frac{\pi}{4}$ about the origin to give point C.
Point C is scaled by a factor of 2 to give point D.

Example 2.85

The complex number $1 - i$ is multiplied by $1 + i$. Write the multiplication as a combination of a scaling and a rotation factor, and determine the final answer in this way only.

$$w = 1 - i = \sqrt{2}\, cis\left(-\frac{\pi}{4}\right)$$
$$z = 1 + i = \sqrt{2}\, cis\left(\frac{\pi}{4}\right)$$

$$Scaling\ Factor = \sqrt{2}$$
$$Rotation\ Factor = \frac{\pi}{4}$$

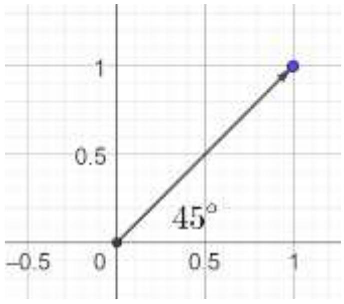
$$wz = \left(\sqrt{2}\right)^2 cis\left(-\frac{\pi}{4}+\frac{\pi}{4}\right) = \left(\sqrt{2}\right)^2 cis\,(0) = 2\, cis\, 0 = 2$$

Example 2.86

If $i^2 = -1$, then $(1+i)^{20} - (1-i)^{20}$ equals (AHSME 1974/17)

$$z = 1 + i = \sqrt{2}\, cis\ 45°$$
$$z^2 = \left(\sqrt{2}\, cis\ 45°\right)^2 = 2\ cis\ 90° = 2i$$

$$z = 1 - i = \sqrt{2}\, cis(-45°)$$
$$z^2 = \left(\sqrt{2}\, cis(-45°)\right)^2 = 2\ cis(-90°) = -2i$$



$$(1+i)^{20} - (1-i)^{20} = (2i)^{10} - (-2i)^{10} = (2i)^{10} - (2i)^{10} = 0$$

2.87: Multiplying Conjugates

The product of a complex number and its conjugate is the square of the distance of the number from the origin.

$$z\bar{z} = |z|^2$$

Suppose

$$z = r \text{ cis } \theta \Rightarrow \bar{z} = r\ cis(-\theta)$$

Then, we can show that:

$$LHS = z\bar{z} = [r \text{ cis } \theta][r\ cis(-\theta)] = r^2\ cis\ (\theta - \theta) = r^2\ cis\ (0)$$

Using the definition of cis:

$$= r^2[\cos 0 + i \sin 0] = r^2[1 + 0] = r^2 = |\bar{z}|^2 = RHS$$

Example 2.88

$$(a + bi)(c + di)$$

C. Division in Polar Form

2.89: Reciprocal of a Number

$$\frac{1}{r\ cis\ \theta} = \frac{cis(-\theta)}{r}$$

$$LHS = \frac{1}{r\ cis\ \theta} = \frac{1}{r} \cdot \frac{1}{\cos\theta + i \sin\theta}$$

Carry out the division by multiplying by the conjugate:

$$= \frac{1}{r} \cdot \frac{1}{\cos\theta + i\sin\theta} \cdot \frac{\cos\theta - i\sin\theta}{\cos\theta - i\sin\theta}$$
$$= \frac{1}{r} \cdot \frac{\cos\theta - i\sin\theta}{\cos^2\theta + \sin^2\theta}$$

Substitute $\cos^2\theta + \sin^2\theta = 1$:

$$= \frac{\cos\theta - i\sin\theta}{r}$$

Substitute $\cos(-\theta) = \cos\theta, \sin(-\theta) = -\sin\theta$

$$= \frac{\cos(-\theta) + i\sin(-\theta)}{r}$$
$$= \frac{cis(-\theta)}{r} = RHS$$

Example 2.90

State the identity below in rectangular form and prove it using rectangular methods:

$$\frac{1}{r\ cis\ \theta} = \frac{cis(-\theta)}{r}$$

Substitute $z = a + bi = r\ cis\ \theta$ in $LHS = \frac{1}{r\ cis\ \theta}$:

$$\frac{1}{a + bi}$$

Multiply by the conjugate to carry out the division:

$$\frac{1}{a + bi} \cdot \frac{a - bi}{a - bi} = \frac{a - bi}{a^2 + b^2}$$

Substitute

$$Magnitude = a^2 + b^2 = r^2$$
$$Conjugate = a - bi = r\ cis(-\theta)$$

To get:

$$\frac{r\ cis(-\theta)}{r^2} = \frac{cis(-\theta)}{r} = RHS$$

2.91: Division in Polar Form

$$\frac{z_1}{z_2} = \frac{r_1 \operatorname{cis} \alpha}{r_2 \operatorname{cis} \beta} = \frac{r_1}{r_2} \operatorname{cis}(\alpha - \beta)$$

$$\frac{r_1 \operatorname{cis} \alpha}{r_2 \operatorname{cis} \beta} = \frac{r_1 \operatorname{cis} \alpha}{r_2} \times \operatorname{cis}(-\beta) = \frac{r_1}{r_2} \operatorname{cis}(\alpha - \beta)$$

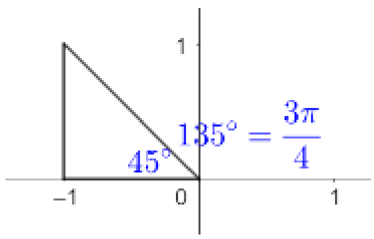
Example 2.92

$$z = \frac{i - 1}{\cos\frac{\pi}{3} + i \sin\frac{\pi}{3}} = r\ cis\ \theta, z \in \mathbb{C}$$

If r is the magnitude and $\theta > 0$ is the angle with the positive direction of x axis, written in radians then find $r\ cis\ \theta$. **(JEE Main, Jan 31, 2023-II, Adapted)**

Convert the numerator to polar form using $r_1 = \sqrt{(-1)^2 + 1^2} = \sqrt{2}, \theta_1 = 135°$

$$z_1 = i - 1 = r_1 cis\ \theta_1 = \sqrt{2}\ cis\left(\frac{3\pi}{4}\right)$$



$$z = \frac{\sqrt{2}\ cis(135°)}{cis(60°)} = \sqrt{2}\ cis(135° - 60°) = \sqrt{2}\ cis(75°) = \sqrt{2}\ cis\left(\frac{5\pi}{12}\right)$$

Example 2.93

Suppose two real numbers a and b are both unity. Let z_1 be a complex number equal to $a + bi$. Let z_2 be the conjugate of z_1. Let z_3 be the quotient of z_1 and z_2. Let z_4 be the n^{th} power of z_3, where n is a natural number. Find the smallest possible value of n such that z_4 is 1.

Algebraic Method

$$z_3 = \frac{z_1}{z_2} = \frac{a+bi}{a-bi} = \frac{1+i}{1-i} = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1-1+2i}{1+1} = \frac{2i}{2} = i$$

We now need to find:

$$z_3^n = 1 \Rightarrow i^n = 1 \Rightarrow Smallest\ n = 4$$

Trigonometry Method

$$z_1 = 1 + i = \sqrt{2}\left(cis\frac{\pi}{4}\right)$$

$$z_2 = \sqrt{2}\left(cis\left(-\frac{\pi}{4}\right)\right)$$

$$z_3 = \frac{z_1}{z_2} = \frac{\sqrt{2}\left(cis\ \frac{\pi}{4}\right)}{\sqrt{2}\left(cis\left(-\frac{\pi}{4}\right)\right)} = cis\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = cis\frac{\pi}{2}$$

$$z_4 = z_3^n = \left(cis\frac{\pi}{2}\right)^n = \underbrace{\left(cis\frac{\pi}{2}\right)\left(cis\frac{\pi}{2}\right)\dots\left(cis\frac{\pi}{2}\right)}_{n\ times} = cis\frac{n\pi}{2}$$

$$cis\ \frac{n\pi}{2} = 1$$

$$cis\frac{n\pi}{2} = cis\ 2\pi$$

Equating the arguments:

$$\frac{n\pi}{2} = 2\pi$$

$$n = 4$$

Note:

$$cis\frac{n\pi}{2} = cis\ 0$$

$$\frac{n\pi}{2} = 0$$

$$n = 0$$

But 0 is not a natural number. So, this solution is not useful for us.

2.94: Division as Rotation and Scaling

Example 2.95

A. Division from rotation and scaling
B. Identify rotation and scaling factor

2.96: Quotient of Conjugates

The product of a complex number and its conjugate is the square of the distance of the number from the origin.

$$z\bar{z} = \frac{r\ \text{cis}\ \theta}{r\ cis(-\theta)} = cis\ 2\theta$$

D. Equality in Polar Form

2.97: Equality in Polar Form

Two complex numbers z and w in polar form are equal if and only if their magnitudes and their arguments are both equal.

Example 2.98

If the non-zero complex numbers $z = a\ cis\ \sqrt{a}$ and $w = \sqrt{a}\ cis\ a$, where $a \in \mathbb{R}$ are equal. Then find zw.

Equate the magnitudes

$$a = \sqrt{a}$$
$$a^2 = a$$
$$a \in \{0,1\}$$

Reject 0.

$$a = 1$$

Equating the arguments also gives us:

$$a = 1$$

$$zw = (1\ cis\ 1)(1\ cis\ 1) = 1\ cis\ 2$$

E. Addition and Subtraction in Polar Form

2.99: Addition and Subtraction

Example 2.100

Convert from polar to rectangular form, add/subtract, and then re-convert

2.101: Adding Conjugates

$$z + \bar{z} = 2\ Re(z)$$

$$z + \bar{z} = r \operatorname{cis} \theta + r\ cis(-\theta) = r[\operatorname{cis} \theta + cis(-\theta)]$$
$$= r[(\cos\theta + i \sin\theta) + (\cos(-\theta) + i\sin(-\theta))]$$

Substitute $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$

$$= r[(\cos\theta + i\sin\theta) + (\cos\theta - i\sin\theta)]$$
$$= 2r\cos\theta$$
$$= 2\ Re(z)$$

$$(a + bi) + (a - bi) = 2a = 2\ Re(z)$$

F. De Moivre's Theorem

2.102: De Moivre's Theorem

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis} n\theta$$

Where

$$n\ is\ an\ integer$$

Example 2.103: Number Theory

If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{i-1}\right)^{\frac{n}{3}} = 1, (m, n \in N)$, then the greatest common divisor of the least values of m and n is (JEE Main

3 Sep. 2020, Shift-I)

$$\frac{1+i}{1-i}=\frac{\sqrt{2}\,cis\,45°}{\sqrt{2}\,cis(-45°)}=cis\,90°$$
$$\frac{1+i}{i-1}=-\left(\frac{1+i}{1-i}\right)=-i=cis(-90°)$$

$$(cis\,90°)^{\frac{m}{2}}=1\,cis\,(0+360k°)$$
$$cis\,45m=1\,cis\,(0+360k°)$$
$$45m=360k$$
$$m=8k$$

$$(cis\,90°)^{\frac{m}{2}}=1\,cis\,(0+360k°),\qquad k\in\mathbb{Z}$$
$$cis\,45m=1\,cis\,(0+360k°)$$
$$45m=360k$$
$$m=8k,\qquad k\in\mathbb{Z}$$

$$\left(cis(-90°)\right)^{\frac{n}{3}}=1\,cis\,(0+360k°)$$
$$cis\,(-30n)=1\,cis\,(0+360k°)$$
$$-30n=360k$$

Since k is an integer, we can absorb the minus sign into k:

$$n=12k,\qquad k\in\mathbb{Z}$$

Smallest possible value of k is 1.

$$GCD\{m,n\}=GCD\{8,12\}=4$$

Example 2.104

Powers

Example 2.105

Conversion from rectangular to polar form

A. $(1+i)^8$

$$(1+i)^8=\left[\sqrt{2}\,cis\frac{\pi}{4}\right]^8=2^{\frac{1}{2}\times 8}cis\left(\frac{\pi}{4}\times 8\right)=2^4\,cis\,2\pi=16$$

Example 2.106

If $i^2=-1$, then $(1+i)^{20}-(1-i)^{20}$ equals (AHSME 1974/17)

$$=\left[\sqrt{2}\,cis\left(\frac{\pi}{4}\right)\right]^{20}-\left[\sqrt{2}\,cis\left(-\frac{\pi}{4}\right)\right]^{20}$$
$$=2^{10}cis(\pi)-2^{10}cis(-\pi)$$
$$=0$$

Example 2.107

$\left(-i+\sqrt{3}\right)^{300}+\left(-i-\sqrt{3}\right)^{300}=$ (AP EAPCET 21 Sep 2020, Shift I)

Convert to polar form:

$$= \left[2\,\mathrm{cis}\left(-\frac{\pi}{6}\right)\right]^{300} + \left[2\,\mathrm{cis}\left(-\frac{5\pi}{6}\right)\right]^{300}$$

Apply De Moivre's Theorem:

$$= 2^{300}\,\mathrm{cis}\left(-\frac{\pi}{6}\times 300\right) + 2^{300}\,\mathrm{cis}\left(-\frac{5\pi}{6}\times 300\right)$$
$$= 2^{300}\,[\mathrm{cis}(-50\pi) + \mathrm{cis}(-250\pi)]$$
$$= 2^{300}\,[\mathrm{cis}(0) + \mathrm{cis}(0)]$$
$$= 2^{300}\,[1+1]$$
$$= 2^{300}\,[2]$$
$$= 2^{301}$$

Example 2.108

Find the value of

$$\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$$

(JEE Main, Sep 02, 2020-I, JEE Main, Jan 24, 2023-II)

Change of Variable

Let $z=\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}$:

$$\left(\frac{1+z}{1+\bar{z}}\right)^3 = \left(\frac{1+z}{1+\frac{1}{z}}\right)^3 = \left(\frac{1+z}{\frac{z+1}{z}}\right)^3 = z^3$$

Where we made use of:

$$|z| = \sqrt{a^2+b^2} = \sqrt{\sin^2\frac{2\pi}{9}+\cos^2\frac{2\pi}{9}} = \sqrt{1} = 1 \Rightarrow z\bar{z} = |z|^2 = 1 \Rightarrow \bar{z} = \frac{1}{z}$$

Change back to the Original Variable

Factor out i from the expression for z:

$$z = \sin\frac{2\pi}{9} + i\cos\frac{2\pi}{9} = i\left(\frac{\sin\frac{2\pi}{9}}{i} + \cos\frac{2\pi}{9}\right)$$

Substitute $\frac{\sin\frac{2\pi}{9}}{i} = \frac{\sin\frac{2\pi}{9}}{i}\times\frac{i}{i} = -i\sin\frac{2\pi}{9}$, and rearrange:

$$= i\left(-i\sin\frac{2\pi}{9}+\cos\frac{2\pi}{9}\right) = i\left(\cos\frac{2\pi}{9} - i\sin\frac{2\pi}{9}\right)$$

Substitute $\cos\frac{2\pi}{9} = \cos -\frac{2\pi}{9}$ and use $-\sin\theta = \sin(-\theta)$:

$$= i\left(\cos\left(-\frac{2\pi}{9}\right)+i\sin\left(-\frac{2\pi}{9}\right)\right) = i\,cis\left(-\frac{2\pi}{9}\right) = cis\left(\frac{\pi}{2}\right)\,cis\left(-\frac{2\pi}{9}\right) = cis\left(\frac{9\pi}{18}-\frac{4\pi}{18}\right) = cis\left(\frac{5\pi}{18}\right)$$

Then:

$$z^3 = \left[cis\left(\frac{5\pi}{18}\right)\right]^3 = cis\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}+\frac{1}{2}i$$

2.109: Purely Real Powers

For a complex number $z = r\ cis\ \theta$, the values of n such that z^n is purely real will be:

$$\theta = , k \in \mathbb{Z}$$

$$z^n \in \mathbb{R}$$
$$(r\ cis\ \theta)^n \in \mathbb{R}$$
$$r^n\ cis\ n\theta \in \mathbb{R}$$

$$n\theta = k\pi, k \in \mathbb{Z}$$

Example 2.110

For what values of $z = 2\ cis \frac{\pi}{3}$ is z^n:

A. Purely real
B. Purely imaginary

$$z^n = \left(2\ cis \frac{\pi}{3}\right)^n = 2^n\ cis\left(\frac{n\pi}{3}\right)$$

For $k \in \mathbb{Z}$:

$$\frac{n\pi}{3} = k\pi$$
$$\frac{n}{3} = k$$
$$n = 3k$$
$$n = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

2.111: Purely Imaginary Powers

$$n\theta = 90 + 180k$$

$$n\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

Example 2.112

For what values of $z = 2\ cis \frac{\pi}{3}$ is z^n:

A. Purely real
B. Purely imaginary

For $k \in \mathbb{Z}$:

$$\frac{n\pi}{3} = \frac{\pi}{2} + k\pi$$
$$\frac{n}{3} = \frac{1}{2} + k$$
$$n = \frac{3}{2} + 3k$$

$$cis \frac{\pi}{3} = \cos\frac{\pi}{3} + i \sin\frac{\pi}{3}$$

For $k \in \mathbb{Z}$:

$$\frac{k\pi}{2} = \frac{\pi}{3}n$$
$$\frac{k}{2} = \frac{1}{3}n$$
$$n = \frac{3k}{2}$$

G. Sequences and Series

Example 2.113

$$cis\ 72 + cis\ 144 + cis\ 216 + cis\ 288 + cis\ 360$$

$$cis\ 72 + cis\ (2 \times 72) + cis\ (3 \times 72) + cis\ (4 \times 72) + cis\ (5 \times 72)$$
$$cis\ 72 + (cis\ 72)^2 + (cis\ 72)^3 + (cis\ 72)^4 + (cis\ 72)^5$$

Substitute $a = cis\ 72, r = cis\ 72$ in the formula for the sum of a finite geometric series:

$$\frac{a(r^n - 1)}{r - 1} = \frac{(cis\ 72)[(cis\ 72)^5 - 1]}{cis\ 72 - 1} = \frac{(cis\ 72)[cis(0) - 1]}{cis\ 72 - 1} = \frac{cis\ 72[1 - 1]}{cis\ 72 - 1} = \frac{0}{cis\ 72 - 1} = 0$$

H. Circle with Endpoints of Diameter Z_1 and z_2

2.114: Equation of Circle

I. Product and Summation Notation

2.115: Converting from Product to Summation Notation

Product and cis notation can be interchanged by converting the product into a summation:

$$\prod_{k=1}^{n} cis\ k = cis\left(\sum_{k=1}^{n} k\right)$$

$$\prod_{k=1}^{n} cis\ k = (cis\ 1)(cis\ 2)(cis\ 3)\dots(cis\ n) = cis\left(\sum_{k=1}^{n} k\right)$$

2.4 Polar Form: Argument

A. Basics

2.116: Argument

- Consider the point P in the complex plane. The angle made by OP with the positive direction of the x-axis is called the argument of P.
- Since the argument is an angle, the variable used is generally θ.

2.117: arg(z)

Notation: The argument of a number z is written arg(z)

2.118: Values of the Argument

If $\theta = \theta_1$ is an argument of the complex number z, then are an infinite number of arguments given by

$$\theta_1 + 2k\pi, \qquad k \in \mathbb{Z}$$

$$\ldots, \theta_1 - 4\pi, \theta_1 - 2\pi, \theta_1, \theta_1 + 2\pi, \theta_1 + 4\pi, \ldots,$$

And this can be written in more compact notation as:

$$\theta_1 + 2k\pi, k \in \mathbb{Z}$$

Example 2.119

Determine the solution set (and its cardinality) for:

A. $z = 1 + \sqrt{3}i, \ -10\pi \le \arg(z) \le 10\pi$
B. $z = 1, \ -100\pi \le \arg(z) \le 100\pi$

Part A

$$\theta = 60° = \frac{\pi}{3}$$

$$\theta \in \left\{\frac{\pi}{3} + 2k\pi\right\}, \qquad -5 \le k \le 4, \qquad k \in \mathbb{Z}$$

$$k \in \{-5, -4, \ldots, 4\} \Rightarrow 4 - (-5) + 1 = 4 + 5 + 1 = 10$$

Part B

$$\theta = \{2k\pi\}, -50 \le k \le 50, k \in \mathbb{Z}$$

$$k \in \{-50, -49, \ldots, 49, 50\} \Rightarrow 50 - (-50) + 1 = 50 + 50 + 1 = 101$$

2.120: Principal Argument

While there are an infinite number of arguments for a complex number z, there is only a single Principal Argument, which lies in the interval

$$-\pi < Arg(z) \le \pi$$

The principal argument is obtained by adding or subtracting multiples of 2π from a given argument until it lies in the interval

$$-\pi < Arg(z) \le \pi$$

Example 2.121

Determine the principal argument for:

A. $(r, \theta) = (3, 710°)$

$$710° = 710° - 720° = -10°$$

B. Properties of Argument

2.122: Argument of Conjugate

The argument of the conjugate of a complex number is the negative of the argument of the complex number.

$$\operatorname{Arg}(\bar{z}) = -\operatorname{Arg}(z)$$

This is because the conjugate is the reflection of a complex number across the x-axis.

Example 2.123

$$\arg(1+\sqrt{3}i) = 60^\circ = \frac{\pi}{3}$$
$$\arg(1-\sqrt{3}i) = -60^\circ = -\frac{\pi}{3}$$

Example 2.124

The argument of a complex number is $\frac{9\pi}{4}$. Find the argument of the conjugate of the complex number as an angle between 0 and 2π.

$$\arg(z) = \frac{9\pi}{4} \Rightarrow \arg(\bar{z}) = -\arg(z) = -\frac{9\pi}{4} + 4\pi = \frac{7\pi}{4}$$

2.125: Argument of a Product

The argument of a product of two complex numbers is equal to the sum of the arguments of the two complex numbers:

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$ [3]

Notes:

1. When we say equal, we mean coterminal. We see an example of this below.

$$LHS = \arg(z_1 z_2)$$

Let $z_1 = r_1 \, cis \, \theta_1, z_2 = r_2 \, cis \, \theta_2$:

$$= \arg[(r_1 \, cis \, \theta_1)(r_2 \, cis \, \theta_2)](\times) = \arg\big(r_1 \, r_2 cis \, (\theta_1 + \theta_2)\big)$$

The argument is the angle given by:

$$\theta_1 + \theta_2 = \arg(z_1) + \arg(z_2) = RHS$$

Example 2.126

$$\arg(-1 \times -1) = \arg(-1) \times \arg(-1)$$

$$\arg(-1 \times -1) = \arg(1) = 0$$
$$\arg(-1 \times -1) = \arg(-1) + \arg(-1) = \pi + \pi = 2\pi$$

And note that

$$2\pi \neq 0, but\ 2\pi\ is\ coterminal\ with\ 0$$

2.127: Argument of a Quotient

For complex numbers z_1 and z_2 with $z_2 \neq 0$:

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

Notes:

1. When we say equal, we mean coterminal. We see an example of this below.

$$LHS = \arg\left(\frac{z_1}{z_2}\right)$$

Let $z_1 = r_1 \, cis \, \theta_1, z_2 = r_2 \, cis \, \theta_2$:

$$= \arg\left[\frac{r_1 \, cis \, \theta_1}{r_2 \, cis \, \theta_2}\right] = \arg\left(\frac{r_1}{r_2} \, cis \, (\theta_1 - \theta_2)\right)$$

[3] This would be written more precisely $\arg(z_1 z_2) \equiv \arg(z_1) + \arg(z_2)$.

The argument is the angle given by:

$$\theta_1 - \theta_2 = \arg(z_1) - \arg(z_2) = RHS$$

2.128: Argument of a Power

$$\arg(z^n) = n \arg(z)$$

Let $z = r\ cis\ \theta \Rightarrow \arg(z) = \theta$

$$LHS = \arg(z^n) = \arg[(r\ cis\ \theta)^n] = \arg(r^n cis\ n\theta) = n\theta = n \arg(z) = RHS$$

C. Finding Arguments

2.129: $\tan^{-1} x$

Range of $y = \tan^{-1} x$ is

$$-\frac{\pi}{2} \le y \le \frac{\pi}{2}$$

In other words, the output of the function $\tan^{-1} x$ gives you an angle between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.
Note that this is Quadrant I and Quadrant IV only.

Hence, $y = \tan^{-1} x$ will never have an output which is in the second or the fourth quadrant.

2.130: Quadrants I and IV

For a complex number $z = a + bi$ in Quadrant I or Quadrant IV:

$$\arg(z) = \theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

$$\tan\theta = \frac{b}{a} \Rightarrow \theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

- A complex number can lie in any of Quadrants I-IV.
- However, the range of $y = \tan^{-1}\left(\frac{b}{a}\right)$ is $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Hence, y lies in Quadrants I and IV only.

Example 2.131

Find the argument of the following complex numbers

A. $3 + 3i$
B. $\sqrt{27} - 3i$

Part A

The point $(3,3)$ lies in Quadrant I.

$$\tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{3}{3}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Part B

The point $\left(3, -\sqrt{27}\right)$ lies in Quadrant IV.

$$\tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(-\frac{3}{\sqrt{27}}\right) = \tan^{-1}\left(-\frac{3}{3\sqrt{3}}\right) = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

2.132: Quadrants II: Formula

If $z = a + bi$ is a complex number in the second quadrant

$$\theta = \pi + \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

If the complex number that you wish to find the argument of is in either Quadrant II or Quadrant III, you need to change the angle, by adding 180, or subtracting 180.

2.133: Quadrants III: Formula

If $z = a + bi$ is a complex number in the third quadrant:

$$\theta = -\pi + \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

Example 2.134

A. $-3 - \sqrt{3}i$

Part A

The point $(-3, -\sqrt{3})$ lies in Quadrant III.

$$\tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{-\sqrt{3}}{-3}\right) = \tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$$

$$\frac{\pi}{6} + \pi = \frac{7\pi}{6}$$

D. Transformations

Example 2.135

Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then, the principal argument of $(w_1 - w_2)$ written in exact form is equal to: **(JEE Main, April 11, 2023-I)**

To rotate 90° anticlockwise about the origin, we multiply by i:

$$w_1 = iz_1 = i(5 + 4i) = 5i - 4 = -4 + 5i$$

Rotating 90° clockwise about the origin is the same as rotating $-90°$ anticlockwise, which is the same as rotating 270° anticlockwise, which is the same as multiplying by $-i$:

$$w_2 = -iz_2 = -i(3 + 5i) = -3i + 5 = 5 - 3i$$

$$w_1 - w_2 = (-4 + 5i) - (5 - 3i) = -9 + 8i$$

The principal argument of the above expression is:

$$Arg(-9 + 8i) = \tan^{-1}\left(-\frac{8}{9}\right) + \pi$$

Where we added π to the tan inverse function since the range of tan inverse is Quadrants I and IV, and the number that we have is in Quadrant II.

2.5 Exponential Form

A. Exponential Form

Along with rectangular form and polar form, exponential form is a third way of writing complex numbers. The properties are very similar to polar form.

2.136: Definition

$$e^{i\theta} = \cos\theta + i\sin\theta = cis\ \theta$$

Deriving this result requires advanced calculus.

2.137: Exponential Form of a Complex Number

$$re^{i\theta} = r(\text{cis}\ \theta)$$

2.138: Euler's Identity

$$e^{i\pi} + 1 = 0$$

- This is the "most famous" identity in mathematics.
- It connects the five most important constants in mathematics $(e, i, \pi, 0, 1)$.

$$e^{i\pi} = cis\ \pi = \cos\pi + i\sin\pi = -1 + i0 = -1$$
$$e^{i\pi} = -1$$
$$e^{i\pi} + 1 = 0$$

B. Multiplication and Division

2.139: Multiplication

Two complex numbers in exponential form can be multiplied by multiplying their magnitudes and adding their exponents:

$$r_1e^{i\alpha} \times r_2e^{i\beta} = r_1r_2e^{i(\alpha+\beta)}$$

Example 2.140

$$2e^{i\frac{\pi}{2}} \cdot 3e^{i\frac{\pi}{3}}$$

$$2e^{i\frac{\pi}{2}} \cdot 3e^{i\frac{\pi}{3}} = (2\cdot 3)e^{i\left(\frac{\pi}{2}+\frac{\pi}{3}\right)} = 6e^{\frac{5\pi}{6}}$$

2.141: Multiplication

Two complex numbers in exponential form can be divided by dividing the magnitude of the first number by the second and subtracting the exponent of the second number from the first:

$$\frac{r_1e^{i\alpha}}{r_2e^{i\beta}} = \frac{r_1}{r_2}e^{i(\alpha-\beta)}$$

Example 2.142

If $a = 3e^{i\frac{\pi}{3}}, b = 2e^{i\frac{\pi}{2}}$, find $\frac{a}{b}$.

$$\frac{a}{b}=\frac{3e^{i\frac{\pi}{3}}}{2e^{i\frac{\pi}{2}}}=\left(\frac{3}{2}\right)e^{i\left(\frac{\pi}{3}-\frac{\pi}{2}\right)}=\left(\frac{3}{2}\right)e^{i\left(-\frac{\pi}{6}\right)}=\left(\frac{3}{2}\right)e^{i\left(\frac{11\pi}{6}\right)}$$

C. Conversion to Exponential Form

2.143: Converting from Polar Form to Exponential Form

$$r\ cis\ \theta=r\cdot e^{i\theta}$$

In both polar and exponential form:

$$Magnitude=r$$

Argument in both polar and exponential

$$=\theta$$

Example 2.144

Write 3 cis 240° in exponential form with the argument in radians.

$$3\ cis\ 240°=3e^{i\frac{4\pi}{3}}$$

2.145: Converting from Rectangular Form to Exponential

$$r=|z|=\sqrt{a^2+b^2}$$

For a complex number in QI or QIV:

$$\theta=\tan^{-1}\left(\frac{b}{a}\right)=\tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

For a complex number in QII or QIII:

$$\theta=180°+\tan^{-1}\left(\frac{b}{a}\right)=\tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

D. Further Properties

2.146: De Moivre's Theorem

$$\left(re^{i\theta}\right)^n=r^ne^{in\theta}$$

Example 2.147

If $i^2=-1$, then $(1+i)^{20}-(1-i)^{20}$ equals (AHSME 1974/17)

$$=\left[\sqrt{2}\,e^{i\frac{\pi}{4}}\right]^{20}-\left[\sqrt{2}\,e^{-i\frac{\pi}{4}}\right]^{20}$$
$$=2^{10}e^{5i\pi}-2^{10}e^{-5i\pi}$$
$$=2^{10}e^{i\pi}-2^{10}e^{i\pi}$$
$$=0$$

Example 2.148

Write in rectangular form

A. $(1+i)^8$
B. $\binom{8}{0}1^8 i^0 + \binom{8}{1}1^7 i^1 + \cdots + \binom{8}{8}1^0 i^8$

Part A
Convert to exponential form

$$z = 1 + i \Rightarrow r = \sqrt{2}, \theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$z = \sqrt{2}e^{i\frac{\pi}{4}}$$
$$z^8 = \left(\sqrt{2}e^{i\frac{\pi}{4}}\right)^8 = \left(\sqrt{2}\right)^8 e^{i\frac{\pi}{4}\times 8} = 16e^{i2\pi} = 16e^{i\times 0} = 16e^0 = 16$$

Part B
Recognize that this is the binomial expansion of
$$(1+i)^8 = 16$$
As we calculated in Part A.

Example 2.149

$\left(-i+\sqrt{3}\right)^{300} + \left(-i-\sqrt{3}\right)^{300} =$ **(AP EAPCET 21 Sep 2020, Shift I)**

$$z_1 = -i + \sqrt{3} \Rightarrow r = 2, \theta = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$
$$\mathrm{z}_2 = -i + \sqrt{3} \Rightarrow r = 2, \theta = \pi + \tan^{-1}\left(\frac{-1}{-\sqrt{3}}\right) = \pi + \frac{\pi}{6} = \frac{7\pi}{6} = -\frac{5\pi}{6}$$

Convert to exponential form:
$$= \left[2e^{-i\frac{\pi}{6}}\right]^{300} + \left[2e^{-i\frac{5\pi}{6}}\right]^{300}$$

Apply De Moivre's Theorem:
$$= 2^{300}\, e^{-i\frac{\pi}{6}\times 300} + 2^{300}\, e^{-i\frac{5\pi}{6}\times 300}$$

Simplify the arguments above:
$$-i\frac{\pi}{6} \times 300 = -i(50\pi) = -i(0) = 0$$
$$-i\frac{5\pi}{6} \times 300 = -i(250\pi) = -i(0) = 0$$

$$= 2^{300}\, e^0 + 2^{300}\, e^0 = 2^{300}\,[1+1] = 2^{300}\,[2] = 2^{301}$$

E. Roots of Unity

2.150: General Solution of n^{th} root of unity

$$z^n - 1 = 0 \Rightarrow z = \underbrace{\angle\left(\frac{2\pi k}{n}\right)}_{Polar\ Form} = e^{\frac{2\pi k i}{n}}, k \in \{1,2,3,\ldots,n\}$$

2.151: Sum of Roots of Unity is Zero

$$\sum_{k=1}^{n} e^{\frac{2\pi k i}{n}} =$$

F. Addition and Subtraction

G. Logarithms

Logarithms of negative numbers are not possible when considering real values. But, when considering complex values, we can take the log, including the log for complex numbers.

2.152: Log of a Complex Number

$$\ln z = \ln r + (i\theta + 2k\pi), k \in \mathbb{Z}$$

$$z = r \cdot e^{i\theta}$$

Take the natural log of both sides:

$$\ln z = \ln\left(r \cdot e^{i\theta}\right)$$

Use the product rule:

$$\ln z = \ln r + \ln e^{i\theta}$$

Use the power rule:

$$\ln z = \ln r + i\theta$$

2.6 Geometry: Circles

A. Circle at the Origin

We will consider geometrical shapes in the complex plane. These geometrical shapes can have a Cartesian version to the equation.
Certain properties are easier to work with in the complex plane, and certain properties in the Cartesian plane.

2.153: Definition of a Circle

Circle is the set of points with distance r from the center, where $r = radius$.

2.154: Equation of a Circle

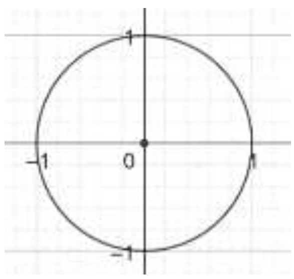
The locus of points on the complex plane that a circle with center at origin and radius r is:

$$|z| = r$$

Logic

$|z|$ is the magnitude of z.
The distance of z from the origin is r. Hence, the locus is the circle with radius r centered at the origin.

Algebra



$$|z| = r$$

Substitute $z = a + bi$:

$$|a + bi| = r$$

Use the definition of magnitude:

$$\sqrt{a^2 + b^2} = r$$

Square both sides:

$$a^2 + b^2 = r^2$$

Let $x = a, y = b$:

$$x^2 + y^2 = r^2$$

Which is the Cartesian equation of a circle with center at origin and radius r.

Example 2.155

A. Write the equation of a circle with radius 5.
B. For a complex number $z = a + bi$, $a^2 + b^2 = 5$. Write this equation using complex numbers in standard form.

Part A

$$|z| = 5$$

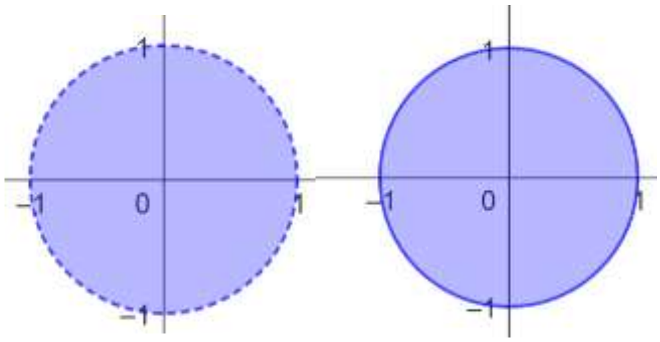
Part B

$$a^2 + b^2 = 5$$
$$\sqrt{a^2 + b^2} = \sqrt{5}$$
$$|z| = \sqrt{5}$$

2.156: Closed Set (Informal)

A closed set includes its "endpoints", or its boundary.

- The interval $[3,5]$ is closed. The interval $(3,5)$ is open.
- The circle $|z| < 1$ is not closed. The boundary is not included.
- The circle $|z| \leq 1$ is closed. The boundary is included.



$$Left\ Diagram: z < 1, Boundary\ not\ included$$
$$Right\ Diagram: z \leq 1, Boundary\ included$$

Example 2.157

Find the area of the region that satisfies:

$$|z| < \sin 30°$$

$$|z| < \frac{1}{2}$$

Area of the circle

$$= \pi r^2 = \pi \left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$$

Example 2.158

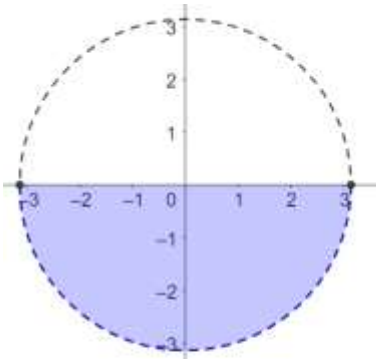
z is a complex number with argument greater than π, where argument is over the domain $[0,2\pi)$ and magnitude less than π. Find the area of the region that z can lie in.

The required area is a semicircle with

$$Radius = \pi$$

Area

$$= \frac{\pi r^2}{2} = \frac{\pi(\pi^2)}{2} = \frac{\pi^3}{2}$$



2.159: Bounded Set (Informal)

A bounded set has an upper, and a lower bound.

- The circles $|z| < 1$ and $|z| \le 1$ are both bounded.

Example 2.160

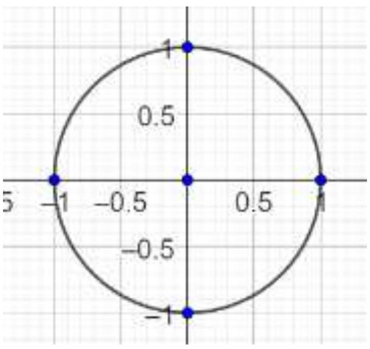
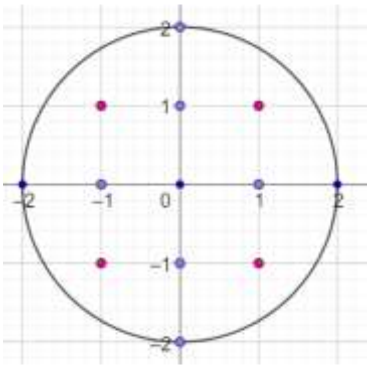
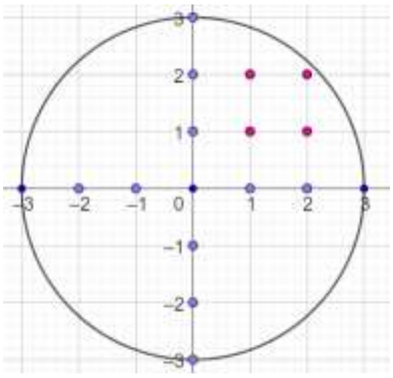
A. $x \ge 2$ on the real plane

$$x \ge 2: Closed, but\ not\ bounded$$

Example 2.161

A complex number $z = a + bi, a, b \in \mathbb{R}$ is a complex lattice point if $a, b \in \mathbb{Z}$. Find the number of complex lattice points in the region:

A. $|z| \le 1$
B. $|z| \le 2$
C. $|z| \le 3$

$$5$$
$$9 + 4 = 13$$
$$4(4) + 3(4) + 1 = 16 + 13 = 29$$

B. Circle with center z_0

2.162: Circle with center at z_0

A circle in the complex plane with radius r and center $z_0 \in \mathbb{C}$ has equation:

$$|z - z_0| = r$$

Let $z = a + bi$ and $z_0 = h + ki$

$$|a + bi - (h + ki)| = r$$

$$|a-h+i(b-k)|=r$$

Use a change of origin. Move the origin by (h,k) to get an equation in the new coordinate system:

$$|a+bi|=r$$

Which we already have shown is the equation of a circle.

Example 2.163

Write the complex version of the equation of a circle with radius 5 and center $(2,-3)$ in the coordinate plane.

$$|z-z_0|=r$$
$$|z-(2-3i)|=5$$
$$|z-2+3i|=5$$

Example 2.164

$$|a-5+i(b+2)|=7$$

A. What is the radius and center of the circle above?
B. What is the area?

$$|z-z_0|=r$$

$$r=Radius=7$$
$$a-5+i(b+2)=a+bi-(5-2i)$$
$$z_0=5-2i$$
$$Center=(5,-2i)$$

$$Area=\pi r^2=49\pi$$

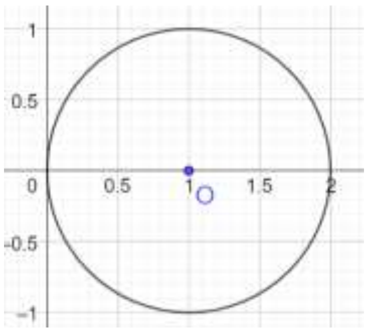
Example 2.165

Geometrically, the set $\{z\in\mathbb{C}:|z-2-2i|\le 1\}$ represents: **(AP EAPCET 20 Sep 2020, Shift-I)**

A closed circular disc with center at (2,2) and radius 1.

Challenge 2.166

What is the area of the region in the complex plane consisting of all points z satisfying both $\left|\frac{1}{z}-1\right|<1$ and $|z-1|<1$? ($|z|$ denotes the magnitude of a complex number, i.e. $|a+bi|=\sqrt{a^2+b^2}$ **(SMT Algebra Tiebreaker 2022/2)**



Condition for a Circle

The second condition is the condition for a disc (interior of a circle) centered at $(a,b)=(1,0)$

$$|z-1|<1$$

First Condition

Substitute $z=a+bi$ in the first condition to get:

$$\left|\frac{1}{a+bi}-1\right|<1$$

$$\frac{1}{a+bi}\cdot\frac{a-bi}{a-bi}=\frac{a-bi}{a^2+b^2}$$

Carry out the division, and rewrite $1 = \frac{a^2+b^2}{a^2+b^2}$:

$$\left|\frac{a-bi}{a^2+b^2} - \frac{a^2+b^2}{a^2+b^2}\right| < 1$$
$$\frac{1}{a^2+b^2}|a - bi - a^2 + b^2| < 1$$
$$|a - bi - a^2 - b^2| < a^2 + b^2$$

Write the complex number in standard form:

$$|a - (a^2+b^2) - bi| < a^2 + b^2$$

Use the definition of magnitude:

$$\sqrt{[a-(a^2+b^2)]^2 + b^2} < a^2 + b^2$$

Square both sides, which we can since $\sqrt{x} > 0, x \in \mathbb{R}$, and expand:

$$a^2 - 2a(a^2+b^2) + (a^2+b^2)^2 + b^2 < (a^2+b^2)^2$$

Add and Rearrange:

$$a^2 + b^2 - 2a(a^2+b^2) + (a^2+b^2)^2 \ < (a^2+b^2)^2$$

Factor $a^2 + b^2$:

$$(a^2+b^2)[1 - 2a + a^2 + b^2] \ < (a^2+b^2)^2$$

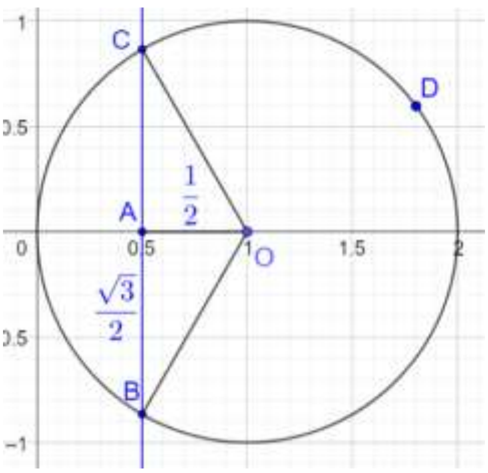
Cancel:

$$1 - 2a + a^2 + b^2 \ < a^2 + b^2$$
$$1 - 2a < 0$$
$$a > \frac{1}{2}$$

Combine the Conditions

Combine the conditions to get the region inside the circle, that is to the right of

$$a = \frac{1}{2}$$



This is given by:

$$Area(\Delta OBC) + Area(Sector\ CBD)$$
$$OA = \frac{1}{2} \Rightarrow AB = \frac{\sqrt{3}}{2} \Rightarrow \Delta OAB\ is\ 30-60-90$$

The area of $A(OBC)$

$$= 2 \times Area(\Delta OAB) = 2 \times \frac{1}{2} \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}$$

The area of $Sector\ CBD$

$$Smaller\ Angle\ \angle COB = 2\angle AOB = 60 \times 2 = 120$$
$$Reflex\ Angle\ \ \angle COB = 360 - 120 = 240$$

$$A(Sector\ CBD) = \frac{\theta}{360}\pi r^2 = \frac{240}{360}\pi(1^2) = \frac{2}{3}\pi$$

And hence, the total area is:

$$\frac{2}{3}\pi+\frac{\sqrt{3}}{4}$$

C. Alternate Equation

2.167: Equation of a Circle (Alternate Form)

Given complex numbers z and B, the equation of a circle is given by:

$$z\bar{z}+B\bar{z}+\bar{B}z+c=0$$

Method I: Complex Variables

Begin with the equation of a circle that we know already:

$$|z-z_0|=r$$

Square both sides of the above:

$$|z-z_0|^2=r^2$$

Use the property $|z|^2=z\bar{z}$:

$$(z-z_0)\overline{(z-z_0)}=r^2$$

Distribute the conjugation:

$$(z-z_0)(\bar{z}-\bar{z_0})=r^2$$

Expand:

$$z\bar{z}-z_0\bar{z}-\bar{z_0}z+z_0\bar{z_0}-r^2=0$$

Substitute $b=-z_0, c=z_0\bar{z_0}-r^2$:

$$z\bar{z}+b\bar{z}+\bar{b}z+c=0$$

Method II: Real Plane

The general equation of a circle on the real plane is:

$$x^2+y^2+2gx+2fy+c=0$$

Let $x=a, y=b$:

$$a^2+b^2+2ga+2fb+c=0$$

Further, let $z=a+bi$. Then

$$\bar{z}=a-bi$$
$$z+\bar{z}=2a$$
$$z-\bar{z}=2bi\Rightarrow 2b=\frac{z-\bar{z}}{i}=i(\bar{z}-z)$$
$$z\bar{z}=|z|^2=a^2+b^2$$

Substitute the above in the equation of a circle:

$$z\bar{z}+g(z+\bar{z})+fi(\bar{z}-z)+c=0$$

Expand:

$$z\bar{z}+gz+g\bar{z}+if\bar{z}-ifz+c=$$

Rearrange and factor:

$$z\bar{z}+(g+if)\bar{z}+(g-if)z+c=00$$

Substitute $B=g+if$:

$$z\bar{z}+B\bar{z}+\bar{B}z+c=0$$

Example 2.168

Mark the correct option

The equation of any _______ in the complex plane is of the form where $(b\in\mathbb{C}, c\in\mathbb{R})$

A. Circle
B. Straight Line
C. Parabola
D. Hyperbola (AP EAPCET 18 Sep 2020, Shift-II)

We assume $z\in\mathbb{C}$. Then, this is exactly the equation of a circle mentioned in the property above.

Option A

D. Regular Polygons

Example 2.169

Product of diagonals of regular polygon in the unit circle (including the sides).

2.7 Geometry: Lines and Triangles

A. Lines

2.170: Vertical Line

$$z + \bar{z} = d$$

Where

$$z \in \mathbb{C}$$
$$\bar{z} \text{ is the conjugate of } z$$
$$d \in \mathbb{R}$$

Let $z = a + bi$

$$z + \bar{z} = (a + bi) + (a - bi) = 2a$$

Let $2a = d$

$$z + \bar{z} = d$$
$$\frac{z + \bar{z}}{2} = a$$

Example 2.171

$$z + \bar{z} = 4$$

$$(2 + 5i) + (2 - 5i) = 4$$

Example 2.172

Given $z \in \mathbb{C}$, find all points that satisfy:

$$z + \bar{z} = 4$$
$$|z| = 5$$

$$\sqrt{a^2 + b^2} = 5$$
$$\sqrt{4 + b^2} = 5$$
$$4 + b^2 = 25$$
$$b^2 = 21$$
$$b = \pm\sqrt{21}$$

$$z \in \left\{2 + \sqrt{21}i, 2 - \sqrt{21}i\right\}$$

2.173: General Line

$$zz_0 + \bar{z}z_0 + C = 0$$

A vertical line in the complex plane is:

$$z + \bar{z} = d$$

Consider $z_0 = r\ cis\ \theta$. Multiplying by z_0 is equivalent to rotating by θ. If the two points on the above line are rotated by θ, we get

$$z \to zz_0$$
$$\bar{z} \to \bar{z}z_0$$

Then, we get:

$$zz_0 + \bar{z}z_0 = rd$$

$$zz_0 + \bar{z}z_0 - rd = 0$$

Substitute $C = -rd$:

$$zz_0 + \bar{z}z_0 + C = 0$$

2.8 Geometry: Triangles

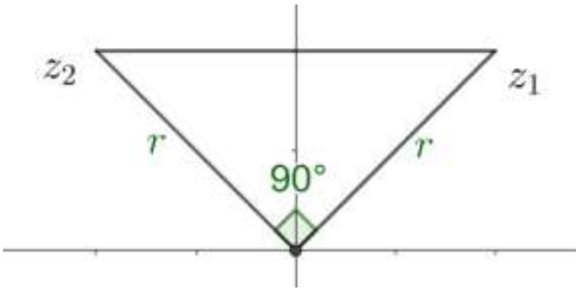
A. Triangles using Rotation

2.174: Isosceles Right Triangle with Vertex at Origin

If the points z_1, z_2 and the origin form a right triangle with the hypotenuse being z_1z_2, then:

$$z_2 = \pm iz_1$$

Since the triangle is isosceles, the lengths of the sides must be equal, and hence the magnitudes of the complex numbers must be equal:



$$z_1 = z_2 = r$$

$$\frac{z_2}{z_1} = \frac{r\ cis\ \theta_1}{r\ cis\ \theta_2} = cis\ (\theta_1 - \theta_2)$$

Since the angle between Oz_1 and Oz_2 is 90°, we must have $\theta_1 - \theta_2 = \pm 90°$

$$\frac{z_2}{z_1} = cis\ (\pm 90°) = \pm i$$

$$z_2 = \pm iz_1$$

2.175: Isosceles Right Triangle with Vertex not at Origin

If the points z_1, z_2 and the origin form a right triangle with the hypotenuse being z_1z_2, then:

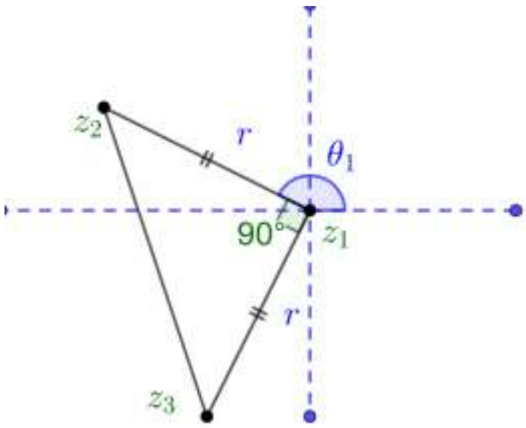
$$\frac{z_3 - z_1}{z_2 - z_1} = \pm i$$

Since the triangle is isosceles, the lengths of the sides must be equal, and hence the magnitudes of the complex numbers must be equal:

$$l(z_2z_1) = l(Oz_1) = r$$

Introduce an origin at the point z_1, which is the vertex at which the right angle occurs:

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{r\ cis\ (\theta_1 + 90°)}{r\ cis\ \theta_1} = cis\ (90°) = \pm i$$



Example 2.176

Mark all correct options

Let O be the origin, and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $Re(z_2) < 0$ such that OAB is a right-angled isosceles triangle with OB as hypotenuse, then which of the following is/are true:

A. $\arg z_2 = \pi - \tan^{-1} 3$
B. $\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$
C. $|z_2| = \sqrt{10}$
D. $|2z_1 - z_2| = 5$ (JEE Main, July 26, 2022-I, Adapted)

B. Triangle

2.177: Angle between two sides of a triangle

If $A = z_1, B = z_2$ and $C = z_3$ are three points that form a triangle in the complex plane, then the angle between the sides BA and BC is given by:

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right)$$

Use a change of origin. Let

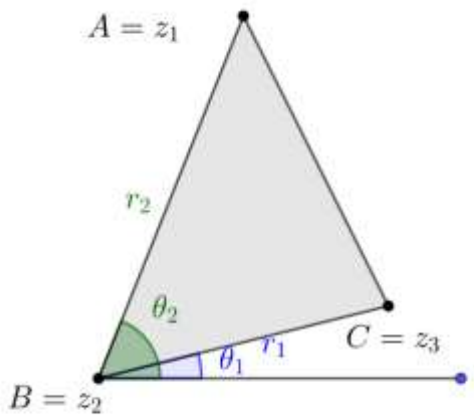
$$z_2 = Origin$$

$$z_1 - z_2 = r_2\ cis\ \theta_2$$
$$z_3 - z_2 = r_1\ cis\ \theta_1$$

$$\frac{AB}{CB} = \frac{z_1 - z_2}{z_3 - z_2} = \frac{r_2\ cis\ \theta_2}{r_1\ cis\ \theta_1} = \frac{r_2}{r_1} cis\ (\theta_2 - \theta_1)$$

Take the argument of the last expression:

$$\arg\left(\frac{r_2}{r_1} cis\ (\theta_2 - \theta_1)\right) = \theta_2 - \theta_1$$



2.178: Right Triangle

For a right ΔABC, right-angled at B, let $A = z_1, B = z_2$ and $C = z_3$ in the complex plane. Then:

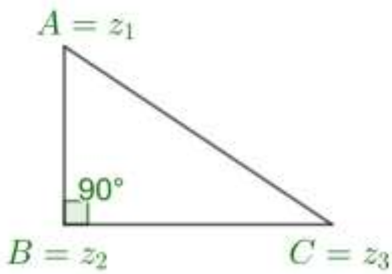
$$\arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = \pm\frac{\pi}{2}$$

The angle between the two legs of the right triangle is $90° = \frac{\pi}{2}$:

$$\arg(AB) - \arg(BC) = \pm\frac{\pi}{2}$$

Use the quotient property:

$$\arg\left(\frac{AB}{BC}\right) = \pm\frac{\pi}{2}$$
$$\arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = \pm\frac{\pi}{2}$$



C. Equilateral Triangles

You can frame the relationship between points that from an equilateral triangle on the complex plane in multiple ways.

2.179: Equilateral Triangle with Vertex at Origin

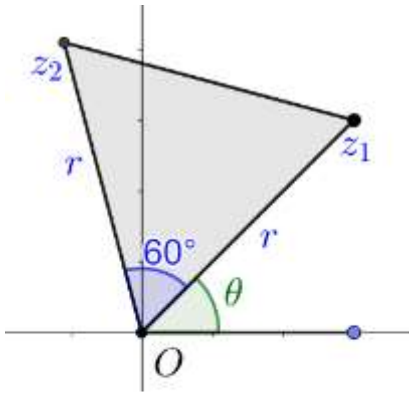
If the points z_1, z_2 and the origin form an equilateral triangle, then:

$$\frac{z_2}{z_1} = \text{cis}\,(\pm 60°)$$

We consider the case where z_1 is to the right of z_2 (see diagram). The other case is similar.

Since the triangle is equilateral, the lengths of the sides must be equal, and hence the magnitudes of the complex numbers must be equal:

$$z_1 = z_2 = r$$



Let the angle between z_1 and the x axis be θ. Then:

$$z_1 = r\text{ cis }\theta$$

Since $\angle z_1 O z_2 = 60°$,

$$z_2 = r\text{ cis }(60 + \theta) = r\text{ cis }\theta\text{ cis }60$$

Substitute $z_1 = r\text{ cis }\theta$:

$$z_2 = z_1\text{ cis }60°$$

2.180: Equilateral Triangle with Vertex at Origin

If $\frac{z_2}{z_1} = \text{cis }60°$ then the points z_1, z_2 and the origin form an equilateral triangle.

Substitute $z_1 = r_1 cis\,\theta_1,\ z_2 = r_2\,cis\,\theta_2,\ \text{cis }60° = 1\text{ cis }60°$:

$$\frac{r_2\,cis\,\theta_2}{r_1 cis\,\theta_1} = 1\text{ cis }60°$$

$$\frac{r_2}{r_1}\cdot cis\,(\theta_2 - \theta_1) = 1\text{ cis }60°$$

Two complex numbers are equal if and only if their magnitudes and their principal arguments are equal.
Equate the magnitudes:

$$\frac{r_2}{r_1} = 1 \Rightarrow r_2 = r_1 \Rightarrow Magnitudes\ are\ same \Rightarrow Lengths\ are\ same$$

Equate the arguments:

$$\theta_2 - \theta_1 = 60°$$
$$\theta_2 = \theta_1 + 60°$$

Hence, the distance from the origin is the same, and the angle between the point is 60°.
Therefore, they form an equilateral triangle.

Example 2.181

Let z_1 and z_2 be two roots of the equation $z^2 + az + b = 0$, z being complex. Further, assume that the origin, z_1 and z_2 form an equilateral triangle. Then, find the value of a^2 in terms of b: (JEE Main 2003, Adapted)

Use of Vieta's Formulas to create a system of equations in the roots:

$$Sum\ of\ Roots\colon \underbrace{z_1 + z_2 = -a}_{Equation\ I}$$

$$Product\ of\ Roots\colon \underbrace{z_1 z_2 = b}_{Equation\ II}$$

Since z_1 and z_2 form an equilateral triangle with the origin, we must have:

$$z_1 = z_2 \operatorname{cis} 60°$$

Substitute $z_1 = z_2 \operatorname{cis} 60°$ in Equation I:

$$\mathit{Equation\ I}: z_2 \operatorname{cis} 60° + z_2 = -a \Rightarrow \underbrace{z_2 = -\frac{a}{1+\operatorname{cis} 60°}}_{\mathbf{Equation\ III}}$$

$$\mathit{Equation\ II}: z_2 \operatorname{cis} 60° z_2 = b \Rightarrow z_2^2 = \frac{b}{\operatorname{cis} 60°} \Rightarrow \underbrace{z_2 = \frac{\sqrt{b}}{\operatorname{cis} 30°}}_{\mathbf{Equation\ IV}}$$

Combine Equations III and IV:

$$-\frac{a}{1+\operatorname{cis} 60°} = \frac{\sqrt{b}}{\operatorname{cis} 30°} \Rightarrow -\frac{a}{\sqrt{b}} = \frac{1+\operatorname{cis} 60°}{\operatorname{cis} 30°}$$

Then, in the above equation:

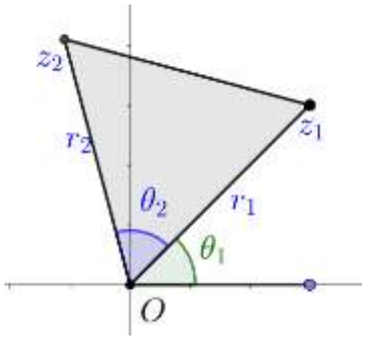
$$RHS = \frac{1}{\operatorname{cis} 30°} + \frac{\operatorname{cis} 60°}{\operatorname{cis} 30°} = \frac{\operatorname{cis} 0°}{\operatorname{cis} 30°} + \operatorname{cis} 30° = \operatorname{cis}(-30°) + \operatorname{cis}(30°) = 2\cos 30° = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

Finally, the expression that we want:

$$-\frac{a}{\sqrt{b}} = \sqrt{3} \Rightarrow a^2 = 3b$$

2.182: Angle between two sides of a triangle

$$z_2 = z_1\left(\frac{r_2}{r_1} cis\ \theta_2\right)$$



2.183: Equilateral Triangle

If complex numbers z_1, z_2 and z_3 satisfy

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

Then the points form an equilateral triangle on the complex plane.

Multiply both sides by 2:

$$z_1^2 + z_2^2 + z_2^2 + z_3^2 + z_3^2 + z_1^2 = 2z_1z_2 + 2z_2z_3 + 2z_3z_1$$

Collate all terms on the LHS:

$$(z_1^2 - 2z_1z_2 + z_2^2) + (z_2^2 - 2z_2z_3 + z_3^2) + (z_3^2 - 2z_3z_1 + z_1^2) = 0$$

Note that each term in the parenthesis forms a perfect square:

$$(z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_1)^2 = 0$$

3. Roots of Unity

3.1 Roots of Unity: Basics

A. Finding Roots of Unity

3.1: Unity
Unity is a way of referring to the number 1. Hence, roots of unity refer to the roots of the number 1.

3.2: n^{th} Root of Unity

$$z^n - 1 = 0, \qquad n \in \mathbb{N}$$

- The n^{th} roots of unity refer to the solutions in the complex number system for the equation given above.
- There are exactly n solutions.

Example 3.3

Mark the Correct Option

The number of n^{th} roots of unity is:

A. Always n
B. Not always n. There are multiple exceptions.
C. Always n, except for prime n
D. Always n, except for when n is unity

Refer the definition given at the beginning of the section.

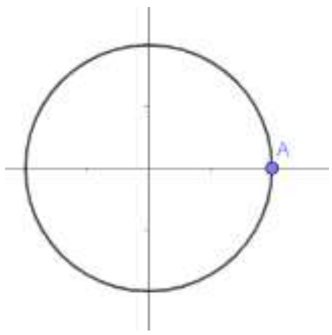
Option A is correct

Example 3.4:

$$z^1 - 1 = 0$$

Algebraic Method

$$z^1 = 1 \Rightarrow z = 1 + 0i$$



Polar Form

$$z^1 = 1$$

Substitute $z = r\sin\theta\,, 1 = 1\sin(2k\pi)\,, k \in \mathbb{Z}$

$$r \operatorname{cis}\theta = 1\operatorname{cis}(2k\pi)\,, k \in \mathbb{Z}$$

For two complex numbers to be equal their magnitudes and their argument must each be equal:

$$Magnitude: r = 1$$

$$Argument: \theta = 2k\pi, n \in \mathbb{Z}$$

Substitute $r = 1, \theta = 2k\pi$

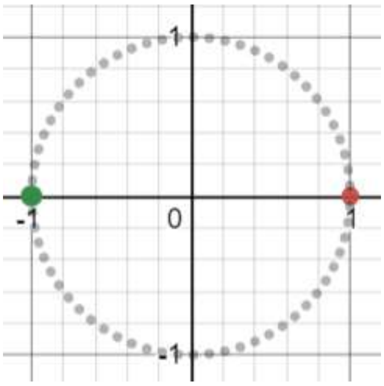
$$r\operatorname{cis}\theta = 1\operatorname{cis}2k\pi = 1\,(\cos 2k\pi + i\sin 2k\pi)$$

Substitute $\cos 2k\pi = 1, \sin 2k\pi = 0$

$$= 1\,(1 + i \cdot 0) = 1\,(1 + 0) = 1$$

Example 3.5

Find the square roots of unity and plot them on the complex plane



Algebraic Method

$$z^2 - 1 = 0$$

This is a quadratic, which can be solved by factoring:

$$(z+1)(z-1) = 0$$

Use the zero-product property to find the roots:

$$z \in \{-1 + 0i, +1 + 0i\}$$

Polar Form

$$z^2 = 1$$

Substitute $z = r \operatorname{cis} \theta, 1 = 1 \operatorname{cis}(2k\pi), k \in \mathbb{Z}$

$$(r \operatorname{cis} \theta)^2 = 1 \operatorname{cis}(2k\pi), k \in \mathbb{Z}$$

Use $(r \operatorname{cis} \theta)^n = r^n \operatorname{cis} n\theta$

$$r^2 \operatorname{cis} 2\theta = 1 \operatorname{cis}(2k\pi), n \in \mathbb{Z}$$

Equate the magnitudes:

$$r^2 = 1 \Rightarrow r = 1$$

Equate the argument:

$$2\theta = 2k\pi, \qquad k \in \mathbb{Z}$$
$$\theta = k\pi, \qquad k \in \mathbb{Z}$$

Substitute $r = 1, \theta = 2n\pi$

$$r \operatorname{cis} \theta = 1 \operatorname{cis} n\pi$$

Substitute:

$$k = 0: 1 \operatorname{cis} 0 = 1$$
$$k = 1: 1 \operatorname{cis} \pi = -1$$
$$k = 2: 1 \operatorname{cis} 2\pi = 1 \operatorname{cis} 0 \Rightarrow \textit{Repeat}$$
$$z \in \{-1 + 0i, +1 + 0i\}$$

Exponential Form

$$z^2 = 1$$

Substitute $z = re^{i\theta}$:

$$\left(re^{i\theta}\right)^2 = 1 \cdot e^{i(2k\pi)}$$
$$r^2 e^{i2\theta} = 1 \cdot e^{i(2k\pi)}$$

Equate the magnitudes:

$$r^2 = 1 \Rightarrow r = 1$$

Equate the argument:

$$2\theta = 2k\pi, k \in \mathbb{Z}$$

Example 3.6: Fourth Roots of Unity

Find the fourth roots of unity and plot them on the complex plane

Algebraic Method

$$z^4 - 1 = 0$$

Factor using difference of squares:

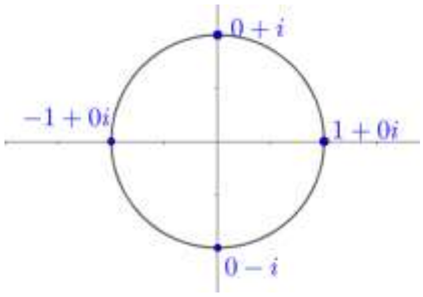
$$(z^2 + 1)(z^2 - 1) = 0$$

Factor using difference of squares:

$$(z^2+1)(z+1)(z-1)=0$$

Use the zero-product property:

$$z-1=0 \Rightarrow z=1$$
$$z+1=0 \Rightarrow z=-1$$
$$z^2+1=0 \Rightarrow z^2=-1 \Rightarrow z=\pm\sqrt{-1}=\pm i$$
$$\textit{Fourth Roots of Unity are } z \in \{-1+0i,+1+0i,0+i,0-i\}$$



Polar Form

$$z^4=1$$

Substitute $z=r\,\mathrm{cis}\,\theta\,,1=1\,\mathrm{cis}(2k\pi)\,,k\in\mathbb{Z}$

$$(r\,\mathrm{cis}\,\theta)^4=1\,\mathrm{cis}(2k\pi)\,,n\in\mathbb{Z}$$

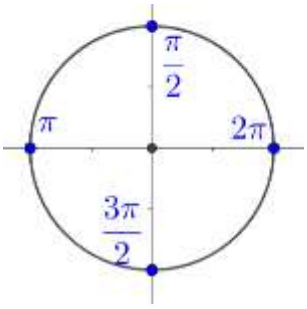
Use $(r\ \mathrm{cis}\ \theta)^n=r^n\mathrm{cis}\ n\theta$

$$r^4\,\mathrm{cis}\,4\theta=1\,\mathrm{cis}(2k\pi)\,,n\in\mathbb{Z}$$

Equate the magnitudes:

$$r^4=1\Rightarrow r=1$$

Equate the argument:



$$4\theta=2kk\pi,k\in\mathbb{Z}$$
$$\theta=k\frac{\pi}{2},n\in\mathbb{Z}$$

Substitute $r=1,\theta=2n\pi$

$$r\,\mathrm{cis}\,\theta=1\,\mathrm{cis}\,n\frac{\pi}{2}$$

Substitute:

$$k=0{:}\,1\,\mathrm{cis}\,0=1$$
$$k=1{:}\,1\,\mathrm{cis}\frac{\pi}{2}=i$$
$$k=2{:}\,1\,\mathrm{cis}\,\pi=-1$$
$$k=3{:}\,1\,\mathrm{cis}\frac{3\pi}{2}=-i$$
$$k=4{:}\,1\,\mathrm{cis}\,2\pi=1\,\mathrm{cis}\,0\Rightarrow \textit{Repeat}$$

$$cis\ 0=\cos 0+i\sin 0=1+0=1$$
$$cis\frac{\pi}{2}=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=0+i=i$$
$$cis\ \pi=\cos\pi+i\sin\pi=-1+0=-1$$
$$cis\frac{3\pi}{2}=\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}=0-i=-i$$

$$\textit{Fourth Roots of Unity are } z \in \{-1+0i,+1+0i,0+i,0-i\}$$

Example 3.7: Fourth Roots of Unity

Find the fourth roots of unity and plot them on the complex plane

A. $z^4-1=0$
B. $z^4=16$

Part B

$$z^4-16=0$$
$$(z^2-4)(z^2+4)=0$$
$$(z+2)(z-2)(z+2i)(z-2i)=0$$

Apply the zero-product property:

$$z \in \{\pm 2, \pm 2i\}$$

3.8: n^{th} root of unity

$$z = \sqrt[n]{1} \Rightarrow r = 1, \theta = \frac{360}{n}$$

$$z = \sqrt[n]{1} \Rightarrow z^n = 1$$

Substitute $z = r \operatorname{cis} \theta, 1 = 1 \operatorname{cis}(2k\pi), k \in \mathbb{Z}$

$$(r \operatorname{cis} \theta)^n = 1 \operatorname{cis}(2k\pi), k \in \mathbb{Z}$$

Use $(r \operatorname{cis} \theta)^n = r^n \operatorname{cis} n\theta$

$$r^n \operatorname{cis} n\theta = 1 \operatorname{cis}(2k\pi), n \in \mathbb{Z}$$

Equate the magnitudes:

$$r^n = 1 \Rightarrow r = 1$$

Equate the argument:

$$n\theta = 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$

3.9: General Solution for θ

$$z^n = 1 \Leftrightarrow z = cis\ \theta, \theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$

$$z^n = 1$$

$$(r\ cis\ \theta)^n = 1$$

$$r^n\ cis\ n\theta = 1\ cis\ 0$$

Equate the magnitude:

$$r^n = 1 \Rightarrow r = 1$$

Equate the argument

$$n\theta = 2k\pi$$

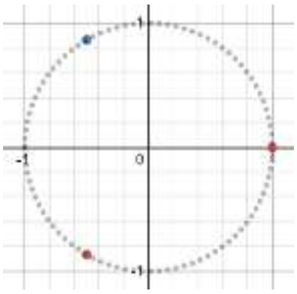
$$\theta = \frac{2k\pi}{n}, \qquad 0 \le k \le n-1, \qquad k \in \mathbb{Z}$$

3.10: Cube Roots of Unity / Omega Notation

ω refers to a cube root of unity. If $z^3 = 1$, then:

$$z \in \left\{1, \underbrace{-\frac{1}{2} + \frac{\sqrt{3}}{2}i}_{\omega}, \underbrace{-\frac{1}{2} - \frac{\sqrt{3}}{2}i}_{\omega^2}\right\}$$

Algebraic Method



Any cube root of unity must satisfy the relation:

$$\omega = \sqrt[3]{1}$$

Cube both sides, and collate all terms on the LHS:

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

Factor the LHS using $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$:

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

Apply the zero-product property.

The first term gives us 1, which we already know is a solution:

$$\omega - 1 = 0 \Rightarrow \omega = 1$$

The second term is a quadratic:

$$\omega^2 + \omega + 1 = 0$$

We can't factor it easily, so apply the quadratic formula with $a = 1, b = 1, c = 1$ to get the other two roots:

$$\omega = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{-3}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$$z \in \left\{1, \underbrace{-\frac{1}{2} + \frac{\sqrt{3}}{2} i}_{\omega}, \underbrace{-\frac{1}{2} - \frac{\sqrt{3}}{2} i}_{\omega^2}\right\}$$

3.11: Cube Roots of Unity / Omega Notation

$$z \in \left\{1\ cis\ 0, \underbrace{1\ cis\ 120°}_{\omega}, \underbrace{1\ cis\ 240°}_{\omega^2}\right\}$$

$$z^3 - 1 = 0$$
$$z^3 = 1$$
$$(r\ cis\ \theta)^3 = \text{cis}(360k°)$$
$$r^3\ cis\ 3\theta = \text{cis}(360k°)$$

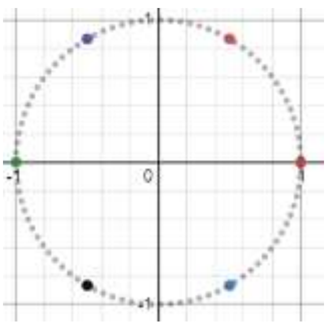
$$r^3 = 1 \Rightarrow r = 1$$
$$3\theta = 360k° \Rightarrow \theta = \frac{360k°}{3} = 120k°$$

$$k = \{0,1,2\} \Rightarrow \theta = \{0°, 120°, 240°\}$$

$$z = \text{cis}(120), \text{cis}(240), \text{cis}(360)$$

Example 3.12: Sixth Roots of Unity

$$z^6 - 1 = 0$$



Factor using a difference of squares:

$$(z^3 + 1)(z^3 - 1) = 0$$

The second term gives us the three cube roots of unity:

$$(z^3 - 1) = 0 \Rightarrow z \in \left\{1, -\frac{1}{2} \pm \frac{\sqrt{3}i}{2}\right\}$$

And we can factor $(z^3 + 1) = 0$ using the formula $a^3 + b^3$:

$$(z + 1)(z^2 - z + 1) = 0$$

Use the zero-product property:

$$z + 1 = 0 \Rightarrow z = -1$$
$$z^2 - z + 1 = 0$$

Apply the quadratic formula with $a = 1, b = -1, c = 1$:

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{1 - 4}}{2} = \frac{1 \pm \sqrt{-3}}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$$(z^3 + 1) = 0 \Rightarrow z \in \left\{-1, \frac{1}{2} \pm \frac{\sqrt{3}i}{2}\right\}$$

And we can combine to get the six sixth roots of unity:

$$z \in \left\{\pm 1, \pm\frac{1}{2} \pm \frac{\sqrt{3}i}{2}\right\}$$

3.2 Roots of Unity: Properties

A. Properties of Roots of Unity

Properties of roots of unity are important, and very deep.

3.13: n Roots of Unity

There are n n^{th} roots of unity.

$$z^n = 1, n \in \mathbb{N}$$
$$z^n - 1 = 0$$

This is a n^{th}degree polynomial.

Every n^{th}degree polynomial has n complex roots. Hence, the above polynomial has n roots. These roots are the n^{th} roots of unity.

$$z^n = 1, n \in \mathbb{N}$$

Example 3.14

What is the number of solutions to:

$$z^{2023} = 1$$

$$2023 \ solutions$$

3.15: Roots of Unity lie on Unit Circle

The n $roots$ of unity lie on the unit circle.

$$z^n = 1$$

Take the modulus (distance) both sides:

$$|z^n| = |1|$$
$$|z^n| = 1$$

Distance of root of unity from the origin is 1.

$$z^n = 1$$
$$(r\ cis\ \theta)^n = 1\ cis\ 0$$
$$r^n\ cis\ n\theta = 1\ cis\ 0$$

$$r = 1$$
$$Magnitude = 1$$

Example 3.16

Let z be a thousandth root of unity. Find the number of values of z such that $Re(z) < 2$ and $Im(z) > 3$.

The roots of unity lie on the unit circle. Hence, they cannot be greater than a distance of 1 from the origin.

$$No\ Values$$

3.17: Zeta Notation

ζ refers to a complex root of unity.

Example 3.18

Show that th

Part A

$$1st\ Root\ of\ Unity = 1 + 0i = 1$$

Part B

$$-1 + 1 = 0$$

Part C

Fourth Roots of Unity

$$1 + i - 1 - i = 0$$

Example 3.19

$$i = i$$
$$i^2 = -1$$
$$i^3 = -i$$
$$i^4 = 1$$

Example 3.20

	1	$\boldsymbol{i}$	$\boldsymbol{i^2}$	$\boldsymbol{i^3}$
1	1	i	i^2	i^3
$\boldsymbol{i}$	i	i^2	i^3	i^4
$\boldsymbol{i^2}$	i^2	i^3	i^4	i^5
$\boldsymbol{i^3}$	i^3	i^4	i^5	i^6

	1	$\boldsymbol{i}$	$\boldsymbol{i^2}$	$\boldsymbol{i^3}$
1	1	i	i^2	i^3
$\boldsymbol{i}$	i	i^2	i^3	1
$\boldsymbol{i^2}$	i^2	i^3	1	i
$\boldsymbol{i^3}$	i^3	1	i	i^2

Note that the first row is:

$$\{1, i, i^2, i^3\}$$

Every other row has the same elements as above in the same order, but starting from a different value.

Example 3.21

	0	**1**	**2**	**3**
0	0	1	2	3
1	1	2	3	4
2	2	3	4	5
3	3	4	5	6

Find the remainder when element in the table is divided by 4:

	0	**1**	**2**	**3**
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Note that each element in this table is the power of i in the previous table.

3.22: Primitive Root of Unity

A n^{th} root of unity that is not a n^{th} root of unity for any value of $n = \{1,2,3, \ldots, n-1\}$ is called a primitive root of unity.

Example 3.23

Identify the primitive roots of unity for $n = 1,2,3,4$.

$$z^n = 1$$

Part A

$$\underbrace{1}_{\substack{First\ Root \\ of\ Unity}}$$

$n = 1$ has one root. Since $n = 1$ is the first value which we consider 1 is a primitive.

Part B

$n = 2$ has two roots.

$$\underbrace{1}_{\substack{First\ Root \\ of\ Unity}}, -1$$

$$1\ is\ not\ primitive\ since\ it\ is\ a\ first\ root\ of\ unity$$
$$-1\ is\ primitive\ since\ we\ see\ it\ for\ the\ first\ time$$
$$One\ primitive\ root$$

Part C

$n = 3$ has three roots.

$$\underbrace{1}_{\substack{First\ Root \\ of\ Unity}}, -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i \Rightarrow -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$$\underbrace{1}_{\substack{First\ Root \\ of\ Unity}}\quad is\ non\ primitive$$

$$, -\frac{1}{2} \pm \frac{\sqrt{3}}{2} i \Rightarrow -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\ are\ primitive$$
$$are\ primitive \Rightarrow Two\ primitive\ Roots$$

Part D

$n = 4$ has four roots.

$$\underbrace{\pm 1}_{\substack{Second\ Roots \\ of\ Unity}}, \pm i \Rightarrow$$

$$\underbrace{\pm 1}_{\substack{\text{Second Roots}\\ \text{of Unity}}} \quad \textit{is non primitive}$$

$$\pm i \textit{ is primitive}$$
$$\textit{Two primitive Roots}$$

3.24: Primitive Roots of Unity

Let ζ be a n^{th} root of unity. If ζ^n is the smallest value of n for which

$$\zeta^n = 1$$

Then ζ is a primitive root of unity.

Example 3.25

Consider the fourth roots of unity

$$Powers\ of\ 1\ are\ 1,1,1,1, \dots . \Rightarrow \{1\}$$
$$Powers\ of - 1\ are - 1,1,-1,1, \dots . \{-1,1\}$$

$$Powers\ of\ i\ are\ \{i, -1, -i, 1\}$$

Example 3.26

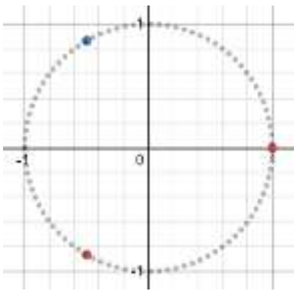
x^k	$k=1$	$k=2$	$k=3$	$k=4$	
1	1^1	1^2	1^3	1^4	Non primitive
−1	$(-1)^1$	$(-1)^2$	$(-1)^3$	$(-1)^4$	
i	i^1	i^2	i^3	i^4	Primitive
$-i$	$(-i)^1$	$(-i)^2$	$(-i)^3$	$(-i)^4$	

x^k	$k=1$	$k=2$	$k=3$	$k=4$	
1	1	1	1	1	Non primitive
−1	-1	1	-1	1	
i	i	-1	$-i$	1	Primitive
$-i$	$-i$	-1	i	i	

3.27: Powers of Roots of Unity

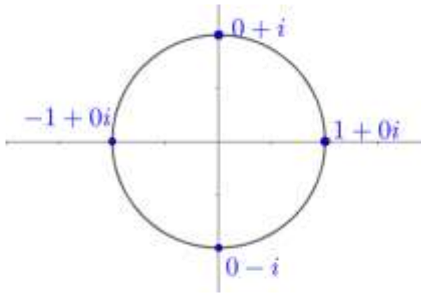
If ζ is a primitive n^{th} root of unity, then the roots of unity are:

$$\{\zeta, \zeta^2, \dots, \zeta^{n-1}, \zeta^n = 1\}$$



$$(Blue\ Dot)^2 = (cis\ 120)^2 = cis\ 240 = Red\ Dot$$
$$(Blue\ Dot)^3 = (cis\ 120)^3 = cis\ 360 = cis\ 0 = 1$$

$$(Red\ Dot)^2 = (cis\ 240)^2 = cis\ 480 = cis\ 120 = Blue\ Dot$$
$$(Red\ Dot)^3 = (cis\ 240)^3 = cis\ 720 = cis\ 0 = 1$$



$i = cis\ 90$

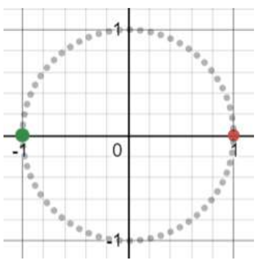
$$i^2 = (cis\ 90)^2 = cis\ 180 = -1$$
$$i^3 = (cis\ 90)^3 = cis\ 270 = -i$$
$$i^4 = (cis\ 90)^4 = cis\ 360 = cis\ 0 = -i$$

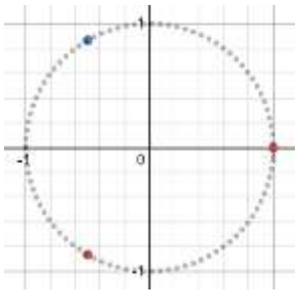
$-i = cis\ 270$

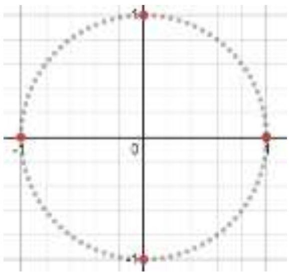
$$(-i)^2 = (cis\ 270)^2 = cis\ 540 = cis\ 180 = -1$$
$$(-i)^3 = (cis\ 270)^3 = cis\ 810 = cis\ 90 = i$$
$$i^4 = (cis\ 270)^4 = cis\ 1080 = cis\ 0 = 1$$

3.28: Roots of Unity lie on a regular polygon in the Complex Plane

n^{th} roots of unity lie on a regular $n - sided$ polygon whose vertices are on the unit circle in the complex plane.







$$(r\ cis\ \theta)^n = 1$$
$$r = 1, \theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$

Because $r = 1$, the roots have magnitude 1 and must lie on the unit circle.
Hence, the polygon is

$$concylic$$

The values of θ are

$$\left\{\frac{2k\pi}{1}, \frac{2k\pi}{2}, \frac{2k\pi}{3}, \dots, \frac{2k\pi}{n}\right\} = 2k\pi\left\{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}\right\}$$

Which divides the unit circle into $n\ equal\ parts$. Hence, the polygon is

$$regular$$

Example 3.29

Determine the degree and radian measures for

A. 4th roots of unity.
B. 5th roots of unity.
C. 8th roots of unity

4th Roots of Unity

$$Degree\ Measures = \{0°, 90°, 180°, 270°\} + n360°, n \in \mathbb{Z}$$
$$Radian\ Measures = \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\} + 2n\pi, n \in \mathbb{Z}$$

5th Roots of Unity

$$Degree\ Measures = \{0°, 72°, 144°, 216°, 288°\} + n360°, n \in \mathbb{Z}$$
$$Radian\ Measures = \left\{0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}\right\} + 2n\pi, n \in \mathbb{Z}$$

8th Roots of Unity

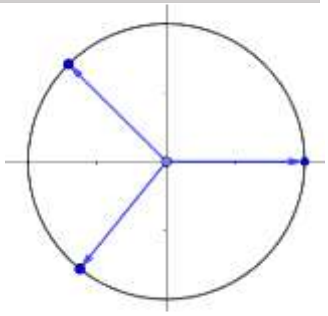
$$\frac{360}{8} = 45°$$
$$Degree\ Measures = \{n45°, \quad n \in \mathbb{Z}\}$$
$$Radian\ Measures = \left\{\frac{n\pi}{4}, \quad n \in \mathbb{Z}\right\}$$

3.30: Sum of Roots of Unity

If $\zeta \neq 1$ is a primitive n^{th} root of unity then:

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1} = 0$$

- The $\zeta \neq 1$ is useful because when $n = 1$, the only root is 1, and the sum is *not* zero.



Proof I: Vectors

Consider the roots of unity as vectors. From the note on vectors, we know that
The sum of the vectors drawn from the center of a regular polygon to the vertices is zero.
Hence, the sum is zero.

Note that since a polygon has n sides, $3 \leq n, n \in \mathbb{N}$, we need to prove the case with $n = 2$ separately:

$$z^2 = 1 \Rightarrow z = \pm 1 \Rightarrow +1 - 1 = 0$$

Proof II: Geometric Series

We consider the sum below as a geometric series:

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1}$$

$first\ term = a = 1, common\ ratio = r = \zeta$, and the sum is given by:

$$S = \frac{a(1 - r^n)}{1 - r} = \frac{(1)(1 - \zeta^n)}{1 - \zeta} = \frac{1 - \zeta^n}{1 - \zeta}$$

Substitute $\zeta^n = 1$ since it an n^{th} root of unity:

$$= \frac{1 - 1}{1 - \zeta} = \frac{0}{1 - \zeta} = 0$$

Proof III: Polynomials

The n^{th} roots of unity satisfy

$$z^n - 1 = 0, n \geq 2$$

Rewrite this as a n^{th} polynomial:

$$z^n + 0z^{n-1} + \cdots - 1 = 0$$

Apply Vieta's Formulas to find the sum of the roots:

$$=\frac{0}{1}=0$$

3.31: Vieta's Formulas

$$a_1x^n+a_2x^{n-1}+a_3x^{n-2}+\cdots+a_{n+1}=0$$
$$Sum\ of\ Roots=Roots\ taken\ 1\ at\ a\ time=-\frac{a_2}{a_1}$$

Example 3.32

The equation $z^6+z^3+1=0$ has complex roots with argument θ between $90°$ and $180°$ in the complex plane. Determine the degree measure of θ. **(AIME 1984/8)**

Use a change of variable on the given equation. Let $Z=z^3$:
$$Z^2+Z+1=0$$
Multiply both sides of the above by $Z-1=0$
$$(Z-1)(Z^2+Z+1)=0$$
$$Z^3-1=0$$
Change back to the original variable:
$$z^9-1=0$$
The above equation has nine solutions which are the nine ninth roots of unity:
$$r=1,\theta=\frac{360}{9}k=40k=\{0°,40°,80°,\mathbf{120°},\mathbf{160°},200°,\dots\}$$

Out of the above solutions, the solutions in the range $90°<\theta<180°$ are:
$$\{120°,160°\}$$
Note that if
$$Z-1=0\Rightarrow z^3-1=0\Rightarrow z=r\ cis\ \theta\Rightarrow r=1,\theta\in\{0°,\mathbf{120°},240°\}$$
Conclusion:
$z^6+z^3+1=0$ has six solutions, which are ninth roots of unity, except for cube roots of unity.

$$(z-1)(z^n+z^{n-1}+\cdots+1)=z^{n+1}-1$$

3.33: Property

The equation
$$z^n=\bar{z}$$
has
$$n+2\ solutions$$

Take the mod:2
$$|z^n|=|\bar{z}|$$
$$|z|^n=|z|$$

$$y^n=y$$
$$y\in\{0,1\}$$

Case I: $y=0$
$$|z|=1\Rightarrow 1\ Solution$$

Case II:$y = |z| = 1$

$$z^n = \bar{z}$$
$$z^{n+1} = z\bar{z} = |z|^2 = 1$$
$$n + 1 \; roots \; of \; unity \; give \; n + 1 \; solutions$$

Total Solutions

$$n + 1 + 1 = n + 2 \; Solutions$$

Example 3.34: Polar Form Method

Find the number of ordered pairs of real numbers (a, b) such that $(a + bi)^{2002} = a - bi$. **(AMC 12A 2002/24)**

Substitute $z = r \; cis \; \theta = a + bi \Rightarrow \bar{z} = r \; cis \; (-\theta) = a - bi$:

$$(r \; cis \; \theta)^{2002} = r \; cis \; (-\theta)$$

Use De Moivre's Theorem:

$$r^{2002} \; cis \; 2002\theta = r \; cis \; (-\theta)$$

For two complex numbers to be equal, their magnitudes and their argument must both be equal:
Equate the magnitudes:

$$r^{2002} = r$$
$$r^{2002} - r = 0$$
$$r(r^{2001} - 1) = 0$$
$$r = \{0,1\}$$

Equate the arguments:

$$2002\theta = -\theta + 360k°$$
$$2003\theta = 360k°$$
$$\theta = \frac{360k°}{2003}$$

When $r = 0$, the complex number is:

$$z = a + bi = 0 \Rightarrow Satisfies \Rightarrow 1 \; Solution$$

When $r = 1$, the complex number is:

$$r = 1, \theta = \frac{360k°}{2003} \Rightarrow 2003 \; Solutions$$

Final Answer

$$1 + 2003 = 2004 \; Solutions$$

Note:

1. We found the unique solutions in polar form. If we were to convert into $a + bi$ form, those solutions would still be unique.
2. If we had considered solutions where the angle is repeated, then we would not have got unique solutions. For example: $r = 1, \theta = 90°$ is the same as $r = 1, \theta = 450°$

Example 3.35: Algebraic Method

Find the number of ordered pairs of real numbers (a, b) such that $(a + bi)^{2002} = a - bi$. **(AMC 12A 2002/24)**

Substitute $z = a + bi \Rightarrow \bar{z} = a - bi$ to write the equation as:

$$z^{2002} = \bar{z}$$

Take the mod both sides:

$$|z^{2002}| = |\bar{z}|$$

Use the property $|z^n| = |z|^n$ on the LHS, and $|z| = |\bar{z}|$ on the RHS:

$$|z|^{2002} = |z|$$

Let $y = |z|$

$$y^{2002} = y$$
$$y^{2002} - y = 0$$
$$y(y^{2001} - 1) = 0$$
$$y = |z| \in \{0,1\}$$

Case I: $|z| = 0 \Rightarrow$z is on the origin

$$a + bi = 0 \Rightarrow 1\ Solution$$

Case II: $|z| = 1 \Rightarrow$ z is on the unit circle

Multiply $z^{2002} = \bar{z}$ on both sides by z:

$$z^{2003} = z\bar{z}$$

Substitute $z\bar{z} = |z|^2 = 1$

$$z^{2003} = 1$$

Hence, the solutions from this case are all the 2003^{rd} roots of unity (which are 2003 in number).
The final answer is:

$$1 + 2003 = 2004$$

3.36: Vieta's Formulas

$$z^n = 1 \Rightarrow z^n - 1 = 0$$

Has solutions

$$z \in \{1, \zeta^1, \zeta^2, \dots, \zeta^{n-1}\}$$

Since the above are the n roots of the polynomial, it can be expressed in factored form as:

$$z^n - 1 = (z-1)(z-\zeta^1)(z-\zeta^2)\dots(z-\zeta^{n-1}) = 0$$

3.3 Roots of Unity: Applications

A. Proving a Trigonometric Formula

Example 3.37

$$\prod_{k=1}^{k=n-1} \sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}, \qquad n \in \mathbb{N}$$

Show that the expression above satisfies $\frac{n}{2^{n-1}}$ for:

A. $n = 2$
B. $n = 3$
C. $n = 4$

Part A
When $n = 2$:

$$\prod_{k=1}^{k=1} \sin\left(\frac{k\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1 = \frac{2}{2} = \frac{2}{2^{2-1}}$$

Part B
When $n = 3$:

$$\prod_{k=1}^{k=2} \sin\left(\frac{k\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right)\sin\left(\frac{2\pi}{3}\right) = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{3}{4} = \frac{3}{2^{3-1}}$$

Part C
When $n = 4$:

$$\prod_{k=1}^{k=3} \sin\left(\frac{k\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{2\pi}{4}\right)\sin\left(\frac{3\pi}{4}\right) = \left(\frac{1}{\sqrt{2}}\right)(1)\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2} = \frac{4}{8} = \frac{4}{2^{4-1}}$$

3.38: Writing sin in cis form

We can write a *sine* function, which is purely trigonometric as a complex number in the form:

$$\sin\left(\frac{k\pi}{n}\right) = \frac{1}{2i} \times cis\left(\frac{k\pi}{n}\right)\left[1 - cis\left(-\frac{2k\pi}{n}\right)\right]$$

$$RHS = \frac{1}{2i}\left[cis\left(\frac{k\pi}{n}\right) - cis\left(-\frac{k\pi}{n}\right)\right]$$

Apply the definition of $cis\ \theta$:

$$\frac{1}{2i}\left\{\left[\cos\left(\frac{k\pi}{n}\right) + i\sin\left(\frac{k\pi}{n}\right)\right] - \left[\cos\left(-\frac{k\pi}{n}\right) + i\sin\left(-\frac{k\pi}{n}\right)\right]\right\}$$

Rearrange and use the identity $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$

$$= \frac{1}{2i}\left\{\cos\left(\frac{k\pi}{n}\right) - \cos\left(\frac{k\pi}{n}\right) + i\left[\sin\left(\frac{k\pi}{n}\right) + \sin\left(+\frac{k\pi}{n}\right)\right]\right\}$$

Simplify:

$$= \frac{2i\left[\sin\left(\frac{k\pi}{n}\right)\right]}{2i} = \sin\left(\frac{k\pi}{n}\right) = LHS$$

3.39: n^{th} Root of Unity

$$cis\ \frac{2k\pi}{n}, \qquad 0 \le k \le n-1, \qquad k \in \mathbb{N}$$

3.40: Roots of Unity

$$\prod_{k=1}^{k=n-1}[x - \zeta^k] = (x - \zeta^1)(x - \zeta^2)\dots(x - \zeta^{n-1}) = x^{n-1} + x^{n-2} + \cdots + x + 1$$

$$\underbrace{x^n - 1 = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1)}_{\textit{Factorization I}}$$

$$\underbrace{x^n - 1 = (x-1)(x-\zeta^1)(x-\zeta^2)\dots(x-\zeta^{n-1})}_{\textit{Factorization II}}$$

From factorizations I and II:

$$(x-1)(x^{n-1} + x^{n-2} + \dots + x + 1) = (x-1)(x-\zeta^1)(x-\zeta^2)\dots(x-\zeta^{n-1})$$

Cancel $x-1$ on both sides:

$$x^{n-1} + x^{n-2} + \dots + x + 1 = (x-\zeta^1)(x-\zeta^2)\dots(x-\zeta^{n-1})$$

The RHS can also be written as a product:

$$x^{n-1} + x^{n-2} + \dots + x + 1 = (x-\zeta^1)(x-\zeta^2)\dots(x-\zeta^{n-1}) = \prod_{k=1}^{k=n-1}[x-\zeta^k]$$

Example 3.41

$$\prod_{k=1}^{k=n-1}[1-\zeta^k] = n$$

$$\prod_{k=1}^{k=n-1}[x-\zeta^k] = x^{n-1} + x^{n-2} + \dots + x + 1$$

Substitute $x = 1$ in the above:

$$\prod_{k=1}^{k=n-1}[1-\zeta^k] = 1^{n-1} + 1^{n-2} + \dots + 1 + 1 = n$$

$$\prod_{k=1}^{k=n-1}[1-\zeta^k] = n$$

Example 3.42[4]

$$\prod_{k=1}^{k=n-1}\sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}, \qquad n \in \mathbb{N}$$

Using the property $\sin\left(\frac{k\pi}{n}\right) = \frac{1}{2i} \times cis\left(\frac{k\pi}{n}\right)\left[1 - cis\left(-\frac{2k\pi}{n}\right)\right]$, split the product:

$$\underbrace{\prod_{k=1}^{k=n-1}\frac{1}{2i}}_{\textit{First Product}} \times \underbrace{\prod_{k=1}^{k=n-1} cis\left(\frac{k\pi}{n}\right)}_{\textit{Second Product}} \times \underbrace{\prod_{k=1}^{k=n-1}\left[1 - cis\left(-\frac{2k\pi}{n}\right)\right]}_{\textit{Third Product}}$$

First Product

$$\prod_{k=1}^{k=n-1}\frac{1}{2i} = \frac{1}{2^{n-1}i^{n-1}}$$

Second Product

[4] The Note on Trigonometry has examples on the use of this property.

Use the property $\prod_{k=1}^{n} cis\ k = cis\left(\sum_{k=1}^{n} k\right)$, simplifying and then using De Moivre's Theorem:

$$\prod_{k=1}^{k=n-1} cis\left(\frac{k\pi}{n}\right) = cis\left(\sum_{k=1}^{n-1} k\frac{\pi}{n}\right) = cis\left(\frac{n(n-1)}{2}\cdot\frac{\pi}{n}\right) = cis\left((n-1)\cdot\frac{\pi}{2}\right) = \left[cis\ \frac{\pi}{2}\right]^{n-1} = i^{n-1}$$

Third Product

We know that $cis\left(-\frac{2k\pi}{n}\right)$ is a root of unity. Substitute $cis\left(-\frac{2k\pi}{n}\right) = \zeta^k$, and use the property:

$$= \prod_{k=1}^{k=n-1}[1-\zeta^k] = n$$

3.4 Roots of Unity: Cube Roots

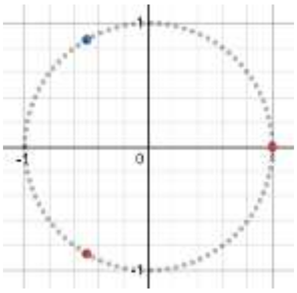
A. Value of the Roots

3.43: Cube Roots of Unity

ω refers to a cube root of unity. If $z^3 = 1$, then:

$$z \in \left\{1, -\frac{1}{2}+\frac{\sqrt{3}}{2}i, -\frac{1}{2}-\frac{\sqrt{3}}{2}i\right\}$$

Algebraic Method



Any cube root of unity must satisfy the relation:

$$\omega = \sqrt[3]{1}$$

Cube both sides, and collate all terms on the LHS:

$$\omega^3 = 1$$
$$\omega^3 - 1 = 0$$

Factor the LHS using $a^3 - b^3 = (a-b)(a^2+ab+b^2)$:

$$(\omega-1)(\omega^2+\omega+1) = 0$$

Apply the zero-product property.
The first term gives us 1, which we already know is a solution:

$$\omega - 1 = 0 \Rightarrow \omega = 1$$

The second term is a quadratic:

$$\omega^2+\omega+1 = 0$$

We can't factor it easily, so apply the quadratic formula with $a = 1, b = 1, c = 1$ to get the other two roots:

$$\omega = \frac{-b\pm\sqrt{b^2-4ac}}{2a} = \frac{-1\pm\sqrt{-3}}{2} = -\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$$

$$z \in \left\{1, \underbrace{-\frac{1}{2}+\frac{\sqrt{3}}{2}i}_{\omega}, \underbrace{-\frac{1}{2}-\frac{\sqrt{3}}{2}i}_{\omega^2}\right\}$$

Example 3.44

If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $(x-1)^3 + 8 = 0$ in terms of ω are: (JEE Adv. 1979, JEE Main 2005)

Let $z = x - 1$

$$z^3 + 8 = 0$$
$$z^3 = -8(1) = (-2)^3(1)$$

The solutions to:

$$Z^3 = 1 \Rightarrow Z \in \{1, \omega, \omega^2\}$$
$$z = (-2)^3 Z^3 = (-2)^3 \Rightarrow z \in \{-2, -2\omega, -2\omega^2\}$$

$$z = -2 \Rightarrow x - 1 = -2 \Rightarrow x = -1$$
$$z = -2\omega \Rightarrow x - 1 = -2\omega \Rightarrow x = 1 - 2\omega$$
$$z = -2\omega^2 \Rightarrow x - 1 = -2\omega^2 \Rightarrow x = 1 - 2\omega^2$$

Example 3.45

Let z_1 and z_2 be n^{th} roots of unity which subtend a right angle at the origin. Then n must be a multiple of which integer: **(JEE Adv. 2001S, Adapted)**

Without loss of generality, let one root be 1. The angle between the t

$$For\ k \in \mathbb{Z}: \frac{360k}{n} = 90 \Rightarrow n = 4k$$

3.46: Polar Form of Cube Roots of Unity

$$z \in \{1\ cis\ 0, 1\ cis\ 120°, 1\ cis\ 240°\}$$

$$z = \sqrt[n]{1} \Rightarrow r = 1, \theta = \frac{2k\pi}{n}, k \in \mathbb{Z}$$
$$\theta = \left(\frac{360}{3}\right)k = (120k)°$$
$$\theta = \{0°, 120°, 240°\}$$

3.47: Omega Notation

ω is any cube root of unity which is not unity.

$$\omega^3 = 1$$
$$\omega \neq 1$$

$$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i = 1\ cis\ 120°$$
$$\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = 1\ cis\ 240°$$

Example 3.48

Show that

A. ω^2 is the square of ω
B. ω is the square of ω^2

Method I: Using Polar Form

$$\omega^2 = (1\ cis\ 120°)^2 = 1^2\ cis\ (2 \times 120) = 1\ cis\ 240° = \omega^2$$
$$(\omega^2)^2 = \omega^4 = 1^4\ cis\ (4 \times 120°) = 1\ cis\ 480° = 1\ cis\ 120° = \omega$$

Method II: Using Algebraic Form

Use $(a+b)^2 = a^2 + 2ab + b^2$:

$$\omega^2 = \left(\underbrace{-\frac{1}{2}}_{a} + \underbrace{\frac{\sqrt{3}}{2}i}_{b}\right)^2 = \underbrace{\frac{1}{4}}_{a^2} + \underbrace{-\frac{3}{4}}_{b^2} + \underbrace{2\left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)i}_{2ab} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$$

B. Cyclicity of the Roots

Example 3.49

If z is a cube root of unity, find the value of

$$z^{2022}$$

$$z^{2022}=(z^3)^{673}=1^{673}=1$$

Note that the number 673 was not important in our arriving at the final answer. We could have simply noted (using the tests of divisibility by 3, or otherwise) that the 2022 is an integer multiple of 3.

3.50: Cyclicity of Powers of Omega

$$\omega^{3k+1}=\omega^{3k}\omega=\omega$$
$$\omega^{3k+2}=\omega^{3k}\omega^2=\omega^2$$
$$\omega^{3k}=1$$

Example 3.51

A. ω^4
B. ω^5
C. ω^6
D. ω^{10}
E. ω^{17}
F. ω^{27}

$$\omega^4=\omega^3\omega=\omega$$
$$\omega^5=\omega^3\omega^2=\omega^2$$
$$\omega^6=\omega^3\omega^3=(1)(1)=1$$
$$\omega^{10}=\omega^9\omega=(\omega^3)^3\omega=\omega$$
$$\omega^{17}=\omega^{15}\omega^2=\omega^{15}\omega^2=\omega^2$$
$$\omega^{27}=(\omega^3)^9=1^9=1=\omega^0$$

Example 3.52

If $i=\sqrt{-1}$, then find the value of the complex number $z=a+bi$, given that

$$z=4+5\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^{334}+3\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^{365}$$ **(JEE Adv. 1999)**

$$z=4+5\omega^{334}+3\omega^{365}$$
$$=4+5\omega+3\omega^2$$
$$=(1+2\omega)+3+3\omega+3\omega^2$$
$$=(1+2\omega)+0$$
$$=1+2\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)$$
$$=1-1+\sqrt{3}i$$
$$=\sqrt{3}i$$

Example 3.53

If $2\alpha = -1 - \sqrt{3}i$ and $2\beta = -1 + \sqrt{3}i$, then $5\alpha^4 + 5\beta^4 + 7\alpha^{-1}\beta^{-1}$ is equal to (AP EAPCET 17 Sep 2020, Shift-I)

Substitute $\alpha = -\frac{1}{2} - \frac{\sqrt{3}}{2}i = \omega$ and $\beta = -\frac{1}{2} + \frac{\sqrt{3}}{2}i = \omega^2$ in $5\alpha^4 + 5\beta^4 + \frac{7}{\alpha\beta}$:

$$5\omega^4 + 5(\omega^2)^4 + \frac{7}{\omega\omega^2}$$

Simplify:

$$5(\omega^4 + \omega^8) + \frac{7}{\omega^3}$$

$$5(\omega\omega^3 + \omega^6\omega^2) + \frac{7}{\omega^3}$$

Substitute $\omega^3 = 1$

$$5(\omega + \omega^2) + \frac{7}{1}$$

Substitute $\omega + \omega^2 = -1$:

$$5(-1) + 7 = 2$$

C. Sum of the Roots

3.54: Sum of Cube Roots of Unity

If $1, \omega, \omega^2$ are the cube roots of unity then:

$$1 + \omega + \omega^2 = 0$$

Method I: Using Vieta's Formulas

$$x = \sqrt[3]{1} \Rightarrow x^3 = 1 \Rightarrow x^3 - 1 = 0 \Rightarrow x^3 + 0x^2 + 0x - 1 = 0$$

$$Sum\ of\ Roots = -\frac{b}{a} = -\frac{0}{1} = -0$$

Method II: Using Addition

$$\underbrace{1}_{z_1} + \underbrace{\left(-\frac{1}{2} + \frac{\sqrt{3}i}{2}\right)}_{z_2} + \underbrace{\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)}_{z_3} = 1 - \frac{1}{2} - \frac{1}{2} + \frac{\sqrt{3}i}{2} - \frac{\sqrt{3}i}{2} = 1 - 1 = 0$$

Example 3.55

If $\omega \neq 1$ is such that $\omega^3 = 1$, then $\sum_{n=900}^{n=1000} \omega^n$ is (JMET 2011/86)

Expand the sum:

$$\sum_{n=900}^{n=1000} \omega^n = \omega^{900} + \omega^{901} + \cdots + \omega^{1000}$$

Note that:

$$\omega^{900} = (\omega^3)^{300} = 1, \qquad \omega^{901} = (\omega^3)^{300}\omega = 1 \times \omega = \omega, \qquad \omega^{902} = (\omega^3)^{300}\omega^2 = 1 \times \omega^2 = \omega$$

Make the substitutions:

$$= 1 + \omega^1 + \omega^2 + \cdots + \omega^{99} + \omega^{100}$$

Note that

$$1 + \omega^1 + \omega^2 = 0, \omega^3 + \omega^4 + \omega^5 = \omega^3(1 + \omega^1 + \omega^2) = \omega^3 \times 0 = 1 \times 0 = 0$$

Hence, successive groups of three terms add up zero:

$$= 0 + 0 + \cdots + 0 + \omega^{99} + \omega^{100}$$
$$= 0 + 0 + \cdots + 0 + 1 + \omega$$
$$= 1 + \omega$$

Example 3.56

If $\omega \neq 1$ is a cube root of unity:

$$1 + \omega + \omega^2 + \cdots + \omega^{2023}$$

$$= 1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \cdots + \omega^{2023}$$
$$= 1 + \omega + \omega^2 + \omega^3(1 + \omega + \omega^2) + \omega^6(1 + \omega + \omega^2) + \cdots + \omega^{2022} + \omega^{2023}$$
$$= 0 + \omega^3(0) + \omega^6(0) + \cdots + 1 + \omega$$
$$= 1 + \omega$$

Example 3.57

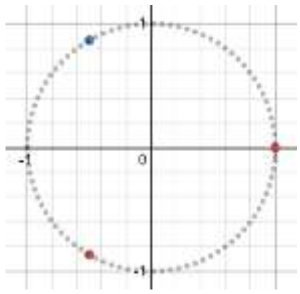
Write $1 - \omega$ in terms of ω and ω^2.

$$1 + \omega + \omega^2 = 0$$
$$1 = -\omega - \omega^2$$
$$1 - \omega = -2\omega - \omega^2$$

3.58: Sum of Cube Roots of Unity

If $1, \omega, \omega^2$ are the cube roots of unity then:

$$1 + \omega = -\omega^2$$



$$\omega = 1\ cis\ 120°$$
$$1 + \omega = 1 + \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = +\frac{1}{2} + \frac{\sqrt{3}}{2}i = 1\ cis\ 60°$$

$$-\omega^2 = -(1\ cis\ 120°)^2 = -1\ cis\ 240°$$

To remove the minus sign, we reflect it across the origin:

$$= 1\ cis\ (240° - 180°) = 1\ cis\ 60°$$

(Important) Example 3.59[5]

If ω is one of the imaginary roots of the equation $x^3 = 1$, then the product $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$ is equal to: **(AHSME 1971/22)**

Note that:

$$1 + \omega = -\omega^2$$
$$1 + \omega^2 = -\omega$$

$$(-\omega - \omega)(-\omega^2 - \omega^2) = (-2\omega)(-2\omega^2) = 4\omega^3 = 4$$

Example 3.60

If $\omega \neq 1$ is a cube root of unity and $(1 + \omega)^7 = A + B\omega$ then A and B are respectively: **(JEE Adv. 1995S, JEE Main 2011)**

[5] The technique used for solving this question is useful in a range of questions.

$$1+\omega+\omega^2=0 \Rightarrow 1+\omega=-\omega^2$$

$$(1+\omega)^7=(-\omega^2)^7$$

Expand using exponent rules:

$$=-\omega^{14}$$

Split:

$$=-\omega^{12}\omega^2=-(\omega^3)^4\omega^2$$

Substitute $\omega^3=1$

$$=-(1)^4\omega^2=-\omega^2=1+\omega=A+B\omega$$

$$A=1, B=1$$

Example 3.61

If $1, \omega, \omega^2$ are the cube roots of unity, then $\frac{1}{1+2\omega}+\frac{1}{2+\omega}-\frac{1}{1+\omega}=$ **(AP EAPCET, 22 April. 2018, Shift-I)**

Add the first two fractions:

$$\frac{2+\omega+1+2\omega}{(1+2\omega)(2+\omega)}-\frac{1}{(1+\omega)}=\frac{3+3\omega}{(1+2\omega)(2+\omega)}-\frac{1}{(1+\omega)}$$

Add the two remaining fractions:

$$=\frac{(3+3\omega)(1+\omega)-(1+2\omega)(2+\omega)}{(1+2\omega)(2+\omega)(1+\omega)}$$

Focus on the numerator:

$$=3+3\omega+3\omega+3\omega^2-(2+\omega+4\omega+2\omega^2)=3+6\omega+3\omega^2-2-\omega-4\omega-2\omega^2=1+\omega+\omega^2=0$$

3.62: Sum of non-real Cube Roots of Unity

$$\omega+\omega^2=-1$$

Method I: Using Addition

$$\omega+\omega^2=\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)+\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)=-1$$

D. Challenging Questions

Example 3.63

How many positive integers n are there such that $3 \le n \le 100$ and $x^{2^n}+x+1$ is divisible by x^2+x+1? **(IOQM 2019/8)**

We want

$$\frac{x^{2^n}+x+1}{x^2+x+1} \ \textit{should have remainder zero}$$

Consider a simple division. Note that we have no remainder precisely when the denominator has the same roots as the numerator:

$$\frac{x^2+5x+6}{x+2}=\frac{(x+2)(x+3)}{x+2}=x+3$$

The solutions to x^2+x+1 are the cube roots of unity which are:

$$\omega \text{ and } \omega^2 \Rightarrow \omega^2+\omega+1=0 \Rightarrow \omega+1=-\omega^2$$

These values must also be roots for $x^{2^n}+x+1$.[6] Substitute $x=\omega$ in $x^{2^n}+x+1$

$$\omega^{2^n}+\omega+1=0$$

Substitute $\omega+1=-\omega^2$:

$$\omega^{2^n}-\omega^2=0$$
$$\omega^{2^n}=\omega^2$$

Since $\omega^{3k+2}=\omega^{3k}\omega^2=\omega^2$:

$$2^n=3k+2$$
$$2^1=2=3(0)+2$$
$$2^2=4=3(1)+1$$
$$2^3=8=3(2)+2$$
$$2^4=16=3(5)+1$$
$$2^5=32=3(10)+2$$

$$n\in\{3,5,\dots,99\}\to\{2,4,\dots,98\}\to\{1,2,\dots,49\}\Rightarrow 49\ Values$$

Example 3.64

If ω is a complex cube root of unity then $\sin\left[(\omega^{10}+\omega^{23})\pi-\frac{\pi}{4}\right]=$ (AP EAPCET, 22 Sep 2020, Shift-II)

Reduce the powers of ω:

$$\sin\left[(\omega+\omega^2)\pi-\frac{\pi}{4}\right]$$

Substitute $\omega+\omega^2=-1$:

$$\sin\left(-\pi-\frac{\pi}{4}\right)$$

Use $\sin(-\theta)=-\sin\theta$:

$$-\sin\left(\pi+\frac{\pi}{4}\right)$$

Use $\sin(\pi+\theta)=-\sin\theta$:

$$\sin\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$$

Example 3.65: Rational Roots Theorem / Polynomial Long Division

If α and β are non-real roots of $x^3-x^2-x-2=0$, then $\alpha^{2020}+\beta^{2020}+\alpha^{2020}\cdot\beta^{2020}=$ (AP EAPCET 18 Sep 2020, Shift-I)

Note: Your answer should be simplified and in terms of both α and β

From the Rational Roots Theorem, the possible rational roots of $P(x)=x^3-x^2-x-2=0$ are the positive and negative factors of 2.

$$Factors\ of\ 2\in\{\pm1,\pm2\}$$

[6] Numerator can and will have more roots than denominator. But roots of denominator must be present in numerator.

Use the Remainder Theorem to check the values from the set above where $P(x) = 0$

$$P(2) = 2^3 - 2^2 - 2 - 2 = 8 - 8 = 0 \Rightarrow x - 2 \ is \ a \ factor$$

By Polynomial Long Division:

$$x^3 - x^2 - x - 2 = (x - 2)(x^2 + x + 1) = 0$$

Recognize by observation that the roots of $x^2 + x + 1$ are the non-real cube roots of unity.
Since $\alpha^{2020} + \beta^{2020} + \alpha^{2020} \cdot \beta^{2020}$ is symmetric, without loss of generality let $\alpha = \omega, \beta = \omega^2$. Substituting:

$$\omega^{2020} + (\omega^2)^{2020} + \omega^{2020} \cdot (\omega^2)^{2020}$$
$$= \omega^{2020} + \omega^{4040} + \omega^{6060}$$

Since $\omega^3 = 1$, check the remainder for each exponent when divided by 3:

$$= \omega + \omega^2 + 1 = \alpha + \beta + 1$$

Example 3.66

$\left(-i + \sqrt{3}\right)^{300} + \left(-i - \sqrt{3}\right)^{300} =$ **(AP EAPCET 21 Sep 2020, Shift I)**

Multiply each term of the expression by $-1 = -1^{75} = -(i^4)^{75} = -i^{300}$:

$$-i^{300}\left(-i + \sqrt{3}\right)^{300} - i^{300}\left(-i - \sqrt{3}\right)^{300}$$

Use the property $a^m b^m = (ab)^m$:

$$= \left[-i\left(-i + \sqrt{3}\right)\right]^{300} + \left[-i\left(-i - \sqrt{3}\right)\right]^{300}$$
$$= \left[-1 - \sqrt{3}i\right]^{300} + \left[-1 + \sqrt{3}i\right]^{300}$$

Factor 2:

$$= \left[2\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\right]^{300} + \left[2\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\right]^{300}$$

Move the 2 out of the brackets, and substitute $\omega = -\frac{1}{2} - \frac{\sqrt{3}i}{2}, \omega^2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$:

$$= 2^{300}\omega^{300} + 2^{300}(\omega^2)^{300}$$

Note 300 and 600 are both divisible by 3, and hence substitute $\omega^{300} = \omega^{600} = 1$:

$$= 2^{300}(1) + 2^{300}(1) = 2^{301}$$

Example 3.67

The triangle formed by joining 1, the origin and $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ on the complex plane is ____.

$$Equilateral$$

3.5 Binomial Theorem

A. Binomial Formula

3.68: Binomial Formula

$$(x+y)^n=\sum_{r=0}^{n}\binom{n}{r}x^{n-r}y^r=\binom{n}{0}x^ny^0+\binom{n}{1}x^{n-1}y^1+\binom{n}{2}x^{n-2}y^2+\cdots+\binom{n}{n}x^0y^n$$

Example 3.69

If $z=2+3i$, then $z^5+(\bar{z})^5$ is equal to:

Write out the binomial expansion:

$$(x+y)^5=\binom{5}{0}x^5y^0+\binom{5}{1}x^4y^1+\binom{5}{2}x^3y^2+\binom{5}{3}x^2y^3+\binom{5}{4}x^1y^4+\binom{5}{5}x^0y^5$$
$$(x-y)^5=\binom{5}{0}x^5y^0-\binom{5}{1}x^4y^1+\binom{5}{2}x^3y^2-\binom{5}{3}x^2y^3+\binom{5}{4}x^1y^4-\binom{5}{5}x^0y^5$$

Add the two to get:

$$(x+y)^5+(x-y)^5=2\left[\binom{5}{0}x^5y^0+\binom{5}{2}x^3y^2+\binom{5}{4}x^1y^4\right]$$

Substitute $x=2, y=3i$ in the above:

$$=2[1\cdot 2^5(3i)^0+10\cdot 2^3\cdot(3i)^2+5\cdot 2^1(3i)^4]$$

$$\begin{aligned}&=2[32+10\cdot 8\cdot(-9)+5\cdot 2\cdot 81]\\&=2[32-720+810]\\&=2[32-720+810]\\&=244\end{aligned}$$

3.70: Sum/Difference of Binomial Conjugate Expansions

$$(x+y)^n+(x-y)^n=2\left[\binom{n}{0}x^ny^0+\binom{n}{2}x^{n-2}y^2+\binom{n}{4}x^{n-4}y^4+\binom{n}{6}x^{n-6}y^6+\cdots\right]$$
$$(x+y)^n-(x-y)^n=2\left[\binom{n}{1}x^{n-1}y^1+\binom{n}{3}x^{n-3}y^3+\binom{n}{5}x^{n-5}y^5+\binom{n}{7}x^{n-7}y^7+\cdots\right]$$

$$(x+y)^n=\binom{n}{0}x^ny^0+\binom{n}{1}x^{n-1}y^1+\binom{n}{2}x^{n-2}y^2+\binom{n}{3}x^{n-3}y^3+\cdots$$
$$(x-y)^n=\binom{n}{0}x^ny^0-\binom{n}{1}x^{n-1}y^1+\binom{n}{2}x^{n-2}y^2-\binom{n}{3}x^{n-3}y^3+\cdots$$

Sum

Add the two equations above and note that the violet terms (even terms) all cancel:

$$(x+y)^n+(x-y)^n=2\left[\underbrace{\binom{n}{0}x^ny^0}_{1st\ Term}+\underbrace{\binom{n}{2}x^{n-2}y^2}_{3rd\ Term}+\underbrace{\binom{n}{4}x^{n-4}y^4}_{5th\ Term}+\underbrace{\binom{n}{6}x^{n-6}y^6}_{7th\ Term}+\cdots\right]$$

$$(x+y)^n+(x-y)^n=2\underbrace{\sum_{r=0}^{n}\binom{n}{r}x^{n-r}y^r}_{\substack{r\ even\\ odd\ r\ cancels}}$$

Difference

Subtract the second equation from the first, and note the violet terms double, and the other terms vanish:

$$(x+y)^n-(x-y)^n=2\left[\underbrace{\binom{n}{1}x^ny^0}_{2nd\ Term}+\underbrace{\binom{n}{3}x^{n-2}y^2}_{4th\ Term}+\underbrace{\binom{n}{5}x^{n-4}y^4}_{6th\ Term}+\underbrace{\binom{n}{7}x^{n-6}y^6}_{8th\ Term}+\cdots\right]$$

$$(x+a)^n - (x-a)^n = 2\underbrace{\sum_{r=0}^{n}\binom{n}{r}x^{n-r}a^r}_{\substack{r\ odd \\ even\ r\ cancels}}$$

3.6 Matrices

Example 3.71

$$(1+i) = 1+i$$
$$(1+i)^2 = 1+2i+i^2 = 2i$$
$$(1+i)^3 = 1+2i+i^2 = -2+2i$$
$$(1+i)^4 = -4$$

A. Matrices as Complex Numbers

Complex numbers can be represented as matrices[7].

3.72: Multiplicative Identity

3.73: Multiplicative Identity (Real Numbers)

In real numbers, the number 1 is special because anything multiplied by 1 equal itself. That is:

$$a \times 1 = 1 \times a = a$$

We give this a name by calling 1 the multiplicative identity in the real numbers.

3.74: Multiplicative Identity (Matrices)

Matrices also have a multiplicative identity, given by

$$\boldsymbol{I_2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The matrix I has similar properties to the number 1 in the real number system:

$$AI = IA = A$$

Example 3.75: Numbers

Since we have a 1, we can multiply it by a real number x to get a matrix corresponding to x. For example

$$5I \rightarrow Matrix\ 5$$

3.76: Iota

We want a matrix that satisfies the identity

$$\boldsymbol{i^2} = -1$$

$$\boldsymbol{i} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Rightarrow i^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\boldsymbol{I_2}$$

Example 3.77: Multiples of Iota

3.78: Complex Number

[7] For example, this article derives the same properties as the ones in this section.

The complex number $a+bi, i^2=-1, a \in \mathbb{R}, b \in \mathbb{R}$ corresponds to the matrix

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$a+bi = a\boldsymbol{I} + b\boldsymbol{i} = a\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} + \begin{bmatrix} 0 & -b \\ b & 0 \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

3.79: Addition and Subtraction

Example 3.80

A. Find the sum
B. Find the difference
C. Find missing values

3.81: Conjugate

Finding the conjugate of a complex number is equivalent to finding the transpose of its matrix equivalent.

Example 3.82

Find the conjugate

3.83: Multiplication

Multiplying two complex numbers is equivalent to multiplying the two matrix equivalents.

3.84: Commutativity

In general, matrix multiplication is not commutative. However, multiplication of complex numbers represented as matrices is commutative.

B. Division

3.85: Inverse of a Matrix

3.86: Reciprocal

3.87: Division

We have been looking at the rectangular form of complex numbers as matrices. We can do the same with the polar and exponential forms.

3.88:

Rotation

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Scaling

$$r$$

3.89:
Combination of Rotation and Scaling

3.7 Further Topics

A. References and Further Resources

Notes 3.90
These notes have an expository style that focuses on explanation. There are a few numericals as well.

Video 3.91
This video is a fast-paced look at solving problems using complex numbers (with some attention paid to properties as well)

92 Examples